

Chapter 1: Introduction to Data Communication

1.1 Data Communication

Definition: Data communication is the exchange of data between two or more devices through some form of transmission medium.

Data can be in any format like text, image, number, audio, and video.

Fundamental characteristics of data Communication system:

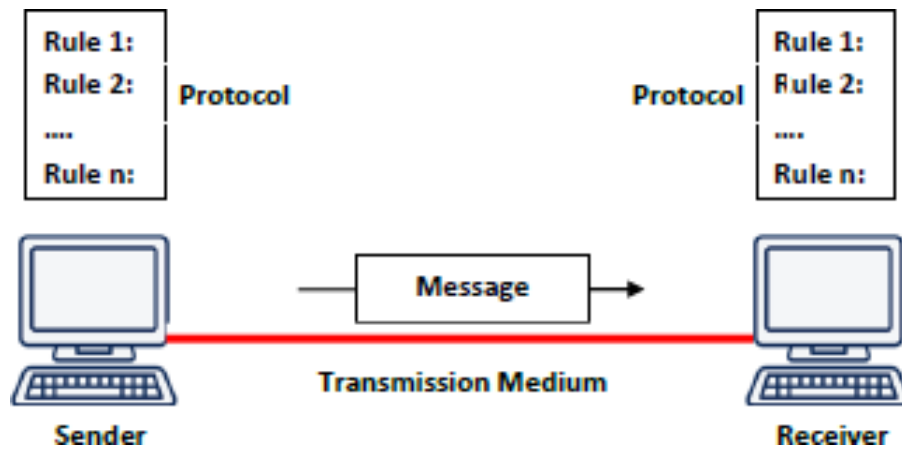
1. **Delivery:** The Data must be delivered to the correct and intended destination.
2. **Accuracy:** The system must deliver the data accurately without any modification.
3. **Timeliness:** The data must be delivered in a timely manner. If the data is delivered late then it becomes useless.
4. **Jitter:** Jitter is the variation in the packet arrival time.

For example, suppose in a live streaming of cricket, video packets are sent for every 8ms. If some of the packets arrive with a delay of 8ms and others with 10ms delay, then it results in an uneven quality in the video.

1.1.1 Components of Data Communication

A data communications system has five components as shown in figure 1.1.

1. **Message:** The message is the data to be transmitted. Data can be text, numbers, pictures, audio, and video.
2. **Sender:** The sender is the device that sends the data. It can be a computer, workstation, telephone handset, video camera, and so on.
3. **Receiver:** The receiver is the device that receives the data. It can be a computer, workstation, telephone handset, television, and so on.
4. **Transmission medium:** The transmission medium is the physical path by which a message travels from sender to receiver. It can be wired or wireless medium.
5. **Protocol:** A protocol is a set of rules that govern data communications. It represents an agreement between the communicating devices.



1.1.2 Data Representation

Data can be represented in different forms like text, numbers, images, audio, and video.

1) Text

- Text is represented as a bit pattern, which is a sequence of bits made up of 0's and 1's.
- Different sets of bit patterns are used to represent each text symbol. These sets of bit patterns are called as code and the process of representing symbols is called coding.
- Most Common coding system is **Unicode** which uses 32 bits to represent a symbol. The **American Standard Code for Information Interchange (ASCII)** constitutes the first 127 characters in Unicode.

2) Numbers

The Numbers are directly represented as bit patterns without using any coding scheme.

3) Images

- Image is composed of a matrix of pixels. Each pixel is represented by a bit pattern. The size and the value of the pattern depend on the image.
- Number of colors in an image is equal to 2^n where n is number of bits per pixel.
- For an image made up of only black and white dots, a 1-bit pattern is enough to represent a pixel.
- Color Images can be represented in several ways. One method is called RGB, where each color is made of a combination of three primary colors: red, green, and blue. Another method is called YCM, in which a color is made of a combination of three other primary colors: yellow, cyan, and magenta.

4) Audio

Audio is different from text, numbers, or images. It is continuous, not discrete.

5) Video

Video can either be produced as a continuous entity, or it can be a combination of images, each a discrete entity, arranged to convey the idea of motion.

Example 1.1:

A color image uses 16 bits to represent a pixel. What is the maximum number of different colors that can be represented?

Solution:

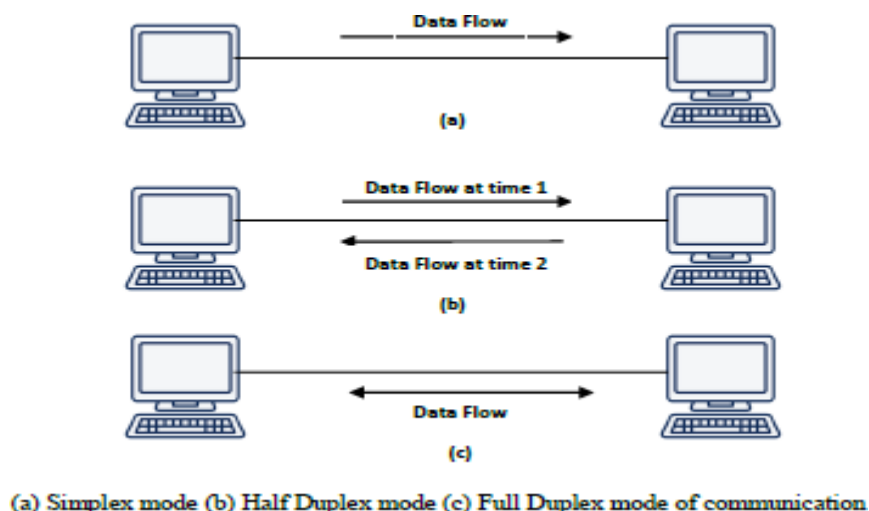
Given: $n = 16\text{bits}$

Maximum Number of colors in a image is $= 2^n = 2^{16} = 65,536$

1.1.4 Data Flow

Communication between two devices can be simplex, half-duplex, or full-duplex.

- **Simplex:** In simplex mode, the communication is unidirectional. Only one of the two devices on a link can transmit; the other can only receive.
- **Example:** Keyboards and traditional monitors.
- **Half-Duplex:** In half-duplex mode, each station can both transmit and receive, but not at the same time. When one device is sending, the other can only receive, and vice versa. The entire capacity of the channel can be utilized for each direction.
- **Example:** Walkie-talkies.
- **Full-Duplex:** In full-duplex mode (also called duplex), both stations can transmit and receive simultaneously. The capacity of the channel is divided between the two directions.
- **Example:** telephone network



1.2 Networks

- A network is a set of devices (*nodes*) connected through communication links. A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.
- Most networks use **distributed processing**, in which a task is divided among multiple computers.

1.2.1 Network Criteria

The networks should meet some criteria, they are:

1) Performance

- Performance can be measured using transit time and response time.
 - **Transit time** is the amount of time required for a message to travel from one device to another.
 - **Response time** is the elapsed time between an inquiry and a response.
- The performance of a network depends on a number of factors:
 - The number of users
 - The type of transmission medium
 - The capabilities of the connected hardware
 - The efficiency of the software.
- Performance can be evaluated using throughput and delay. Throughput should be more and delay should be less.

2) Reliability

Network reliability is measured by the frequency of failure, the time taken by a link to recover from a failure, and the network's robustness in a catastrophe.

3) Security

Network security includes protecting data from unauthorized access, and implementing policies and procedures for recovery from breaches and data losses.

1.2.2 Physical Structures

Type of Connection:

For communication to occur, two or more devices must be connected in some way. There are two possible types of connections they are point-to-point and multipoint connection.

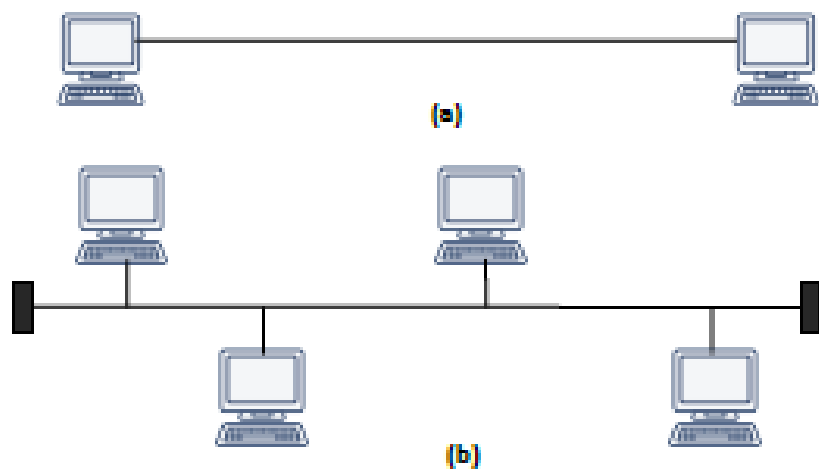
1) Point-to-Point

- A point-to-point connection provides a dedicated link between two devices.

- The entire capacity of the link is reserved for transmission between those two devices.

2) Multipoint

- A multipoint connection is one in which more than two specific devices share a single link.
- In a multipoint environment, the capacity of the channel is shared, either spatially or temporally.



(a) Point to point connection (b) Multipoint connection

1.2.3 Physical Topology

The term physical topology refers to the way in which a network is laid out physically. The topology of a network is the geometric representation of the relationship of all the links and linking devices to one another.

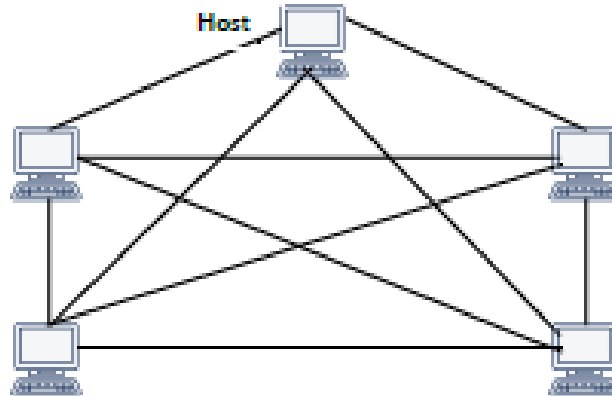
There are four basic topologies: mesh, star, bus, and ring.

1) Mesh Topology

- In a mesh topology, every device has a dedicated point-to-point link to every other device.

Consider a fully connected mesh network with n nodes in which each node must be connected to every other node. Node 1 must be connected to $n - 1$ nodes, node 2 must be connected to $n - 1$ nodes, and finally node n must be connected to $n - 1$ nodes. Hence $n(n - 1)$ physical links are required. If each physical link allows communication in both directions (duplex mode), we need $n(n - 1) / 2$ duplex-mode links.

- To accommodate many links, every device on the network must have $n - 1$ input/output (I/O) ports to be connected to the other $n - 1$ stations.



Advantages

- The use of dedicated links guarantees that each connection can carry its own data load, thus eliminating the traffic problems that can occur when links are shared between multiple devices.
- A mesh topology is robust. If one link becomes unusable, it does not affect the entire system.
- There is the advantage of privacy or security. When every message travels along a dedicated line, only the intended recipient sees it. Physical boundaries prevent other users from gaining access to messages.
- Point-to-point links make fault identification and fault isolation easy.

Disadvantages

- The amount of cabling and the number of I/O ports required is more because every device must be connected to every other device, installation and reconnection are difficult.
- The sheer bulk of the wiring can be greater than the available space (in walls, ceilings, or floors) can accommodate.
- The hardware required to connect each link (I/O ports and cable) is expensive.

Example 1.2

Assume six devices are arranged in a mesh topology. How many cables are needed?

How many ports are needed for each device?

Solution:

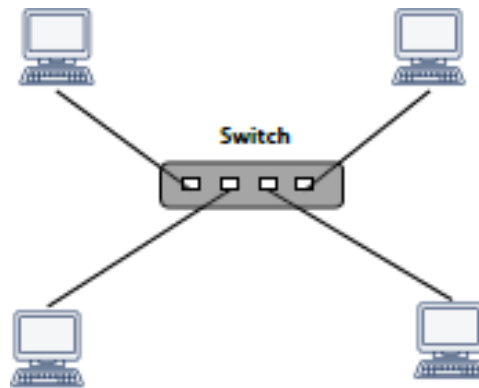
Number of cables = $n(n - 1) / 2 = 6(6 - 1) / 2 = 15$

Number of ports = $(n - 1) = 5$ ports needed per device

2) Star Topology

In a star topology, each device has a dedicated point-to-point link only to a central controller, called as switch.

The devices are not directly connected to one another. If one device wants to send data to another, it sends the data to the switch, which then relays the data to the other connected device.



Advantages:

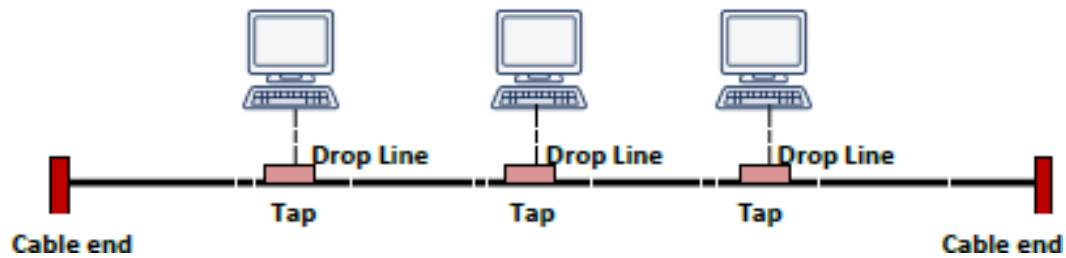
- A star topology is less expensive than a mesh topology.
- In a star, each device needs only one link and one I/O port to connect it to any number of others. This factor also makes it easy to install and reconfigure.
- The robustness. If one link fails, only that link is affected. All other links remain active.
- Fault isolation and identification is easy.
- As long as the hub is working, it can be used to monitor link problems and bypass defective links.

Disadvantages:

- One big disadvantage of a star topology is the dependency of the whole topology on one single point, the hub. If the hub goes down, the whole system is dead.

3) Bus Topology

A **bus topology** is multipoint. All the devices in a network are connected through a single link, which acts as a backbone.



- Nodes are connected to the bus cable by drop lines and taps.
- A drop line is a connection running between the device and the main cable.
- A tap is a connector that either splices into the main cable or punctures the sheathing of a cable to create a contact with the metallic core.
- As a signal travels along the backbone, some of its energy is transformed into heat. Therefore, it becomes weaker and weaker as it travels farther and farther.

Advantages:

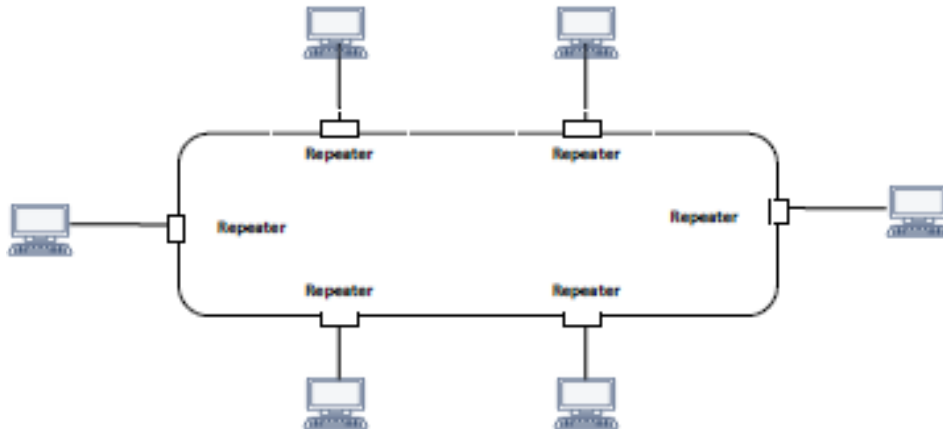
- Ease of Installation.
- A bus uses less cabling than mesh or star topologies.

Disadvantage:

- Difficult reconnection and fault isolation.
- A bus is usually designed to be optimally efficient at installation. It can therefore be difficult to add new devices.
- Signal reflection at the taps can cause degradation in quality.
- A fault or break in the bus cable stops all transmission, even between devices on the sameside of the problem

4) Ring Topology

- In a ring topology, each device has a dedicated point-to-point connection with only the two devices on either side of it.
- A signal is passed along the ring in one direction until it reaches its destination.
- Each device in the ring uses a repeater. When a device receives a signal intended for another device, its repeater regenerates the bits and passes them along



Advantages:

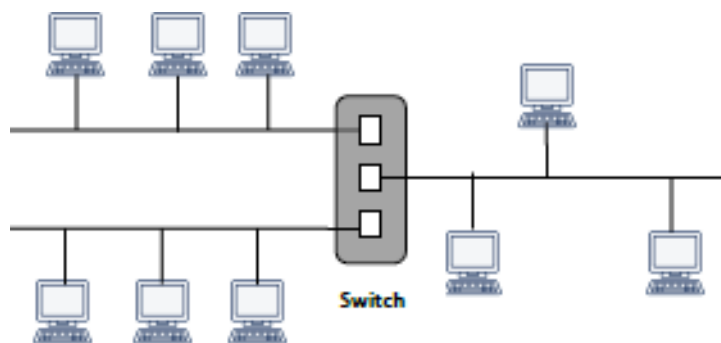
- A ring is relatively easy to install and reconfigure.
- Each device is linked to only its immediate neighbors. Hence adding or deleting a device requires changing only two connections.
- Fault Isolation is simplified.

Disadvantages:

- Unidirectional traffic.
- In a simple ring, a break in the ring can disable the entire network.

5) Hybrid Topology

A network can be hybrid. Two or more different kinds of topologies can be combined to form hybrid topology. For example, we can have a main star topology with each branch connecting several stations in a bus topology as shown in figure 1.8.

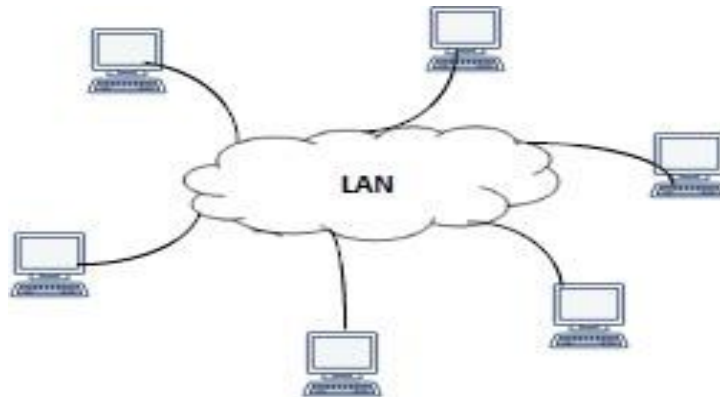


1.3 Network Types

1) Local Area Network

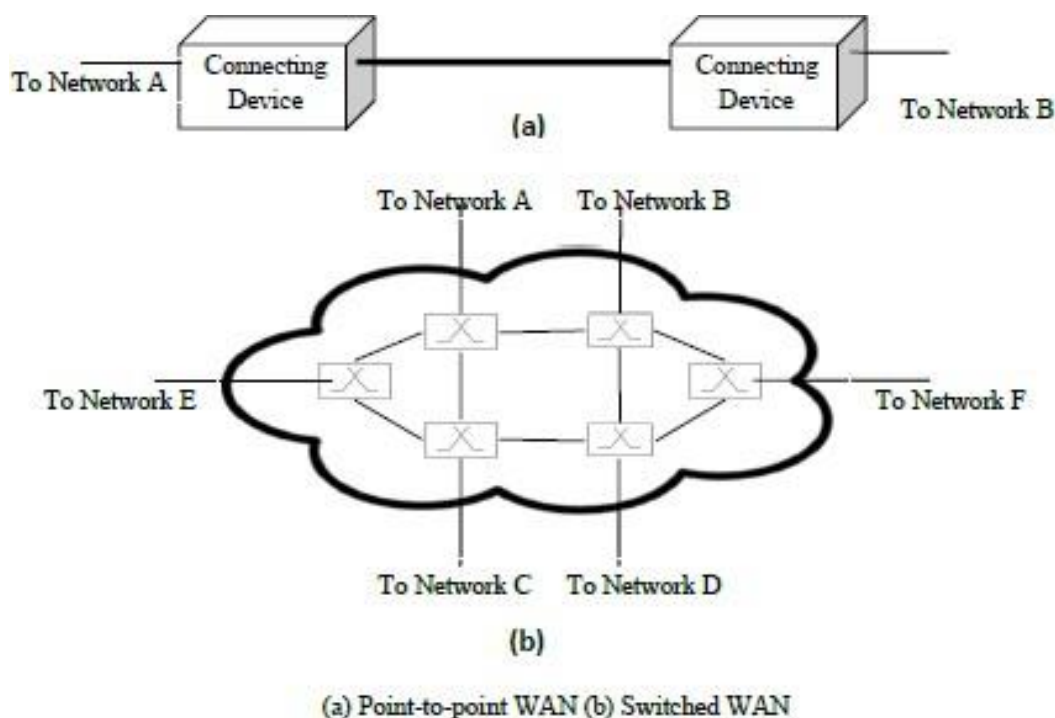
- A local area network (LAN) is usually privately owned.
- LAN connects the devices in a single office, building, or campus.

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- Depending on the needs of an organization and the type of technology used, a LAN can be as simple with just two PCs and a printer; or it can extend throughout a company and include audio and video peripherals.
 - LAN size is limited to a few kilometers.
 - The most common LAN topologies are bus, ring, and star. LAN speeds are normally 100 or 1000 Mbps.



2) Wide Area Network

- A wide area network (WAN) provides long-distance transmission of data, image, audio, and video information over large geographic areas that may comprise a country, a continent, or even the whole world.
- The switched WAN connects the end systems, which usually comprise a router that connects to another LAN or WAN.
- The point-to-point WAN is normally a line leased from a telephone or cable TV provider that connects a home computer or a small LAN to an Internet service provider (ISP). This type of WAN is often used to provide Internet access.



3) Metropolitan Area Networks

A metropolitan area network (MAN) is a network with a size between a LAN and a WAN. It normally covers the area inside a town or a city. It is designed for customers who need a high-speed connectivity, normally to the Internet, and have endpoints spread over a city or part of city.

Example: high-speed DSL line, cable TV network.

1.3.1 Internetwork

Connection of two or more networks is called as internetwork or internet.

1.3.2 Switching

An internet is a switched network in which a switch connects at least two links together. The two most common types of switched networks are circuit-switched and packet-switched networks.

Circuit-Switched Network

- In a **circuit-switched network**, a dedicated connection, called a circuit, is always available between the two end systems; the switch can only make it active or inactive.
- Example: telephone network
- Circuit switched networks requires resources to be reserved for each pair of end users. The reservation of the network resources for each user results in an inefficient use of bandwidth for applications in which information transfer is busty or if the information is small.

Packet-Switched Network

- In a computer network, the communication between the two ends is done in blocks of data called **packets**.
- Packet switched networks are the building blocks of computer communication systems in which data units known as packets flow across the networks.
- It provides flexible communication in handling all kinds of connections for a wide range of applications e.g. telephone calls, video conferencing, distributed data processing etc...

1.4 The Internet

An internet (note the lowercase *i*) is two or more networks that can communicate with each other. The most notable internet is called the **Internet** (uppercase *I*), and is composed of thousands of interconnected networks.

1.4.1 Accessing the Internet

In order to access the Internet the user needs to be physically connected to an ISP (Internet service provider). The physical connection is normally done through a point-to-point WAN.

- **Using Telephone Networks**
 - **Dial-up service.** A modem is added to telephone line. which converts data to voice. The software installed on the computer dials the ISP and imitates making a telephone connection. The dial-up service is very slow, and when the line is used for Internet connection.
 - **DSL Service.** The DSL service also allows the line to be used simultaneously for voice and data communication.
- **Using Cable Networks**
- A residence or a small business can be connected to the Internet by using cable network service. It provides a higher speed connection, but the speed varies depending on the number of neighbors that use the same cable.
- **Using Wireless Networks**
- Wireless connectivity has recently become increasingly popular. A household or a small business can use a combination of wireless and wired connections to access the Internet.
- **Direct Connection to the Internet**

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- A large organization or a large corporation can itself become a local ISP and be connected to the Internet. This can be done if the organization or the corporation leases a high-speed WAN from a carrier provider and connects itself to a regional ISP. For example, a company with several branches can create an internetwork and then connect the internetwork to the Internet.

1.4.2 Brief History

- In the mid-1960s, mainframe computers in research organizations were standalone devices. Computers from different manufacturers were unable to communicate with one another.
- The Advanced Research Projects Agency (ARPA) in the Department of Defense (DoD) was interested in finding a way to connect computers so that the researchers they funded could share their findings, thereby reducing costs and eliminating duplication of effort.
- In 1967, at an Association for Computing Machinery (ACM) meeting, ARPA presented its ideas for ARPANET, a small network of connected computers. The idea was that each host computer would be attached to a specialized computer, called an *interface message processor* (IMP). The IMPs, in turn, would be connected to one another. Each IMP had to be able to communicate with other IMPs as well as with its own attached host.
- By 1969, ARPANET was a reality. Four nodes, at the University of California at Los Angeles (UCLA), the University of California at Santa Barbara (UCSB), Stanford Research Institute (SRI), and the University of Utah, were connected via the IMPs to form a network. Software called the *Network Control Protocol* (NCP) provided communication between the hosts.
- In 1972, Vint Cerf and Bob Kahn, both of whom were part of the core ARPANET group, collaborated on what they called the *Internetting Project*¹. Cerf and Kahn's landmark 1973 paper outlined the protocols to achieve end-to-end delivery of packets. This paper on Transmission Control Protocol (TCP) included concepts such as encapsulation, the datagram, and the functions of a gateway.
- Authorities made a decision to split TCP into two protocols: Transmission Control Protocol (TCP) and Internetworking Protocol (IP). IP would handle datagram routing while TCP would be responsible for higher-level functions such as segmentation, reassembly, and error detection. The internetworking protocol became known as TCP/IP.

1.4.3 The Internet Today

The Internet today is made up of many wide- and local-area networks joined by connecting

devices and switching stations. Today most end users who want Internet connection use the services of Internet service providers (ISPs). There are international service providers, national service providers, regional service providers, and local service providers.

1.5 Standards and Administration

1.5.1 Internet Standards

An **Internet standard** is a thoroughly tested specification that is useful to and adhered to by those who work with the Internet.

There is a strict procedure by which a specification attains Internet standard status.

- A specification begins as an Internet draft. An **Internet draft** is a working document with no official status and a six-month lifetime.
- Upon recommendation from the Internet authorities, a draft may be published as a **Request for Comment (RFC)**.
- Each RFC is edited, assigned a number, and made available to all interested parties.
- RFCs go through maturity levels and are categorized according to their requirement level.

Maturity Levels

An RFC, during its lifetime, falls into one of six *maturity levels*:

- **Proposed Standard.** A proposed standard is a specification that is stable, well understood, and of sufficient interest to the Internet community. At this level, the specification is usually tested and implemented by several different groups.
- **Draft Standard.** A proposed standard is elevated to draft standard status after at least two successful independent and interoperable implementations. Barring difficulties, a draft standard, with modifications if specific problems are encountered, normally becomes an Internet standard.
- **Internet Standard.** A draft standard reaches Internet standard status after demonstrations of successful implementation.
- **Historic.** The historic RFCs are significant from a historical perspective. They either have been superseded by later specifications or have never passed the necessary maturity levels to become an Internet standard.
- **Experimental.** An RFC classified as experimental describes work related to an experimental situation that does not affect the operation of the Internet. Such an RFC should not be implemented in any functional Internet service.

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- **Informational.** An RFC classified as informational contains general, historical, or tutorial information related to the Internet. It is usually written by someone in a non-Internet organization, such as a vendor.

Requirement Levels

RFCs are classified into five *requirement levels*:

- **Required.** An RFC is labeled *required* if it must be implemented by all Internet systems to achieve minimum conformance.
- **Recommended.** An RFC labeled *recommended* is not required for minimum conformance; it is recommended because of its usefulness.
- **Elective.** An RFC labeled *elective* is not required and not recommended. However, a system can use it for its own benefit.
- **Limited Use.** An RFC labeled *limited use* should be used only in limited situations. Most of the experimental RFCs fall under this category.
- **Not Recommended.** An RFC labeled *not recommended* is inappropriate for general use. Normally a historic (deprecated) RFC may fall under this category.

1.5.2 Internet Administration

Various groups and organization coordinates the Internet issues. Few are listed below.

- **ISOC**

The **Internet Society (ISOC)** is an international, nonprofit organization formed in 1992 to provide support for the Internet standards process. ISOC also promotes research and other scholarly activities relating to the Internet.

- **IAB**

The **Internet Architecture Board (IAB)** is the technical advisor to the ISOC. The main purposes of the IAB are to oversee the continuing development of the TCP/IP Protocol Suite. IAB accomplishes this through its two primary components, the Internet Engineering Task Force (IETF) and the Internet Research Task Force (IRTF). Another responsibility of the IAB is the editorial management of the RFCs

- **IETF**

The **Internet Engineering Task Force (IETF)** is a forum of working groups managed by the Internet Engineering Steering Group (IESG). IETF is responsible for identifying operational problems and proposing solutions to these problems. IETF also develops and

reviews specifications intended as Internet standards.

- **IRTF**

The **Internet Research Task Force (IRTF)** is a forum of working groups managed by the Internet Research Steering Group (IRSG). IRTF focuses on long-term research topics related to Internet protocols, applications, architecture, and technology.

Chapter 2: Network Model

2.1 Protocol Layering

In data communication and networking, a protocol defines the rules that both the sender and receiver and all intermediate devices need to follow to be able to communicate effectively.

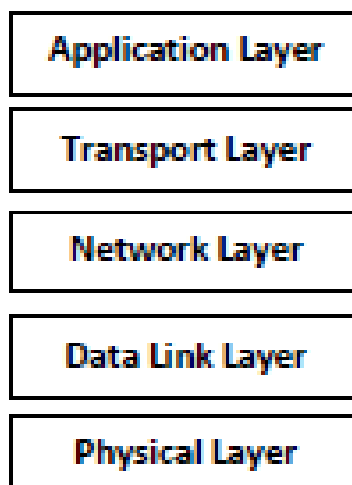
When communication is simple, we may need only one simple protocol; when the communication is complex, we may need to divide the task between different layers, in which case we need a protocol at each layer, or protocol layering.

2.1.1 Principles of Protocol Layering

- The first principle dictates that if we want bidirectional communication, we need to make each layer so that it is able to perform two opposite tasks, one in each direction.
- The second principle that we need to follow in protocol layering is that the two objects under each layer at both sites should be identical.

2.2 TCP/IP Protocol Suite

- TCP/IP is a protocol suite (a set of protocols organized in different layers) used in the Internet today.
- It is a hierarchical protocol made up of interactive modules, each of which provides a specific functionality. The term hierarchical means that each upper level protocol is supported by the services provided by one or more lower level protocols.



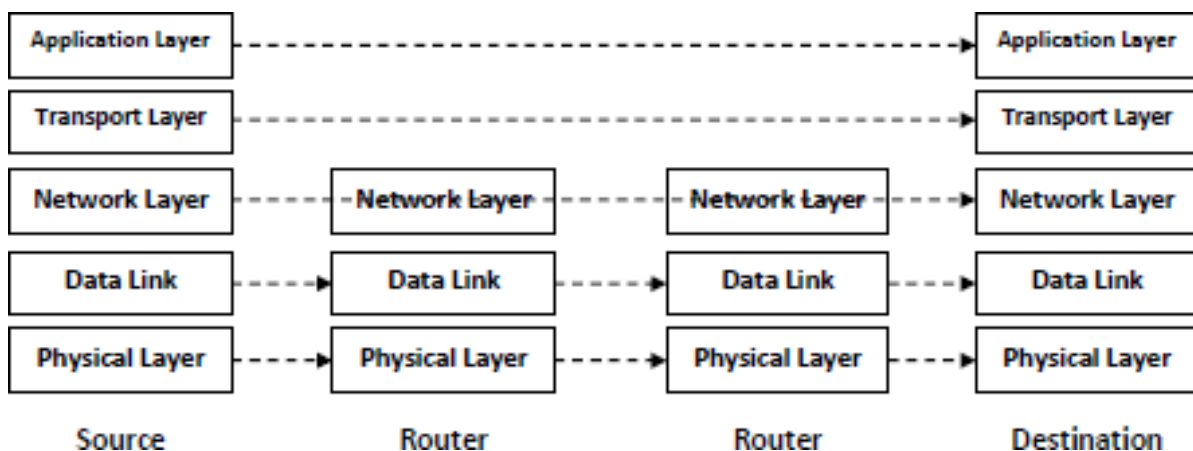
2.2.1 Layered Architecture

- Each device is involved with a set of layers depending on the role of the device in the internet. The two hosts (source and destination) are involved in all five layers.

- The source creates message in the application layer and send it down the layers so that it is physically sent to the destination. The destination host needs to receive the communication at the physical layer and then deliver it through the other layers to the application layer.
- The router is involved in only three layers: Network layer, Data Link layer, physical layer. There is no transport or application layer in a router as long as the router is used only for routing.
- A link-layer switch in a link, however, is involved only in two layers, data-link and physical.

2.2.2 Layers in the TCP/IP Protocol Suite

As the figure shows, the duty of the application, transport, and network layers is end-to-end. However, the duty of the data-link and physical layers is hop-to-hop, in which a hop is a host or router.



2.2.3 Description of Each Layer

1) Physical Layer

- Physical layer is responsible for carrying individual bits in a frame across the link.
- The communication between two devices at the physical layer is still a logical communication because there transmission media under the physical layer. Two devices are connected by a transmission medium (cable or air).
- The transmission medium does not carry bits; it carries electrical or optical signals. So the bits received in a frame from the data-link layer are transformed and sent through the transmission media.

2) Data-link Layer

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- There may be several overlapping sets of links that a datagram can travel from the host to the destination. The routers are responsible for choosing the best links.
 - The link can be a wired LAN with a link-layer switch, a wireless LAN, a wired WAN, or a wireless WAN. We can also have different protocols used with any link type. In each case, the data-link layer is responsible for moving the packet through the link.
 - TCP/IP does not define any specific protocol for the data-link layer. It supports all the standard and proprietary protocols. Any protocol that can take the datagram and carry it through the link suffices for the network layer. The data-link layer takes a datagram and encapsulates it in a packet called a frame.
 - Each link-layer protocol may provide a different service. Some link-layer protocols provide complete error detection and correction, some provide only error correction.

3) Network Layer

- The network layer is responsible for creating a connection between the source and destination.
- The network layer is responsible for host-to-host communication and routing the packet through possible routes.
- The network layer in the Internet includes the main protocol, Internet Protocol (IP), that defines the format of the packet, called a datagram at the network layer. IP also defines the format and the structure of addresses used in this layer.
- IP is also responsible for routing a packet from its source to its destination.
- IP is a connectionless protocol that provides no flow control, no error control, and no congestion control services. If any of these services is required for an application, the application should rely only on the transport-layer protocol.
- The network layer also has some auxiliary protocols that help IP in its delivery and routing tasks.
 - The Internet Control Message Protocol (ICMP) helps IP to report some problems when routing a packet.
 - The Internet Group Management Protocol (IGMP) is another protocol that helps IP in multitasking.
 - The Dynamic Host Configuration Protocol (DHCP) helps IP to get the network-layer address for a host.
 - The Address Resolution Protocol (ARP) is a protocol that helps IP to find the link-layer address of a host or a router when its network-layer address is given.

4) Transport Layer

- The transport layer at the source host gets the message from the application layer, encapsulates it in a transport layer packet called a segment and sends it, through the logical connection, to the transport layer at the destination host.
- The transport layer is responsible for giving services to the application layer: to get a message from an application program running on the source host and deliver it to the corresponding application program on the destination host.

There are a few transport-layer protocols in the Internet, each designed for some specific task.

Transmission Control Protocol (TCP):

TCP is a connection-oriented protocol that first establishes a logical connection between transport layers at two hosts before transferring data.

TCP provides:

- Flow control that is matching the sending data rate of the source host with the receiving data rate of the destination host to prevent overwhelming the destination
- Error control to guarantee that the segments arrive at the destination without error and resending the corrupted ones.
- Congestion control to reduce the loss of segments due to congestion in the network.

User Datagram Protocol (UDP):

UDP is a connectionless protocol that transmits user datagrams without first creating a logical connection.

UDP is a simple protocol that does not provide flow, error, or congestion control.

Stream Control Transmission Protocol (SCTP):

SCTP is designed to respond to new applications that are emerging in the multimedia.

5) Application Layer

- The application layer enables the user to access the network.
- Communication at the application layer is between two processes. To communicate, a process sends a request to the other process and receives a response. Process-to-process communication is the duty of the application layer.
- It provides user interfaces and support for services such as electronic mail, remote file access and transfer, shared database management, and other types of distributed information services.
- Protocols:

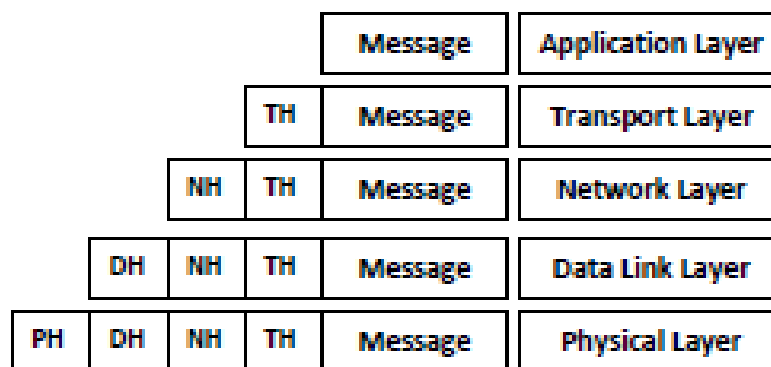
- The Hypertext Transfer Protocol (HTTP) is used for accessing the World Wide Web (WWW).
- The Simple Mail Transfer Protocol (SMTP) is the main protocol used in electronic mail (e-mail) service.
- The File Transfer Protocol (FTP) is used for transferring files from one host to another.
- The Terminal Network (TELNET) and Secure Shell (SSH) are used for accessing a site remotely.
- The Simple Network Management Protocol (SNMP) is used by an administrator to manage the Internet at global and local levels.
- The Domain Name System (DNS) is used by other protocols to find the network-layer address of a computer.
- The Internet Group Management Protocol (IGMP) is used to collect membership in a group.

2.2.4 Encapsulation and Decapsulation

One of the important concepts in protocol layering in the Internet is encapsulation/decapsulation.

1. At the application layer, the data to be exchanged is referred to as a message. A message normally does not contain any header or trailer, but if it does, we refer to the whole as the message.

The message is passed to the transport layer.



2. The transport layer takes the message as the payload, the load that the transport layer should take care of. It adds the transport layer header to the payload, which contains the identifiers of the source and destination application programs that want to communicate plus some more information that is needed for the end-to end delivery of the message, such as information needed for flow, error control, or congestion control. The result is the transport-layer packet, which is

called the segment (in TCP) and the user datagram (in UDP). The transport layer then passes the packet to the network layer.

3. The network layer takes the transport-layer packet as data or payload and adds its own header to the payload. The header contains the addresses of the source and destination hosts and some more information used for error checking of the header, fragmentation information, and so on. The result is the network-layer packet, called a datagram. The network layer then passes the packet to the data-link layer.

4. The data-link layer takes the network-layer packet as data or payload and adds its own header, which contains the link-layer addresses of the host or the next hop (the router). The result is the link-layer packet, which is called a frame. The frame is passed to the physical layer for transmission.

Decapsulation and Encapsulation at the Router

At the router, we have both decapsulation and encapsulation because the router is connected to two or more links.

1. After the set of bits are delivered to the data-link layer, this layer decapsulates the datagram from the frame and passes it to the network layer.
2. The network layer only inspects the source and destination addresses in the datagram header and consults its forwarding table to find the next hop to which the datagram is to be delivered. The contents of the datagram should not be changed by the network layer in the router unless there is a need to fragment the datagram if it is too big to be passed through the next link. The datagram is then passed to the data-link layer of the next link.
3. The data-link layer of the next link encapsulates the datagram in a frame and passes it to the physical layer for transmission.

Decapsulation at the Destination Host

At the destination host, each layer only decapsulates the packet received, removes the payload, and delivers the payload to the next-higher layer protocol until the message reaches the application layer. It is necessary to say that decapsulation in the host involves error checking.

2.2.5 Addressing

Four levels of addresses are used in an internet employing the *TCP/IP* protocols:

- physical (link) addresses
- logical (IP) addresses
- port addresses

-
- specific addresses

Physical Addresses

The physical address, also known as the link address, is the address of a node as defined by its LAN or WAN. It is included in the frame used by the data link layer. It is the lowest-level address.

Logical Addresses

A logical address in the Internet is currently a 32-bit address that can uniquely define a host connected to the Internet. No two publicly addressed and visible hosts on the Internet can have the same IP address.

Port Addresses

Port numbers are local addresses that distinguish between several programs running at the same time. A port address is a 16-bit address represented by one decimal number

Specific Addresses

Some applications have user-friendly addresses that are designed for that specific address.

Example: e-mail address and the Universal Resource Locator (URL)

2.2.6 Multiplexing and Demultiplexing

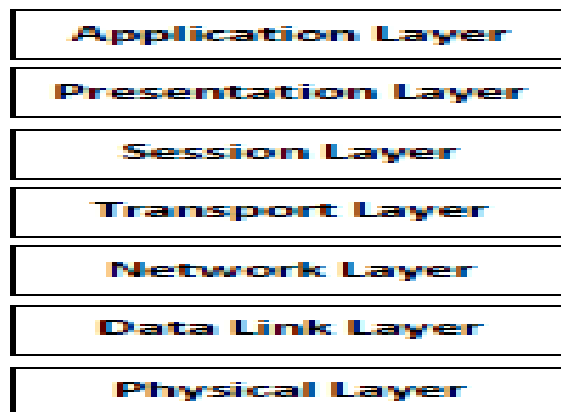
- Multiplexing means that a protocol at a layer can encapsulate a packet from several next-higher layer protocols, demultiplexing means that a protocol can decapsulate and deliver a packet to several next-higher layer protocols.
- To be able to multiplex and demultiplex, a protocol needs to have a field in its header to identify to which protocol the encapsulated packets belong.
- At the transport layer, either UDP or TCP can accept a message from several application- layer protocols.
- At the network layer, IP can accept a segment from TCP or a user datagram from UDP. IP can also accept a packet from other protocols such as ICMP, IGMP, and so on.
- At the data-link layer, a frame may carry the payload coming from IP or other protocols such as ARP.

2.3 The OSI Model

- An ISO standard that covers all aspects of network communications is the Open Systems Interconnection model.
- It was first introduced in the late 1970s.
- An open system is a set of protocols that allows any two different systems to communicate

regardless of their underlying architecture.

- The purpose of the OSI model is to show how to facilitate communication between different systems without requiring changes to the logic of the underlying hardware and software.
- The OSI model is a layered framework for the design of network systems that allows communication between all types of computer systems.
- The OSI model is a layered framework for the design of network systems that allows communication between all types of computer systems. It consists of seven separate but related layers, each of which defines a part of the process of moving information across a network.



2.3.1 OSI versus TCP/IP

Session and presentation, are missing from the TCP/IP protocol suite. The application layer in the suite is usually considered to be the combination of three layers in the OSI model. Some of the functionalities of the session layer are available in some of the transport-layer protocols. If some of the functionalities mentioned in the session and presentation layers are needed for a particular application, they can be included in the development of that piece of software.

2.3.2 Lack of OSI Model's Success

- First, OSI was completed when TCP/IP was fully in place and a lot of time and money had been spent on the suite; changing it would cost a lot.
- Second, some layers in the OSI model were never fully defined.
- Third, when OSI was implemented by an organization in a different application, it did not show a high enough level of performance to entice the Internet authority to switch from the TCP/IP protocol suite to the OSI model.

Chapter 3: Physical Layer - 1

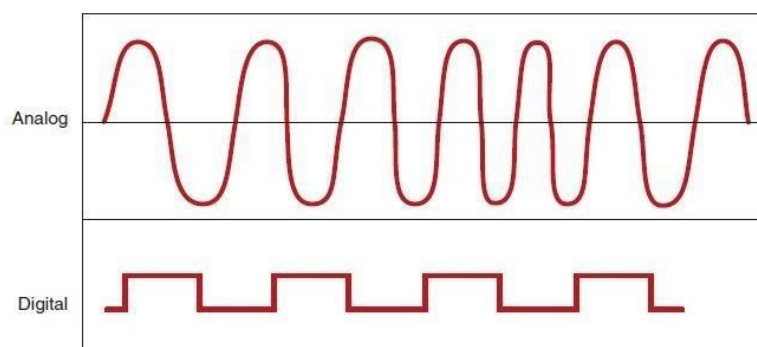
3.1 Data and Signals

3.1.1 Analog and Digital Data

- Data can be analog or digital. The term **analog data** refers to information that is continuous; **digital data** refers to information that has discrete states.
- **Example:** an analog clock that has hour, minute, and second hands gives information in a continuous form; the movements of the hands are continuous. On the other hand, a digital clock that reports the hours and the minutes will change suddenly from 8:05 to 8:06.
- Analog data, such as the sounds made by a human voice, take on continuous values. When someone speaks, an analog wave is created in the air. This can be captured by a microphone and converted to an analog signal or sampled and converted to a digital signal.
- Digital data take on discrete values. For example, data are stored in computer memory in the form of 0s and 1s. They can be converted to a digital signal or modulated into an analog signal for transmission across a medium.

3.1.2 Analog and Digital Signals

- Like the data they represent, signals can be either analog or digital.
- An analog signal has infinitely many levels of intensity over a period of time. As the wave moves from value *A* to value *B*, it passes through and includes an infinite number of values along its path.
- A digital signal, on the other hand, can have only a limited number of defined values. Although each value can be any number, it is often as simple as 1 and 0.

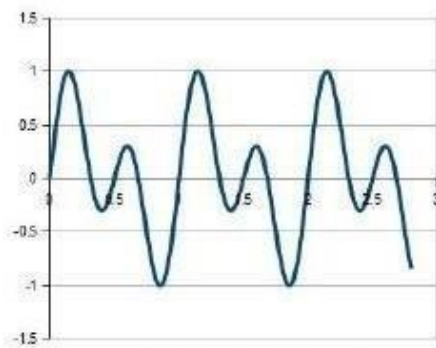


3.1.3 Periodic and Nonperiodic Signals

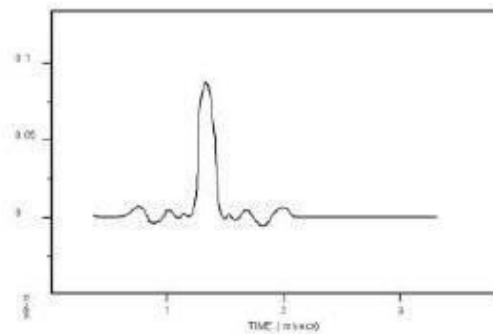
- Both analog and digital signals can take one of two forms: *periodic* or *nonperiodic*
- A **periodic signal** completes a pattern within a measurable time frame, called a **period**, and

repeats that pattern over subsequent identical periods.

- The completion of one full pattern is called a **cycle**.
- A **nonperiodic** signal changes without exhibiting a pattern or cycle that repeats over time.
- In data communications, we commonly use periodic analog signals and nonperiodic digital signals.



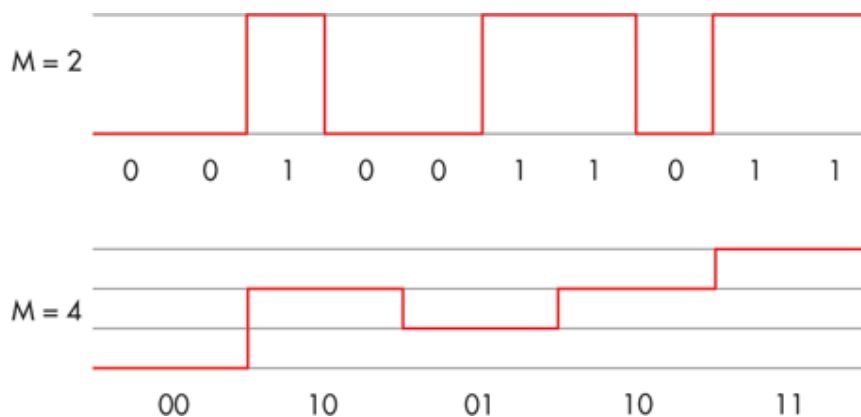
Periodic signal



Non-periodic signal

Digital Signals

In addition to being represented by an analog signal, information can also be represented by a digital signal. For example, a 1 can be encoded as a positive voltage and a 0 as zero voltage. A digital signal can have more than two levels. In this case, we can send more than 1 bit for each



level. If a signal has M levels, each level needs $\log_2 M$ bits.

3.1.4 Bit Rate and Bit Length

The bit rate is the number of bits sent in 1s, expressed in bits per second.

The bit length is the distance one bit occupies on the transmission medium.

$$\text{Bit length} = \text{propagation speed} \times \text{bit duration}$$

Example 3.1:

A digital signal has eight levels. How many bits are needed per level?

Solution:

Number of bits per level = $\log_2 8 = 3$

Example 3.2:

Assume we need to download text documents at the rate of 100 pages per minute. What is the required bit rate of the channel?

Solution: A page is an average of 24 lines with 80 characters in each line. If we assume that one character requires 8 bits, the bit rate is $100 \times 24 \times 80 \times 8 = 1,536,000 \text{ bpm} = 25,600 \text{ bps} = 25 \text{ Kbps}$

Example 3.3:

A digitized voice channel, as we will see in Chapter 4, is made by digitizing a 4-kHz bandwidth analog voice signal. We need to sample the signal at twice the highest frequency (two samples per hertz). We assume that each sample requires 8 bits. What is the required bit rate?

Solution:

Bit rate is $= 2 \times 4000 \times 8 = 64 \text{ kbps}$

Example 3.4:

What is the bit rate for each of the following signals?

- a. A signal in which 1 bit lasts 0.001 s
- b. A signal in which 1 bit lasts 2 ms
- c. A signal in which 10 bits last 20 μs

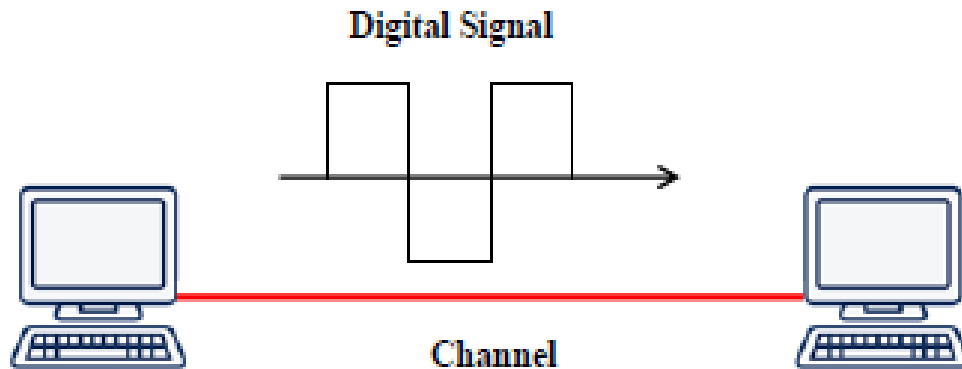
Solution:

- a. bit rate = $1 / (\text{bit duration}) = 1 / (0.001 \text{ s}) = 1000 \text{ bps} = 1 \text{ Kbps}$
- b. bit rate = $1 / (\text{bit duration}) = 1 / (2 \text{ ms}) = 500 \text{ bps}$
- c. bit rate = $1 / (\text{bit duration}) = 1 / (20 \mu\text{s} / 10) = 1 / (2 \mu\text{s}) = 500 \text{ Kbps}$

3.1.5 Transmission of Digital Signals

1) Baseband Transmission

-
- Baseband transmission means sending a digital signal over a channel without changing the digital signal to an analog signal.



- Baseband transmission uses **low-pass channel**, which is a channel with a bandwidth that starts from zero.
- For example, the entire bandwidth of a cable connecting two computers is one single channel.
- Baseband transmission of a digital signal that preserves the shape of the digital signal is possible only if we have a low-pass channel with an infinite or very wide bandwidth.
- In baseband transmission, the required bandwidth is proportional to the bit rate; if we need to send bits faster, we need more bandwidth.

2) Broadband Transmission

- Broadband transmission or modulation means changing the digital signal to an analog signal for transmission. Modulation allows us to use a **bandpass channel**-a channel with a bandwidth that does not start from zero. This type of channel is more available than a low-pass channel.
- If the available channel is a bandpass channel we cannot send the digital signal directly to the channel; we need to convert the digital signal to an analog signal before transmission.

Example 3.5:

We have a low-pass channel with bandwidth 100 kHz. What is the maximum bit rate of this channel?

Solution:

The bit rate is 2 times the available bandwidth.

Therefore Maximum bit rate of a low pass channel with bandwidth 100kHz = $2 \times 100 = 200\text{kbps}$

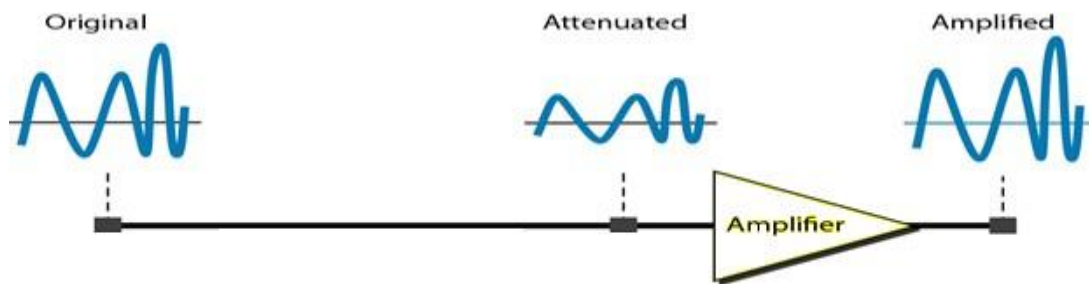
3.4 Transmission Impairment

Signals travel through transmission media, which are not perfect. The imperfection causes signal impairment.

Three causes of impairment are attenuation, distortion, and noise.

3.4.1 Attenuation

- Attenuation means a loss of energy.
- When a signal, simple or composite, travels through a medium, it loses some of its energy in overcoming the resistance of the medium.
- To compensate for this loss, amplifiers are used to amplify the signal.



Decibel:

The decibel (dB) measures the relative strengths of two signals or one signal at two different points. The decibel is negative if a signal is attenuated and positive if a signal is amplified.

$$\text{dB} = 10 \log_{10} (P_2/P_1)$$

Variables P_1 and P_2 are the powers of a signal at points 1 and 2, respectively.

Example 3.7:

A signal travels through an amplifier, and its power is increased 10 times. Calculate the amplification.

Solution:

$$\text{Amplification} = 10 \log_{10} (P_2/P_1) = 10 \log_{10} (10 P_1 / P_1) = 10 \log_{10} 10 = 10(1) = 10\text{dB}$$

Example 3.8:

A signal travels from point A to point B. At point A, the signal power is 100 W. At point B, the power is 90 W. What is the attenuation in decibels?

Solution:

$$\text{Given: } p_1 = 100\text{W } p_2 = 90\text{W}$$

$$\text{dB} = 10 \log_{10}(p_1/p_2)$$

$$\text{dB} = 10 \log_{10} (90 / 100) = -0.46 \text{ dB}$$

Example 3.9:

If the signal at the beginning of a cable with -0.3 dB/km has a power of 2 mW, what is the power of the signal at 5 km?

Solution:

The loss in the cable in decibels is $5 \times (-0.3) = -1.5 \text{ dB}$.

We can calculate the power as

$$\text{dB} = 10 \log_{10} (P_2/P_1)$$

$$-1.5 = 10 \log_{10} (P_2/P_1)$$

$$\log_{10} (P_2/P_1) = -1.5/10 = -$$

$$0.15 \quad P_2/P_1 = 10^{-0.15}$$

$$P_2/P_1 = 0.71$$

$$P_2 = 0.71 P_1 = 0.7 \times 2 \text{ mW}$$

$$P_2 = 1.4 \text{ mW}$$

Example 3.10:

The attenuation of a signal is -10 dB. What is the final signal power if it was originally 5 W?

Solution:

Given: $\text{dB} = -10\text{dB}$ $p_1 = 5\text{W}$

$$\text{dB} = 10 \log_{10}(p_1/p_2)$$

$$-10 = 10 \log_{10} (P_2 / 5)$$

$$\log_{10} (P_2 / 5) = -1$$

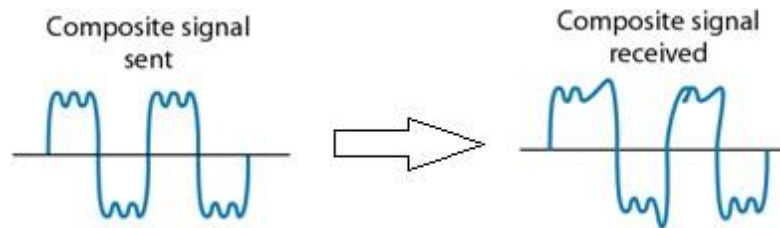
$$(P_2 / 5) = 10^{-1}$$

$$P_2 = 0.5 \text{ W}$$

3.4.2 Distortion

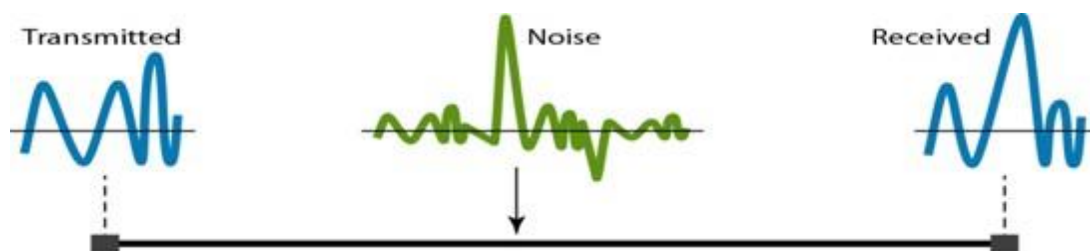
- **Distortion** means that the signal changes its form or shape.
- Distortion can occur in a composite signal made of different frequencies.
- Each signal component has its own propagation speed (see the next section) through a medium and, therefore, its own delay in arriving at the final destination.
- Differences in delay may create a difference in phase if the delay is not exactly the same as the period duration.

- In other words, signal components at the receiver have phases different from what they had at the sender. The shape of the composite signal is therefore not the same.



3.4.3 Noise

- Noise is another cause of impairment.
- Several types of noise, such as thermal noise, induced noise, crosstalk, and impulse noise, may corrupt the signal.
- Thermal noise is the random motion of electrons in a wire which creates an extra signal not originally sent by the transmitter.
- Induced noise comes from sources such as motors and appliances. These devices act as a sending antenna, and the transmission medium acts as the receiving antenna.
- Crosstalk is the effect of one wire on the other. One wire acts as a sending antenna and the other as the receiving antenna.
- Impulse noise is a spike (a signal with high energy in a very short time) that comes from power lines, lightning, and so on.



Signal-to-Noise Ratio (SNR):

$$\text{SNR} = \frac{\text{average signal power}}{\text{average noise power}}$$

SNR is actually the ratio of what is wanted (signal) to what is not wanted (noise).

A high SNR means the signal is less corrupted by noise; a low SNR means the signal is more

corrupted by noise. Because SNR is the ratio of two powers, it is often described in decibel units, SNR_{dB}, defined as

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR}$$

Example 3.11:

The power of a signal is 10 mW and the power of the noise is 1 μW; what are the values of SNR and SNR_{dB} ?

Solution:

SNR = average signal power / average noise power

$$= 10,000 \mu\text{W} / 1 \mu\text{W}$$

$$= 10,000$$

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR}$$

$$= 10 \log_{10} 10,000$$

$$= 10 \log_{10} 10^4$$

$$= 40$$

Example 3.12:

A signal with 200 milliwatts power passes through 10 devices, each with an average noise of 2 microwatts. What is the SNR? What is the SNR_{dB}?

Solution:

SNR = (Average Signal power / Average noise power)

$$\text{SNR} = (200 \text{ mW}) / (10 \times 2 \mu\text{W})$$

$$\text{SNR} = 10,000$$

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR} = 40$$

3.5 Data Rate Limits

Data rate depends on three factors:

1. The bandwidth available
2. The level of the signals we use
3. The quality of the channel (the level of noise)

3.5.1 Noiseless Channel: Nyquist Bit Rate

For a noiseless channel, the Nyquist bit rate formula defines the theoretical maximum bit rate

$$\text{BitRate} = 2 \times \text{bandwidth} \times \log_2 L$$

In this formula, bandwidth is the bandwidth of the channel, L is the number of signal levels used to represent data, and BitRate is the bit rate in bits per second.

Increasing the levels of a signal may reduce the reliability of the system.

Example 3.13:

Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels. Calculate the maximum bit rate.

Solution:

Using Nyquist Bit rate formula:

$$\text{Bit rate} = 2 \times \text{Bandwidth} \times \log_2 L$$

$$\begin{aligned}\text{Bit rate} &= 2 \times 3000 \times \log_2 2 \\ &= 6000 \text{ bps}\end{aligned}$$

Example 3.14:

Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with four signal levels. Calculate the maximum bit rate.

Solution:

Using Nyquist Bit rate formula:

$$\text{Bit rate} = 2 \times \text{Bandwidth} \times \log_2 L$$

$$\begin{aligned}\text{Bit rate} &= 2 \times 3000 \times \log_2 4 \\ &= 12,000 \text{ bps}\end{aligned}$$

Example 3.15:

We need to send 265 kbps over a noiseless channel with a bandwidth of 20 kHz. How many signal levels do we need?

Solution:

Using Nyquist Bit rate formula:

$$\text{Bit rate} = 2 \times \text{Bandwidth} \times \log_2 L$$

$$265,000 = 2 \times 20,000 \times \log_2 L$$

$$\log_2 L = 6.625$$

$$L = 26.625$$

$$L = 98.7 \text{ Levels}$$

3.5.2 Noisy Channel: Shannon Capacity

In reality, we cannot have a noiseless channel; the channel is always noisy. In 1944, Claude Shannon introduced a formula, called the Shannon capacity, to determine the theoretical highest data rate for a noisy channel:

$$\text{Capacity} = \text{bandwidth} \times \log_2 (1 + \text{SNR})$$

In this formula, bandwidth is the bandwidth of the channel, SNR is the signal-to noise ratio, and capacity is the capacity of the channel in bits per second.

Example 3.16:

Consider an extremely noisy channel in which the value of the signal-to-noise ratio is almost zero. Calculate the Shannon capacity.

Solution:

$$\begin{aligned}\text{Capacity} &= \text{bandwidth} \times \log_2 (1 + \text{SNR}) \\ &= \text{bandwidth} \times \log_2 (1 + 0) \\ &= \text{bandwidth} \times 0 \quad \text{Capacity} = 0\end{aligned}$$

Example 3.17:

A telephone line normally has a bandwidth of 3000. The signal-to-noise ratio is usually 3162. Calculate the Shannon capacity.

Solution:

$$\begin{aligned}\text{Capacity} &= \text{bandwidth} \times \log_2 (1 + \text{SNR}) \\ &= 3000 \log_2 (1 + 3162) \\ &= 3000 \times 11.62 \\ \text{Capacity} &= 34,860 \text{ bps}\end{aligned}$$

Example 3.18:

Assume that SNR_{dB} = 36 and the channel bandwidth is 2 MHz. Calculate the Shannon capacity.

Solution:

$$\begin{aligned}\text{SNR}_{\text{dB}} &= 10 \log_{10} \text{SNR} \quad \text{SNR} = 10^{(\text{SNR}_{\text{dB}}/10)} \\ \text{SNR} &= 10^{(36/10)} = 10^{3.6} \quad \text{SNR} = 3981 \\ \text{Capacity} &= \text{bandwidth} \times \log_2 (1 + \text{SNR}) \\ &= 2 \times 10^6 \times \log_2 (1 + 3982) \\ \text{Capacity} &= 24 \text{ Mbps}\end{aligned}$$

Example 3.19:

We have a channel with a 1-MHz bandwidth. The SNR for this channel is 63. What are the appropriate bit rate and signal level?

Solution:

First, we use the Shannon formula to find the upper limit.

$$\text{Capacity} = \text{bandwidth} \times \log_2 (1 + \text{SNR})$$

$$= 10^6 \times \log_2 (1 + 63) = 6 \text{ Mbps}$$

The Shannon formula gives us 6 Mbps, the upper limit. For better performance we choose something lower, 4 Mbps, for example. Then we use the Nyquist formula to find the number of signal levels.

$$4\text{Mbps} = 2 \times 1\text{MHz} \times$$

$$\log_2 LL = 4$$

Example 3.20:

A line has a signal-to-noise ratio of 1000 and a bandwidth of 4000 KHz. What is the maximum data rate supported by this line?

Solution:

Given: $B = 4000\text{KHz}$

$$\text{SNR} = 1000$$

To find: Shanon capacity Formula: $C = B \times \log_2(1 + \text{SNR})$

$$C = 4,000 \log_2 (1 + 1,000) \approx 40 \text{ Kbps}$$

Example 3.21:

We measure the performance of a telephone line (4 KHz of bandwidth). When the signal is 10 V, the noise is 5 mV. What is the maximum data rate supported by this telephone line?

Solution:

Given: $B = 4\text{KHz}$, Average Signal power = 10V, Average noise power = 5mV

Data Rate = Shanon Capacity = ?

$\text{SNR} = (\text{Average Signal power} / \text{Average noise power})$

$$= 10\text{V} / 5\text{mV}$$

Data Rate = $B \times \log_2(1 + \text{SNR})$

$$= 4,000 \log_2 (1 + 10 / 0.005) = 43,866 \text{ bps}$$

3.6 Performance

3.6.1 Bandwidth

One characteristic that measures network performance is bandwidth. However, the term can be used in two different contexts with two different measuring values: bandwidth in hertz and bandwidth in bits per second.

Bandwidth in Hertz

Bandwidth in hertz is the range of frequencies contained in a composite signal or the range of frequencies a channel can pass.

For example, we can say the bandwidth of a subscriber telephone line is 4 kHz.

Bandwidth in Bits per Seconds

The term bandwidth can also refer to the number of bits per second that a channel, a link, or even a network can transmit.

For example, one can say the bandwidth of a Fast Ethernet network (or the links in this network) is a maximum of 100 Mbps. This means that this network can send 100 Mbps.

Relationship

There is an explicit relationship between the bandwidth in hertz and bandwidth in bits per seconds. Basically, an increase in bandwidth in hertz means an increase in bandwidth in bits per second.

3.6.2 Throughput

The throughput is a measure of how fast we can actually send data through a network.

The bandwidth is a potential measurement of a link; the throughput is an actual measurement of how fast we can send data.

For example, we may have a link with a bandwidth of 1 Mbps, but the devices connected to the end of the link may handle only 200 kbps. This means that we cannot send more than 200 kbps through this link.

Example 3.22:

A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

Solution:

Number of bits = 12,000 frames x 10,000 bits

= 120 M bits

Throughput = 120 M bits/60 sec = 2 Mbps

3.6.3 Latency (Delay)

The latency or delay defines how long it takes for an entire message to completely arrive at the destination from the time the first bit is sent out from the source.

Latency is made of four components: propagation time, transmission time, queuing time and processing delay.

Latency = propagation time + transmission time + queuing time + processing delay

Example 3.23:

What is the total delay (latency) for a frame of size 5 million bits that is being sent on a link with 10 routers each having a queuing time of $2\mu\text{s}$ and a processing time of $1\mu\text{s}$. The length of the link is 2000 Km. The speed of light inside the link is 2×10^8 m/s. The link has a bandwidth of 5 Mbps. Which component of the total delay is dominant? Which one is negligible?

Solution:

Latency = processing time + queuing time + transmission time + propagation time

Processing time = $10 \times 1 \mu\text{s} = 10 \mu\text{s} = 0.000010 \text{ s}$

Queuing time = $10 \times 2 \mu\text{s} = 20 \mu\text{s} = 0.000020 \text{ s}$

Transmission time = $5,000,000 / (5 \text{ Mbps}) = 1 \text{ s}$
Propagation time = $(2000 \text{ Km}) / (2 \times 10^8) = 0.01 \text{ s}$

Latency = $0.000010 + 0.000020 + 1 + 0.01 = 1.01000030 \text{ s}$

The transmission time is dominant here because the packet size is huge.

Propagation Time

Propagation time measures the time required for a bit to travel from the source to the destination.

The propagation time is calculated by dividing the distance by the propagation speed.

Propagation time = Distance / Propagation speed

Example 3.24:

What is the propagation time if the distance between the two points is 12,000 km? Assume the propagation speed to be 2.4×10^8 m/s in cable.

Solution:

Propagation time = Distance / Propagation speed

$= 12000 \times 1000 / 2.4 \times 10^8$

$= 50 \text{ ms}$

Transmission time

The time required for transmission of a message depends on the size of the message and the bandwidth of the channel.

Transmission time = Message size / Bandwidth

Example 3.25:

What is the transmission time for a 2.5-kbyte message if the bandwidth of the network is 1 Gbps?

Solution:

Transmission time = Message size/ Bandwidth

$$\begin{aligned} &= 2.5 \text{ KB} / 1 \text{ Gbps} \\ &= (2500 \times 8\text{bits}) / 10^9 \\ &= 0.020 \text{ ms} \end{aligned}$$

Queuing Time

The third component in latency is the queuing time, the time needed for each intermediate or end device to hold the message before it can be processed. The queuing time is not a fixed factor; it changes with the load imposed on the network. When there is heavy traffic on the network, the queuing time increases. An intermediate device, such as a router, queues the arrived messages and processes them one by one. If there are many messages, each message will have to wait.

Bandwidth Delay Product

The bandwidth~delay product defines the number of bits that can fill the link.

Jitter

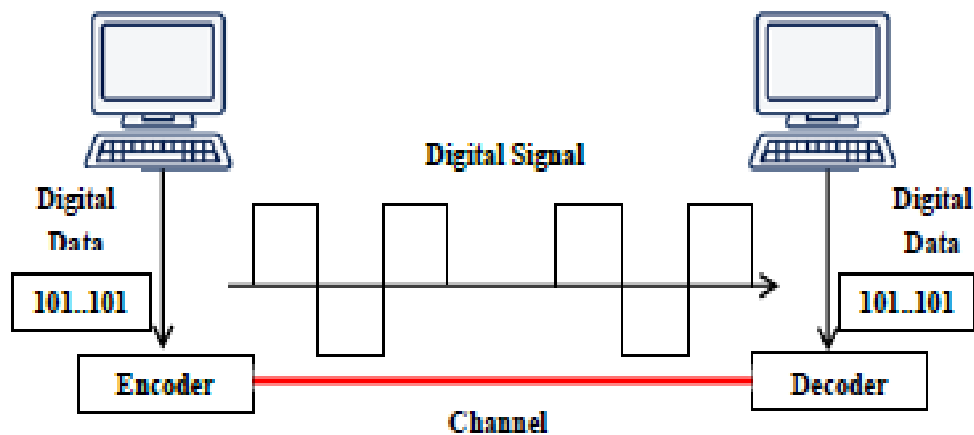
Jitter is a variation in packet arrival time.

For ex, If the delay for the first packet is 20 ms, for the second is 45 ms, and for the third is 40 ms, then the real-time application that uses the packets endures jitter.

3.7 Digital to Digital Conversion

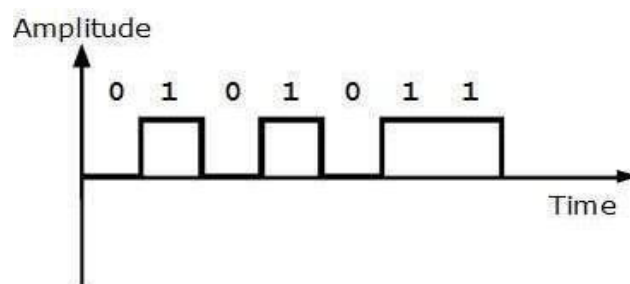
3.7.1 Line Coding

Line coding is the process of converting digital data to digital signals. Line coding converts a sequence of bits to a digital signal. At the sender, digital data are encoded into a digital signal; at the receiver, the digital data are recreated by decoding the digital signal.



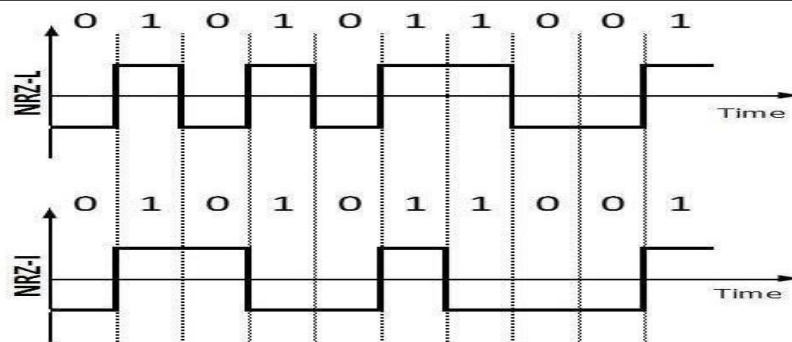
Line Coding Schemes

Unipolar Scheme In a unipolar scheme, all the signal levels are on one side of the time axis, either above or below. A unipolar scheme was designed as a non-return-to-zero (NRZ) scheme in which the positive voltage defines bit 1 and the zero voltage defines bit 0. It is called NRZ because the signal does not return to zero at the middle of the bit.



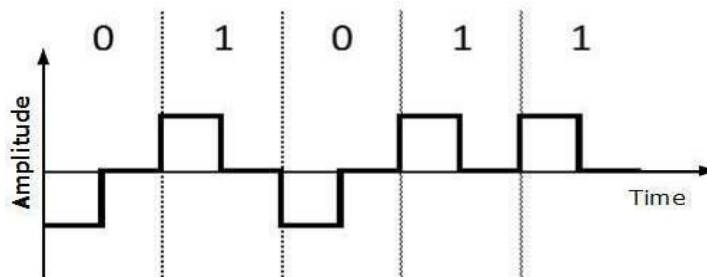
Polar Schemes

- In polar schemes, the voltages are on the both sides of the time axis. For example, the voltage level for 0 can be positive and the voltage level for 1 can be negative.
- **Non-Return-to-Zero (NRZ)**: In polar NRZ encoding, we use two levels of voltage amplitude. We can have two versions of polar NRZ: NRZ-L and NRZ-I
- In **NRZ-L** the level of the voltage determines the value of the bit.
- In **NRZ-I** the inversion or the lack of inversion determines the value of the bit.



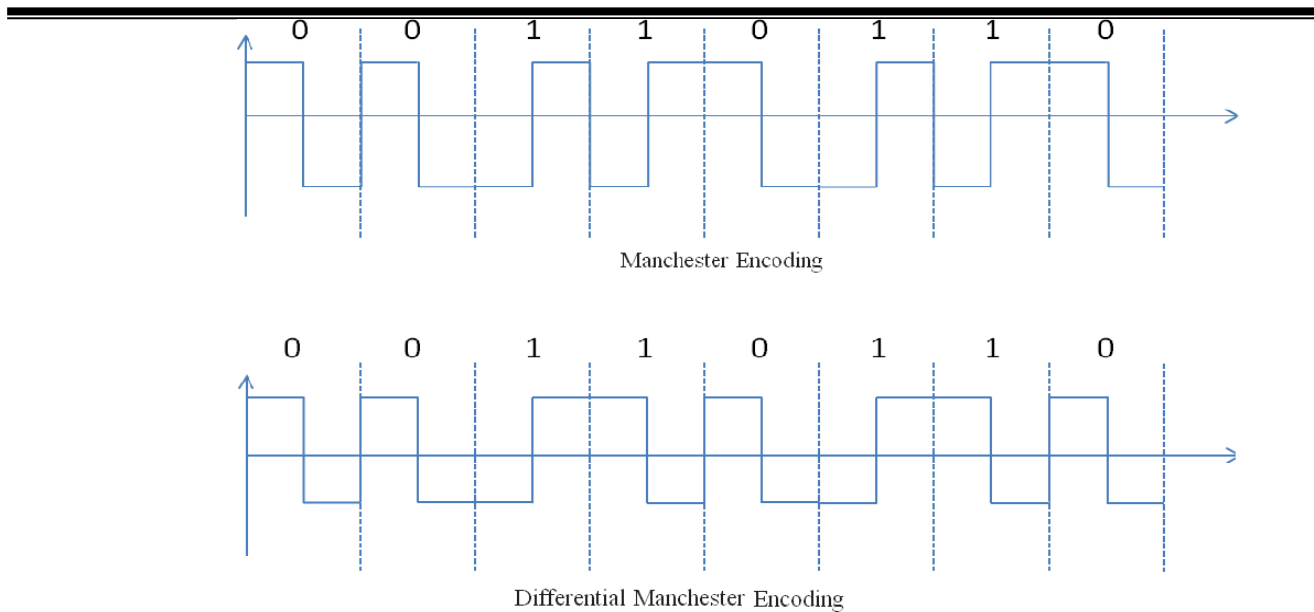
Return to Zero (RZ):

- The main problem with NRZ encoding occurs when the sender and receiver clocks are not synchronized. The receiver does not know when one bit has ended and the next bit is starting.
- One solution is the return-to-zero (RZ) scheme, which uses three values: positive, negative, and zero.
- In RZ, the signal changes not between bits but during the bit. In the below Figure we see that the signal goes to 0 in the middle of each bit. It remains there until the beginning of the next bit.
- The main disadvantage of RZ encoding is that it requires two signal changes to encode a bit and therefore occupies greater bandwidth.



Manchester and Differential Manchester:

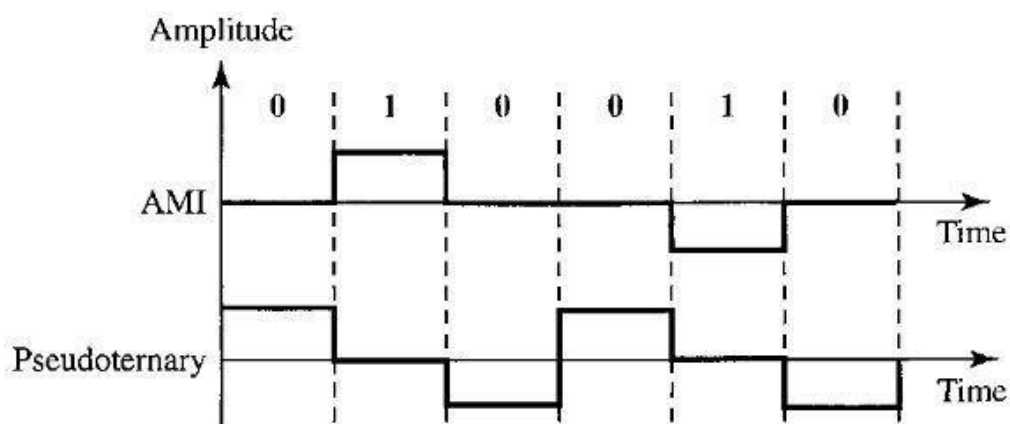
- In Manchester encoding, the duration of the bit is divided into two halves. The voltage remains at one level during the first half and moves to the other level in the second half. The transition at the middle of the bit provides synchronization.
- Differential Manchester on the other hand, combines the ideas of RZ and NRZ-I. There is always a transition at the middle of the bit, but the bit values are determined at the beginning of the bit. If the next bit is 0, there is a transition; if the next bit is 1, there is none.



Bipolar Schemes

In bipolar encoding (sometimes called *multilevel binary*), there are three voltage levels: positive, negative, and zero. The voltage level for one data element is at zero, while the voltage level for the other element alternates between positive and negative.

AMI and Pseudoternary: A common bipolar encoding scheme is called bipolar alternate mark inversion (AMI). In the term alternate mark inversion, the word mark comes from telegraphy and means 1. So AMI means alternate I inversion. A neutral zero voltage represents binary 0. Binary 1s are represented by alternating positive and negative voltages. A variation of AMI encoding is called pseudoternary in which the 1 bit is encoded as a zero voltage and the 0 bit is encoded as alternating positive and negative voltages.



Chapter 4: Physical Layer – 2

Digital to Digital Conversion

Line Coding

Line coding is the process of converting digital data to digital signals. Line coding converts a sequence of bits to a digital signal. At the sender, digital data are encoded into a digital signal; at the receiver, the digital data are recreated by decoding the digital signal.

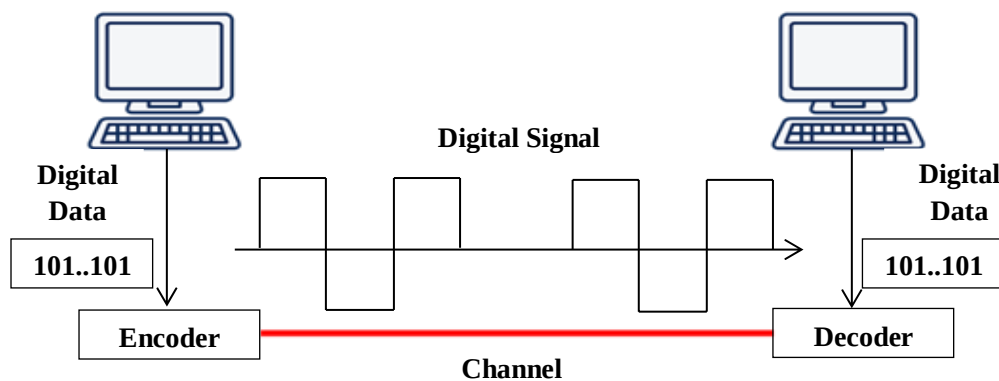


Figure 3.7: Line Coding

Line Coding Schemes

Unipolar Scheme

In a unipolar scheme, all the signal levels are on one side of the time axis, either above or below.

A unipolar scheme was designed as a non-return-to-zero (NRZ) scheme in which the positive voltage defines bit 1 and the zero voltage defines bit 0. It is called NRZ because the signal does not return to zero at the middle of the bit.

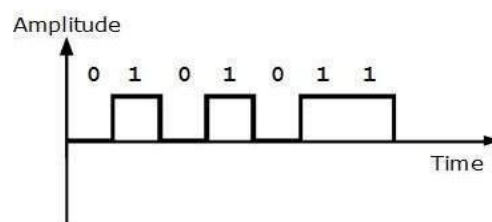


Figure 3.8: Unipolar Scheme

Polar Schemes

- In polar schemes, the voltages are on the both sides of the time axis. For example, the voltage level for 0 can be positive and the voltage level for 1 can be negative.
- **Non-Return-to-Zero (NRZ):** In polar NRZ encoding, we use two levels of voltage amplitude. We

can have two versions of polar NRZ: NRZ-L and NRZ-I

- In **NRZ-L** the level of the voltage determines the value of the bit.
- In **NRZ-I** the inversion or the lack of inversion determines the value of the bit.

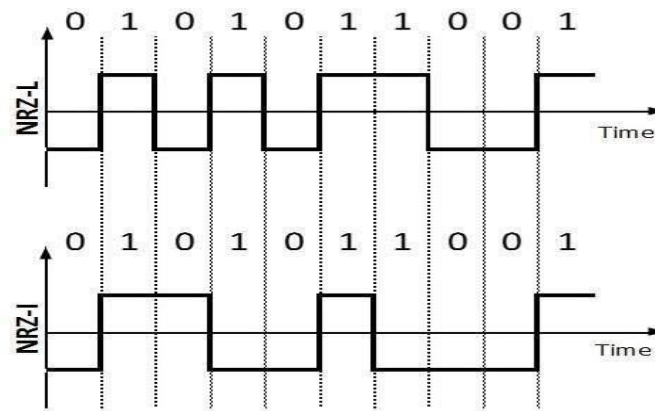


Figure 3.9: Polar Non Return to Zero

Return to Zero (RZ):

- The main problem with NRZ encoding occurs when the sender and receiver clocks are not synchronized. The receiver does not know when one bit has ended and the next bit is starting.
- One solution is the return-to-zero (RZ) scheme, which uses three values: positive, negative, and zero.
- In RZ, the signal changes not between bits but during the bit. In the below Figure we see that the signal goes to 0 in the middle of each bit. It remains there until the beginning of the next bit.
- The main disadvantage of RZ encoding is that it requires two signal changes to encode a bit and therefore occupies greater bandwidth.

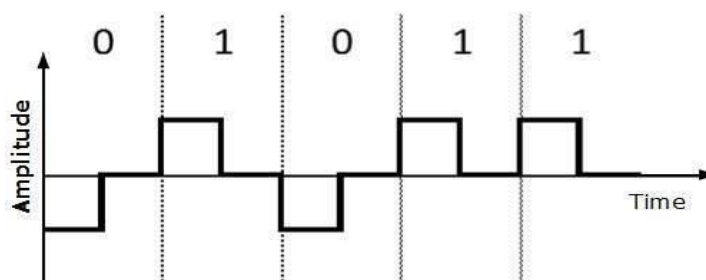


Figure 3.10: Polar Return to Zero

Manchester and Differential Manchester:

- In Manchester encoding, the duration of the bit is divided into two halves. The voltage remains at one level during the first half and moves to the other level in the second half. The transition at the

middle of the bit provides synchronization.

- Differential Manchester on the other hand, combines the ideas of RZ and NRZ-I. There is always a transition at the middle of the bit, but the bit values are determined at the beginning of the bit. If the next bit is 0, there is a transition; if the next bit is 1, there is none.

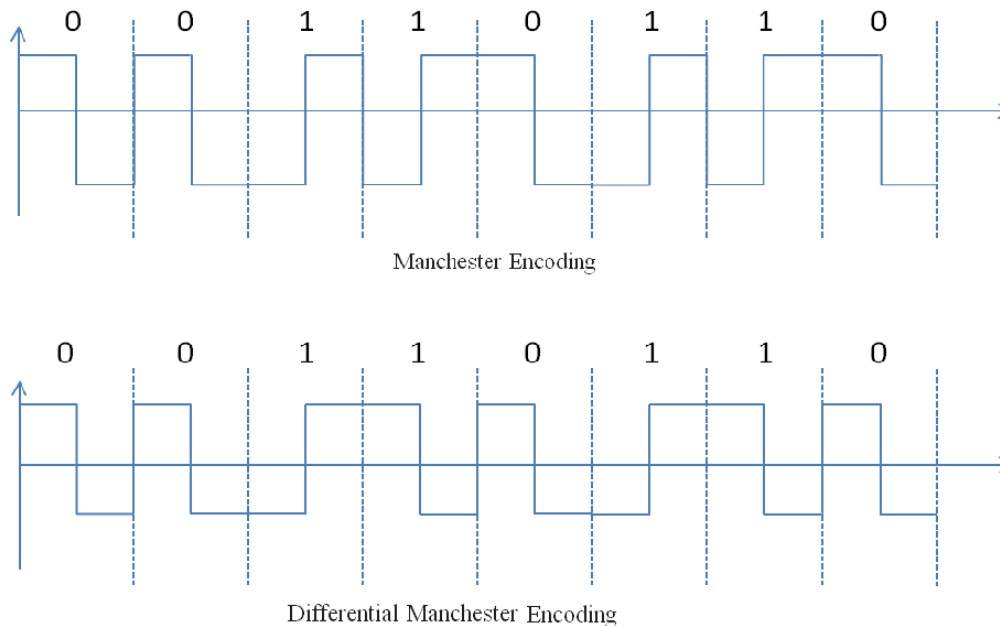


Figure 3.11: Manchester and Differential Manchester

Bipolar Schemes

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AMI and Pseudoternary: A common bipolar encoding scheme is called bipolar alternate mark inversion (AMI). In the term alternate mark inversion, the word mark comes from telegraphy and means 1. So AMI means alternate 1 inversion. A neutral zero voltage represents binary 0. Binary 1s are represented by alternating positive and negative voltages. A variation of AMI encoding is called pseudoternary in which the 1 bit is encoded as a zero voltage and the 0 bit is encoded as alternating positive and negative voltages.

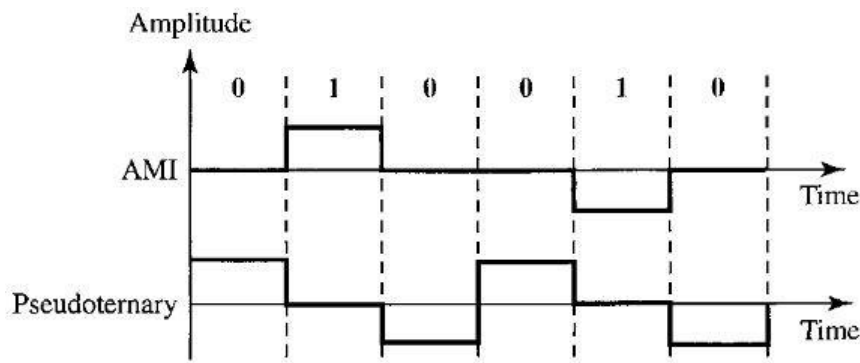


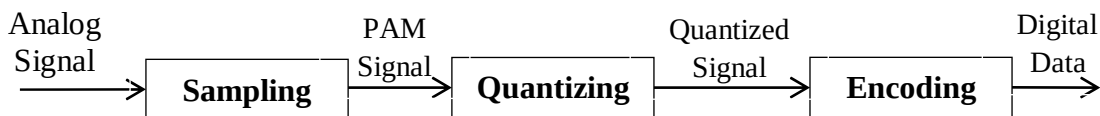
Figure 3.12: AMI and Pseudoternary

4.1 Analog-To-Digital Conversion

4.1.1 Pulse Code Modulation (PCM)

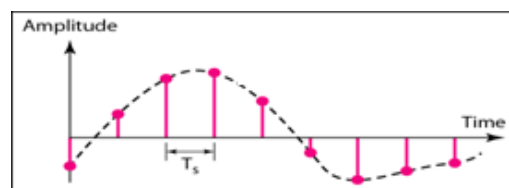
The most common technique to change an analog signal to digital data (digitization) is called pulse code modulation (PCM). A PCM encoder has three processes.

1. Sampling.
2. Quantization.
3. Encoding.



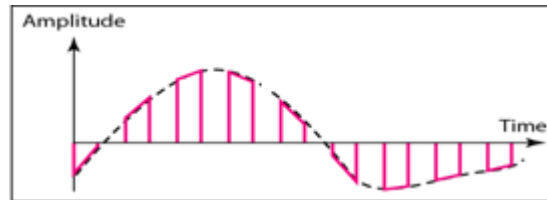
Sampling:

- The first step in PCM is sampling. The analog signal is sampled every T_s s, where T_s is the sample interval or period.
- The inverse of the sampling interval is called the sampling rate or sampling frequency and denoted by f_s , where $f_s = 1/T_s$.
- There are three sampling methods-ideal, natural, and flat-top.
 1. In **Ideal sampling**, pulses from the analog signal are sampled.

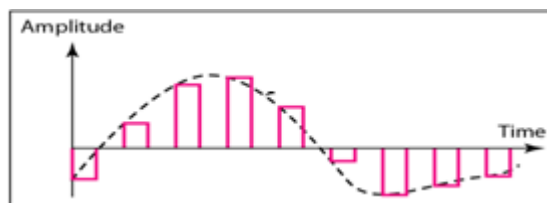


2. In **natural sampling**, a high-speed switch is turned on for only the small period of time when

the sampling occurs. The result is a sequence of samples that retains the shape of the analog signal.



3. The most common sampling method, called **sample and hold**, however, creates **flat-top samples** by using a circuit.



- The sampling process is sometimes referred to as **pulse amplitude modulation (PAM)**.
- **Sampling Rate:** According to the Nyquist theorem, to reproduce the original analog signal, one necessary condition is that the sampling rate be at least twice the highest frequency in the original signal.

Quantization

- The result of sampling is a series of pulses with amplitude values between the maximum and minimum amplitudes of the signal.
- The set of amplitudes can be infinite with non integral values between the two limits. These values cannot be used in the encoding process.

The following are the steps in quantization:

1. We assume that the original analog signal has instantaneous amplitudes between V_{min} and V_{max} .
2. We divide the range into L zones, each of height Δ (delta).

$$\Delta = (V_{max} - V_{min}) / L$$

3. We assign quantized values of 0 to $L - 1$ to the midpoint of each zone.
4. We approximate the value of the sample amplitude to the quantized values.

Quantization Levels: The choice of L , the number of levels, depends on the range of the amplitudes of the analog signal and how accurately we need to recover the signal.

Quantization Error: One important issue is the error created in the quantization process. Quantization is an approximation process. The input values to the quantizer are the real values; the output values are

the approximated values. The output values are chosen to be the middle value in the zone. If the input value is also at the middle of the zone, there is no quantization error; otherwise, there is an error.

The quantization error changes the signal-to-noise ratio of the signal, which in turn reduces the upper limit capacity according to Shannon.

It can be proven that the contribution of the quantization error to the SNR_{dB} of the signal depends on the number of quantization levels L , or the bits per sample n_b as shown in the following formula:

$$SNR_{dB} = 6.02n_b + 1.76 \text{ dB}$$

Encoding:

The last step in PCM is encoding. After each sample is quantized and the number of bits per sample is decided, each sample can be changed to an n_b -bit code word.

Example: A quantization code of 2 is encoded as 010; 5 is encoded as 101; and so on.

Number of bits for each sample is determined from the number of quantization levels. If the number of quantization levels is L , the number of bits is $n_b = \log_2 L$. The bit rate can be found from the formula

$$\text{Bit rate} = \text{sampling rate} \times \text{number of bits per sample} = f_s \times n_b$$

Example 4.1:

What is the Nyquist sampling rate for each of the following signals?

a. A low-pass signal with bandwidth of 200 KHz?

b. A band-pass signal with bandwidth of 200 KHz if the lowest frequency is 100 KHz?

Solution:

a. In a low-pass signal, the minimum frequency 0. Therefore, we have
 $f_{\max} = 0 + 200 = 200 \text{ KHz}$. $\rightarrow f_s = 2 \times 200,000 = \mathbf{400,000 \text{ samples/s}}$

b. In a bandpass signal, the maximum frequency is equal to the minimum frequency plus the bandwidth. Therefore, we have

$f_{\max} = 100 + 200 = 300 \text{ KHz}$. $\rightarrow f_s = 2 \times 300,000 = \mathbf{600,000 \text{ samples /s}}$

4.2 Transmission Modes

- The transmission of binary data across a link can be accomplished in either parallel or serial mode.
- In parallel mode, multiple bits are sent with each clock tick.
- In serial mode, 1 bit is sent with each clock tick. While there is only one way to send parallel data, there are three subclasses of serial transmission: asynchronous, synchronous, and isochronous.

4.2.1 Parallel Transmission

- Binary data, consisting of 1s and 0s, may be organized into groups of n bits each. Computers produce and consume data in groups of bits. By grouping, we can send data n bits at a time instead of 1. This is called parallel transmission.
- The mechanism for parallel transmission is a conceptually simple one: Use n wires to send n bits at one time. That way each bit has its own wire, and all n bits of one group can be transmitted with each clock tick from one device to another. Figure shows how parallel transmission works for $n = 8$. Typically, the eight wires are bundled in a cable with a connector at each end.



- The advantage of parallel transmission is speed. All else being equal, parallel transmission can increase the transfer speed by a factor of n over serial transmission.
- But there is a significant disadvantage: cost. Parallel transmission requires n communication lines (wires in the example) just to transmit the data stream. Because this is expensive, parallel transmission is usually limited to short distances.

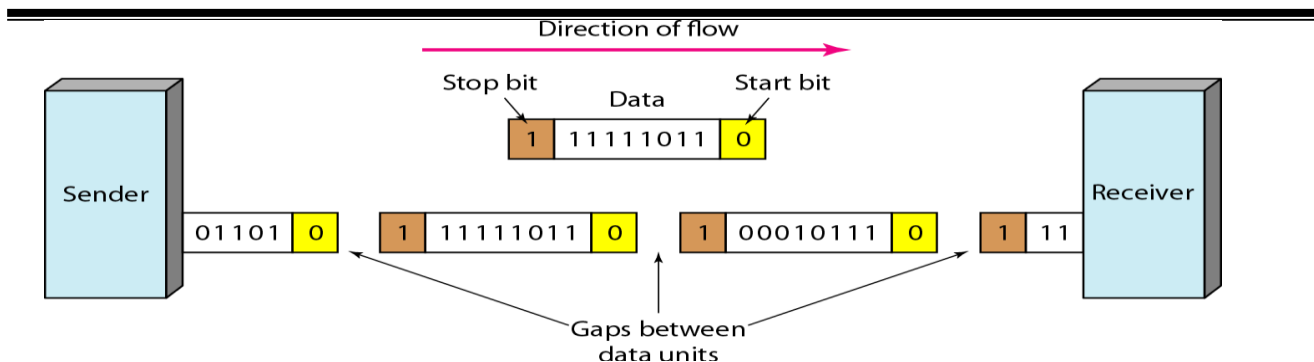
4.2.2 Serial Transmission

- In serial transmission one bit follows another, so we need only one communication channel rather than n to transmit data between two communicating devices.
- The advantage of serial over parallel transmission is that with only one communication channel, serial transmission reduces the cost of transmission over parallel by roughly a factor of n .
- Since communication within devices is parallel, conversion devices are required at the interface between the sender and the line (parallel-to-serial) and between the line and the receiver (serial-to-parallel).
- Serial transmission occurs in one of three ways: asynchronous, synchronous, and isochronous.



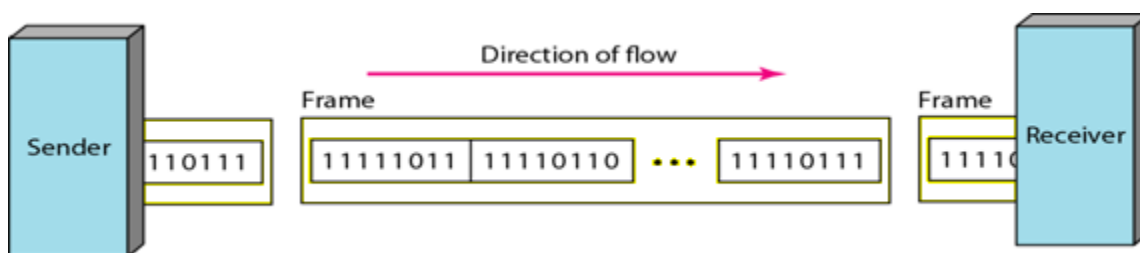
1) Asynchronous Transmission

- Asynchronous transmission is so named because the timing of a signal is unimportant. Instead, information is received and translated by agreed upon patterns.
- As long as those patterns are followed, the receiving device can retrieve the information without regard to the rhythm in which it is sent.
- Patterns are based on grouping the bit stream into bytes. Each group, usually 8 bits, is sent along the link as a unit. The sending system handles each group independently, relaying it to the link whenever ready, without regard to a timer.
- Without synchronization, the receiver cannot use timing to predict when the next group will arrive.
- To alert the receiver to the arrival of a new group, therefore, an extra bit is added to the beginning of each byte. This bit, usually a 0, is called the start bit.
- To let the receiver know that the byte is finished, 1 or more additional bits are appended to the end of the byte. These bits, usually 1 s, are called stop bits.
- By this method, each byte is increased in size to at least 10 bits, of which 8 bits is information and 2 bits or more are signals to the receiver.
- In addition, the transmission of each byte may then be followed by a gap of varying duration. This gap can be represented either by an idle channel or by a stream of additional stop bits.
- This mechanism is called *asynchronous* because, at the byte level, the sender and receiver do not have to be synchronized. But within each byte, the receiver must still be synchronized with the incoming bit stream.
- The addition of stop and start bits and the insertion of gaps into the bit stream make asynchronous transmission slower than forms of transmission that can operate without the addition of control information. But it is cheap and effective.



2) Synchronous Transmission

- In synchronous transmission, the bit stream is combined into longer "frames," which may contain multiple bytes.
- Each byte, however, is introduced onto the transmission link without a gap between it and the next one. It is left to the receiver to separate the bit stream into bytes for decoding purposes.



- The advantage of synchronous transmission is speed. With no extra bits or gaps to introduce at the sending end and remove at the receiving end, and, by extension, with fewer bits to move across the link, synchronous transmission is faster than asynchronous transmission.
- Although there is no gap between characters in synchronous serial transmission, there may be uneven gaps between frames.

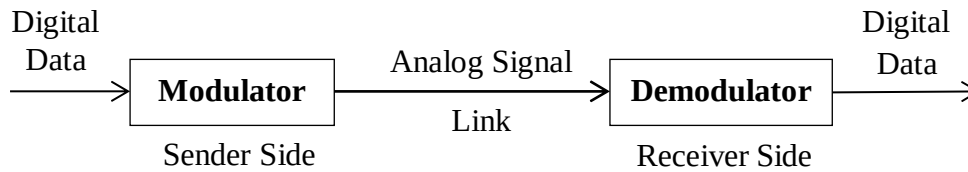
3) Isochronous

In real-time audio and video, in which uneven delays between frames are not acceptable, synchronous transmission fails. For example, TV images are broadcast at the rate of 30 images per second; they must be viewed at the same rate. If each image is sent by using one or more frames, there should be no delays between frames. For this type of application, synchronization between characters is not enough; the entire stream of bits must be synchronized. The isochronous transmission guarantees that the data arrive at a fixed rate.

4.3 Digital-To-Analog Conversion

Digital-to-analog conversion is the process of changing one of the characteristics of an analog signal

based on the information in digital data.



- Sine wave is defined by three characteristics: amplitude, frequency, and phase.
- When we vary anyone of these characteristics, we create a different version of that wave. So, by changing one characteristic of a simple electric signal, we can use it to represent digital data.
- Any of the three characteristics can be altered in this way, giving us at least three mechanisms for modulating digital data into an analog signal: amplitude shift keying (ASK), frequency shift keying (FSK), and phase shift keying (PSK).
- In addition, there is a fourth (and better) mechanism that combines changing both the amplitude and phase, called quadrature amplitude modulation (QAM). QAM is the most efficient of these options and is the mechanism commonly used today.

4.3.1 Aspects of Digital-to-Analog Conversion

- **Data Element versus Signal Element:** Data element is the smallest piece of information to be exchanged, that is the bit. Signal element is the smallest unit of a signal that is constant.
- **Data Rate Versus Signal Rate:** The relationship between them is

$$S = N \times 1/r \text{ baud}$$

Where, N is the data rate (bps), r is the number of data elements carried in one signal element. $r = \log_2 L$, where L is the type of signal element.

Example 4.2:

An analog signal carries 4 bits per signal element. If 1000 signal elements are sent per second, find the bit rate.

Solution:

$$S = N \times 1/r$$

$$N = S \times r$$

$$N = 1000 \times 4$$

$$N = 4000\text{bps}$$

Example 4.3:

An analog signal has a bit rate of 6000 bps and a baud rate of 1000 baud. How many data elements are carried by each signal element? How many signal elements do we need?

Solution:

We know that:

$$S = N \times 1/r$$

$$r = N/S$$

$$r = 6000/1000$$

No of data elements carried by signal element $r = 6\text{bits/ baud}$

$$r = \log_2 L$$

$$L = 2^r$$

$$L = 2^6$$

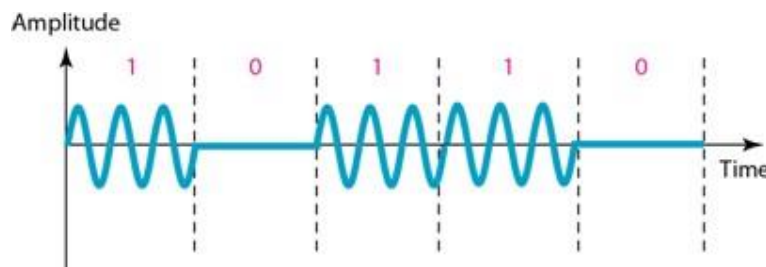
No. of signal elements $L = 64$

4.3.2 Amplitude Shift Keying

In amplitude shift keying, the amplitude of the carrier signal is varied to create signal elements. Both frequency and phase remain constant while the amplitude changes.

Binary ASK (BASK)

Although we can have several levels (kinds) of signal elements, each with a different amplitude, ASK is normally implemented using only two levels. This is referred to as binary amplitude shift keying or on-off keying (OOK). The peak amplitude of one signal level is 0; the other is the same as the amplitude of the carrier frequency.



Bandwidth of ASK is

$$B = (1 + d) \times S$$

Where S is the signal rate and the B is the bandwidth. The d depends on the modulation and filtering process. The value of d is between 0 and 1.

Multilevel ASK

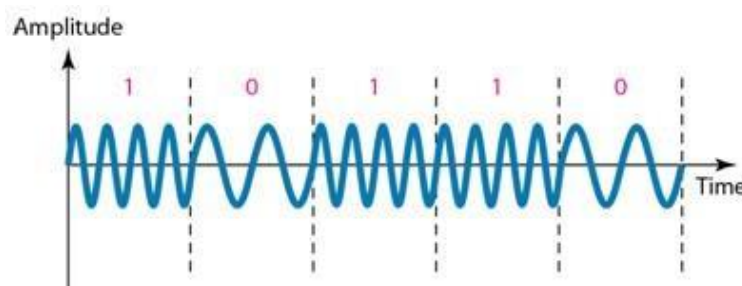
We can have multilevel ASK in which there are more than two levels. We can use 4, 8, 16, or more different amplitudes for the signal and modulate the data using 2, 3, 4, or more bits at a time.

4.3.3 Frequency Shift Keying

In frequency shift keying, the frequency of the carrier signal is varied to represent data. The frequency of the modulated signal is constant for the duration of one signal element, but changes for the next signal element if the data element changes. Both peak amplitude and phase remain constant for all signal elements.

Binary FSK (BFSK)

One way to think about binary FSK (or BFSK) is to consider two carrier frequencies. In Figure, we have selected two carrier frequencies, f_1 and f_2 . We use the first carrier if the data element is 0; we use the second if the data element is 1. However, note that this is an unrealistic example used only for demonstration purposes. Normally the carrier frequencies are very high, and the difference between them is very small.



Bandwidth for BFSK:

We can think of FSK as two ASK signals, each with its own carrier frequency (f_1 or f_2). If the difference between the two frequencies is $2\Delta f$, then the required bandwidth is

$$B = (1 + d) \times S + 2\Delta f$$

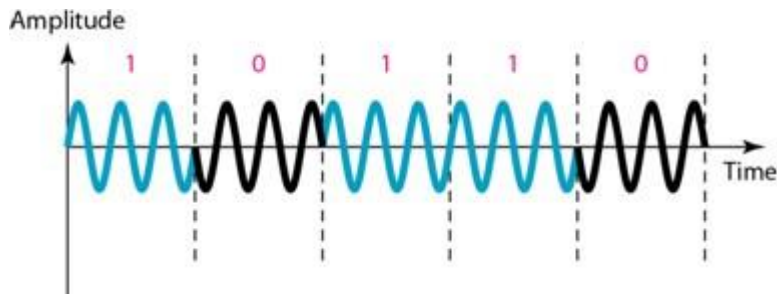
4.3.4 Phase Shift Keying

In phase shift keying, the phase of the carrier is varied to represent two or more different signal elements. Both peak amplitude and frequency remain constant as the phase changes.

Binary PSK (BPSK)

The simplest PSK is binary PSK, in which we have only two signal elements, one with a phase of 0° , and the other with a phase of 180° . Below Figure gives a conceptual view of PSK. Binary PSK is as simple as binary ASK with one big advantage—it is less susceptible to noise. In ASK, the criterion for bit detection is the amplitude of the signal; in PSK, it is the phase. Noise can change the amplitude easier than it can change the phase. In other words, PSK is less susceptible to noise than ASK. PSK is superior to FSK because we do not need two carrier signals.

Bandwidth:



$$B = (1 + d) \times S$$

Where S is the signal rate and the B is the bandwidth. The d depends on the modulation and filtering process. The value of d is between 0 and 1.

Quadrature PSK (QPSK):

The simplicity of BPSK enticed designers to use 2 bits at a time in each signal element, thereby decreasing the baud rate and eventually the required bandwidth. The scheme is called quadrature PSK or QPSK because it uses two separate BPSK modulations; one is in-phase, the other quadrature (out-of-phase). The incoming bits are first passed through a serial-to-parallel conversion that sends one bit to one modulator and the next bit to the other modulator. If the duration of each bit in the incoming signal is T , the duration of each bit sent to the corresponding BPSK signal is $2T$. This means that the bit to each BPSK signal has one-half the frequency of the original signal.

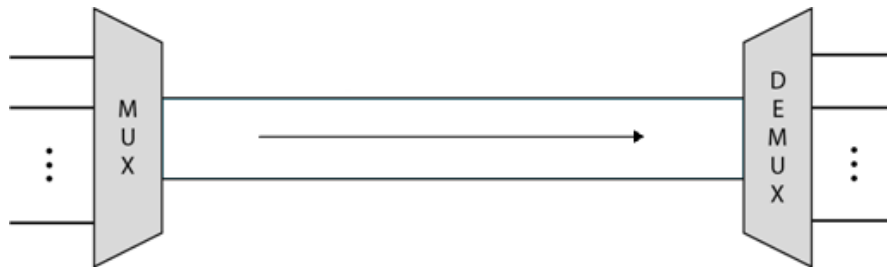
4.3.5 Quadrature Amplitude Modulation

Quadrature amplitude modulation is a combination of ASK and PSK. The idea of using two carriers, one in-phase and the other quadrature, with different amplitude levels for each carrier is the concept behind quadrature amplitude modulation (QAM).

Chapter 5: Bandwidth Utilization

5.1 Multiplexing

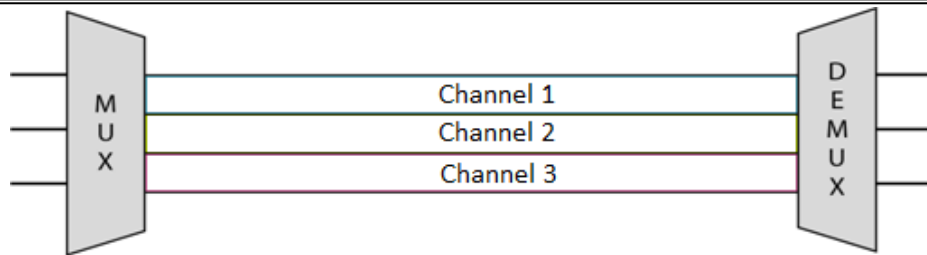
- Multiplexing is the set of techniques that allows the simultaneous transmission of multiple signals across a single data link.
- In a multiplexed system, n lines share the bandwidth of one link.
- Below figure shows the basic format of a multiplexed system.



- The lines on the left direct their transmission streams to a multiplexer (MUX), which combines them into a single stream (many-to one).
- At the receiving end, that stream is fed into a demultiplexer (DEMUX), which separates the stream back into its component transmissions (one-to-many) and directs them to their corresponding lines.
- In the figure, the word link refers to the physical path. The word channel refers to the portion of a link that carries a transmission between a given pair of lines. One link can have many (n) channels.
- There are three basic multiplexing techniques: frequency-division multiplexing, wavelength-division multiplexing, and time-division multiplexing.

5.1.1 Frequency-Division Multiplexing

- Frequency-division multiplexing (FDM) is an analog technique that can be applied when the bandwidth of a link (in hertz) is greater than the combined bandwidths of the signals to be transmitted.
- In FDM, signals generated by each sending device modulate different carrier frequencies.
- These modulated signals are then combined into a single composite signal that can be transported by the link.
- Carrier frequencies are separated by sufficient bandwidth to accommodate the modulated signal. These bandwidth ranges are the channels through which the various signals travel. Channels can be separated by strips of unused bandwidth-guard bands-to prevent signals from overlapping.
- In addition, carrier frequencies must not interfere with the original data frequencies.



Multiplexing Process

Each source generates a signal of a similar frequency range. Inside the multiplexer, these similar signals modulates different carrier frequencies (f_1, f_2 and f_3). The resulting modulated signals are then combined into a single composite signal that is sent out over a media link that has enough bandwidth to accommodate it.

Demultiplexing Process

The demultiplexer uses a series of filters to decompose the multiplexed signal into its constituent component signals. The individual signals are then passed to a demodulator that separates them from their carriers and passes them to the output lines.

Examples 5.1:

Five channels, each with a 100-kHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10kHz between the channels to prevent interference?

Solution:

For five channels, we need at least four guard bands. This means that the required bandwidth is at least $5 \times 100 + 4 \times 10 = 540$ kHz

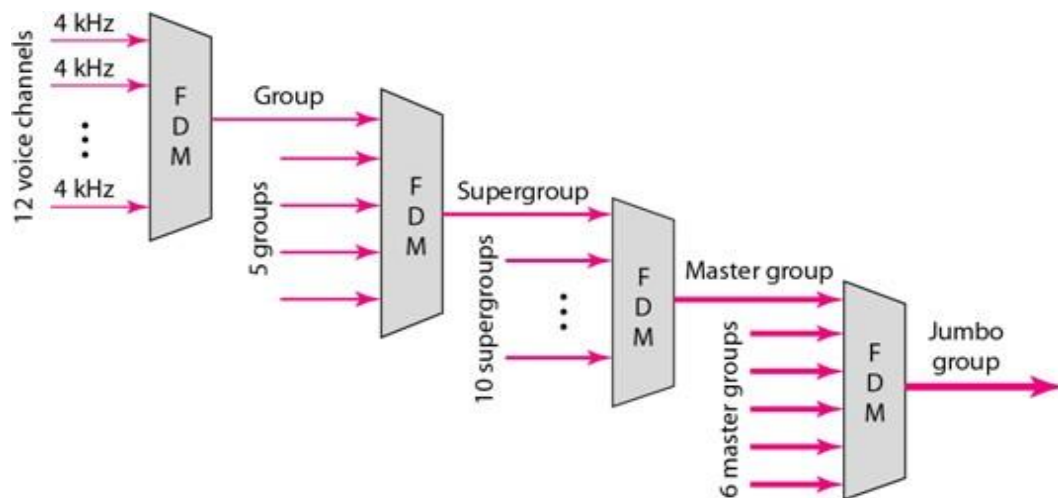
Application:

The Analog Carrier System

- To maximize the efficiency of infrastructure, telephone companies have traditionally multiplexed signals from lower-bandwidth lines onto higher-bandwidth lines. In this way, many switched or leased lines can be combined into fewer but bigger channels. For analog lines, FDM is used.
- One of these hierarchical systems used by AT&T is made up of groups, supergroups, master groups, and jumbo groups.
- In this analog hierarchy, 12 voice channels are multiplexed onto a higher-bandwidth line to create

a group. A group has 48 kHz of bandwidth and supports 12 voice channels.

- At the next level, up to five groups can be multiplexed to create a composite signal called a supergroup. A supergroup has a bandwidth of 240 kHz and supports up to 60 voice channels. Supergroups can be made up of either five groups or 60 independent voice channels.
- At the next level, 10 supergroups are multiplexed to create a master group. A master group must have 2.40 MHz of bandwidth, but the need for guard bands between the supergroups increases the necessary bandwidth to 2.52 MHz. Master groups support up to 600 voice channels.
- Finally, six master groups can be combined into a jumbo group. A jumbo group must have 15.12 MHz (6×2.52 MHz) but is augmented to 16.984 MHz to allow for guard bands between the master groups.



Other Applications of FDM

• AM and FM radio broadcasting.

Radio uses the air as the transmission medium. A special band from 530 to 1700 kHz is assigned to AM radio. All radio stations need to share this band. Each AM station needs 10 kHz of bandwidth. Each station uses a different carrier frequency, which means it is shifting its signal and multiplexing. The signal that goes to the air is a combination of signals. A receiver receives all these signals, but filters (by tuning) only the one which is desired. Without multiplexing, only one AM station could broadcast to the common link, the air. However, we need to know that there is physical multiplexer or demultiplexer here. Multiplexing is done at the data link layer.

The situation is similar in FM broadcasting. However, FM has a wider band of 88 to 108 MHz because each station needs a bandwidth of 200 kHz.

• Television broadcasting.

Each TV channel has its own bandwidth of 6 MHz.

- **The first generation of cellular telephones.**

Each user is assigned two 30-kHz channels, one for sending voice and the other for receiving. The voice signal, which has a bandwidth of 3 kHz (from 300 to 3300 Hz), is modulated by using FM.

Example 5.2:

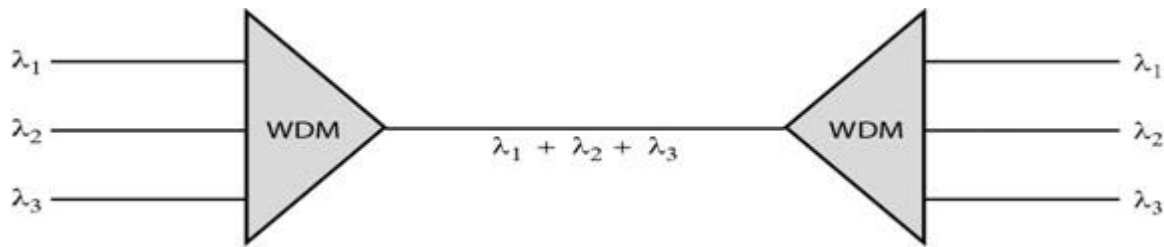
The Advanced Mobile Phone System (AMPS) uses two bands. The first band of 824 to 849 MHz is used for sending, and 869 to 894 MHz is used for receiving. Each user has a bandwidth of 30 kHz in each direction. The 3-kHz voice is modulated using FM, creating 30 kHz of modulated signal. How many people can use their cellular phones simultaneously?

Solution:

Each band is 25 MHz. If we divide 25 MHz by 30 kHz, we get 833.33. In reality, the band is divided into 832 channels. Of these, 42 channels are used for control, which means only 790 channels are available for cellular phone users.

5.1.2 Wavelength-Division Multiplexing

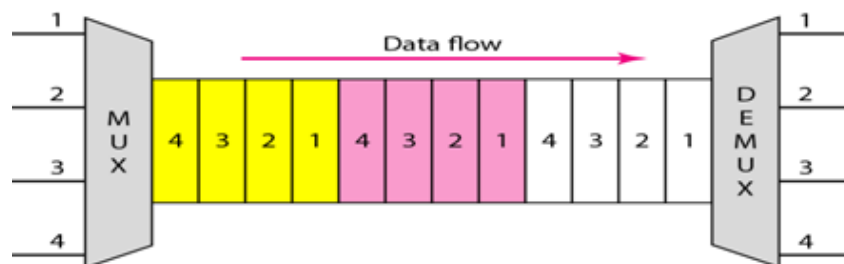
- Wavelength-division multiplexing (WDM) is designed to use the high-data-rate capability of fiber-optic cable.
- The optical fiber data rate is higher than the data rate of metallic transmission cable. Using a fiber-optic cable for one single line wastes the available bandwidth. Multiplexing allows us to combine several lines into one.
- WDM is conceptually the same as FDM, except that the multiplexing and demultiplexing involve optical signals transmitted through fiber-optic channels.
- Very narrow bands of light from different sources are combined to make a wider band of light. At the receiver, the signals are separated by the demultiplexer.
- The combining and splitting of light sources are easily handled by a prism. Prism bends a beam of light based on the angle of incidence and the frequency. Using this technique, a multiplexer can be made to combine several input beams of light, each containing a narrow band of frequencies, into one output beam of a wider band of frequencies. A demultiplexer can also be made to reverse the process.



- One application of WDM is the SONET network in which multiple optical fiber lines are multiplexed and demultiplexed.
- A new method, called dense WDM (DWDM), can multiplex a very large number of channels by spacing channels very close to one another. It achieves even greater efficiency.

5.1.3 Synchronous Time-Division Multiplexing

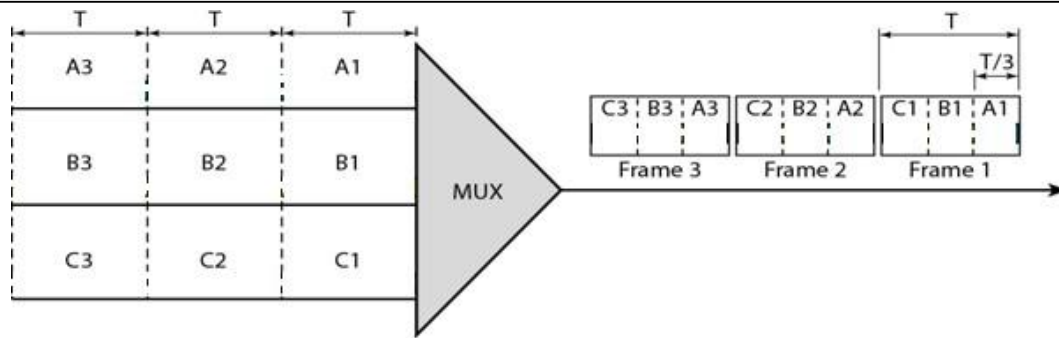
- Time-division multiplexing (TDM) is a digital process that allows several connections to share the high bandwidth of a link.
- Instead of sharing a portion of the bandwidth as in FDM, time is shared. Each connection occupies a portion of time in the link.
- TDM is a digital multiplexing technique for combining several low-rate channels into one high-rate one.



TDM is divided into two different schemes: synchronous and statistical

Time Slots and Frames

In synchronous TDM, the data flow of each input connection is divided into units, where each input occupies one input time slot. A unit can be 1 bit, one character, or one block of data. Each input unit becomes one output unit and occupies one output time slot. However, the duration of an output time slot is n times shorter than the duration of an input time slot. If an input time slot is T s, the output time slot is Tn s, where n is the number of connections.

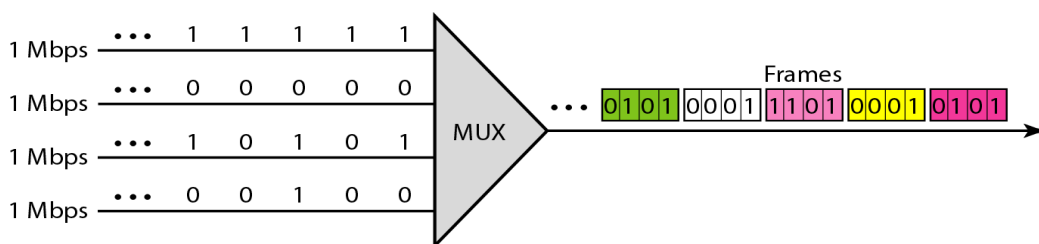


In synchronous TDM, a round of data units from each input connection is collected into a frame. If we have n connections, a frame is divided into n time slots and one slot is allocated for each unit, one for each input line. If the duration of the input unit is T , the duration of each slot is T/n and the duration of each frame is T .

In synchronous TDM, the data rate of the link is n times faster, and the unit duration is n times shorter. Time slots are grouped into frames. A frame consists of one complete cycle of time slots, with one slot dedicated to each sending device. In a system with n input lines, each frame has n slots, with each slot allocated to carrying data from a specific input line.

Example:

Figure shows synchronous TDM with a data stream for each input and one data stream for the output. The unit of data is 1 bit. Find (a) the input bit duration, (b) the output bit duration, (c) the output bit rate, and (d) the output frame rate.



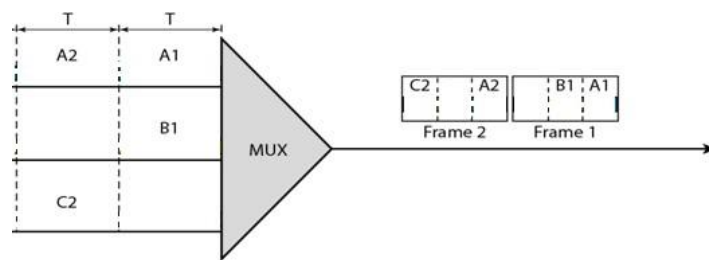
Solution

- The input bit duration is the inverse of the bit rate: $1/1 \text{ Mbps} = 1 \mu\text{s}$.
- The output bit duration is one-fourth of the input bit duration, or $1/4 \mu\text{s}$.
- The output bit rate is the inverse of the output bit duration or $1/4 \mu\text{s}$, or 4 Mbps. This can also be deduced from the fact that the output rate is 4 times as fast as any input rate; so the output rate $= 4 \times 1 \text{ Mbps} = 4 \text{ Mbps}$.
- The frame rate is always the same as any input rate. So the frame rate is 1,000,000 frames per

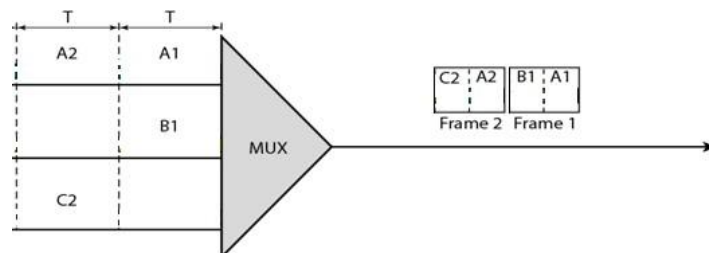
second. Because we are sending 4 bits in each frame, we can verify the result of the previous question by multiplying the frame rate by the number of bits per frame.

5.1.4 Statistical Time-Division Multiplexing

In statistical time-division multiplexing, slots are dynamically allocated to improve bandwidth efficiency. Only when an input line has a slot's worth of data to send is it given a slot in the output frame. In statistical multiplexing, the number of slots in each frame is less than the number of input lines. The multiplexer checks each input line in round robin fashion; it allocates a slot for an input line if the line has data to send; otherwise, it skips the line and checks the next line.



(a) Synchronous TDM

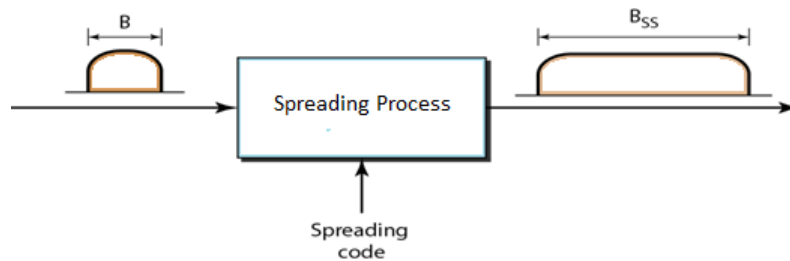


(b) Statistical TDM

5.2 Spread Spectrum

- Multiplexing combines signals from several sources to achieve bandwidth efficiency; the available bandwidth of a link is divided between the sources.
- In spread spectrum, we also combine signals from different sources to fit into a larger bandwidth.
- Spread spectrum is designed to be used in wireless applications (LANs and WANs).
- In wireless applications, all stations use air (or a vacuum) as the medium for communication. Stations must be able to share this medium without interception by an eavesdropper and without being subject to jamming from a malicious intruder (in military operations, for example).
- If the required bandwidth for each station is B , spread spectrum expands it to B_{ss} such that $B_{ss} \gg B$. The expanded bandwidth allows the source to wrap its message in a protective envelope for a

more secure transmission.



Spread spectrum achieves its goals through two principles:

1. The bandwidth allocated to each station needs to be, by far, larger than what is needed. This allows redundancy.
2. The expanding of the original bandwidth B to the bandwidth B_{ss} must be done by a process that is independent of the original signal. In other words, the spreading process occurs after the signal is created by the source.

After the signal is created by the source, the spreading process uses a spreading code and spreads the bandwidth. The figure shows the original bandwidth B and the spreaded bandwidth B_{ss} . The spreading code is a series of numbers that look random, but are actually a pattern.

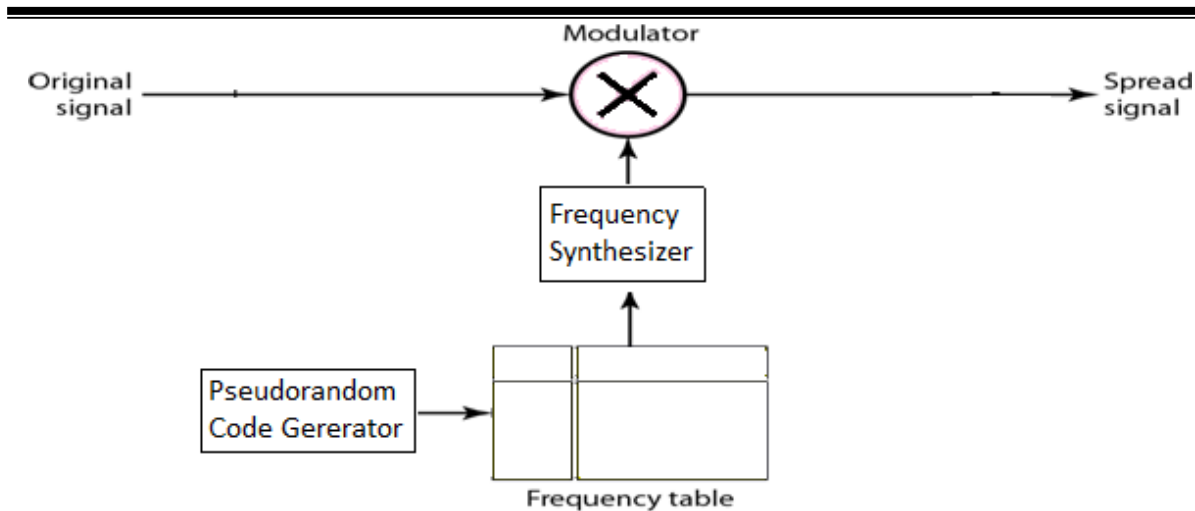
There are two techniques to spread the bandwidth:

1. Frequency hopping spread spectrum (FHSS)
2. Direct sequence spread spectrum (DSSS)

5.2.1 Frequency Hopping Spread Spectrum (FHSS)

- The frequency hopping spread spectrum (FHSS) technique uses M different carrier frequencies that are modulated by the source signal.
- At one moment, the signal modulates one carrier frequency; at the next moment, the signal modulates another carrier frequency. Although the modulation is done using one carrier frequency at a time, M frequencies are used in the long run.
- The bandwidth occupied by a source after spreading is $B_{FHSS} \gg B$.

The general layout for FHSS is shown below:



- A pseudorandom code generator, called pseudorandom noise (PN), creates a k -bit pattern for every hopping period T_h .
- The frequency table uses the pattern to find the frequency to be used for this hopping period and passes it to the frequency synthesizer.
- The frequency synthesizer creates a carrier signal of that frequency, and the source signal modulates the carrier signal.
- Suppose we have decided to have eight hopping frequencies. This is extremely low for real applications and is just for illustration. In this case, M is 8 and k is 3. The pseudorandom code generator will create eight different 3-bit patterns. These are mapped to eight different frequencies in the frequency table.

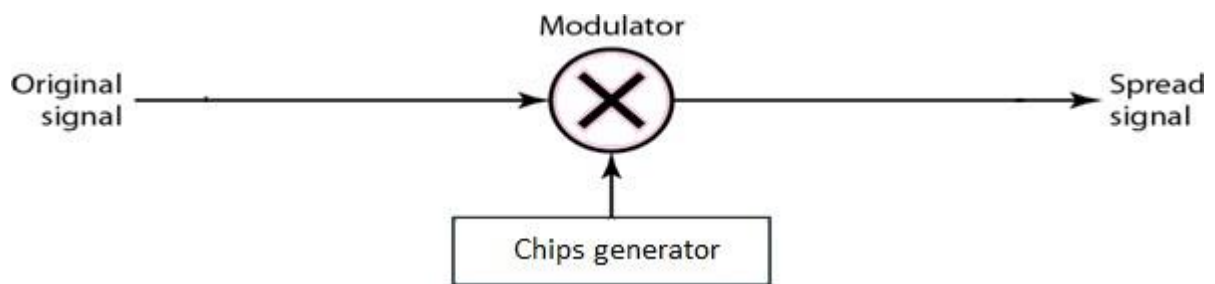
Bandwidth Sharing

If the number of hopping frequencies is M , we can multiplex M channels into one by using the same Bss bandwidth. This is possible because a station uses just one frequency in each hopping period; $M - 1$ other frequencies can be used by other $M - 1$ stations. In other words, M different stations can use the same Bss if an appropriate modulation technique such as multiple FSK (MFSK) is used.

5.2.2 Direct Sequence Spread Spectrum

The direct sequence spread spectrum (DSSS) technique also expands the bandwidth of the original signal, but the process is different.

In DSSS, we replace each data bit with 11 bits using a spreading code. In other words, each bit is assigned a code of 11 bits, called **chips**, where the chip rate is 11 times that of the data bit.



Chapter 6: Switching

6.1 Introduction

- A switched network consists of a series of interlinked nodes, called switches.
- Switches are devices capable of creating temporary connections between two or more devices linked to the switch.
- In a switched network, some of these nodes are connected to the end systems. Others are used only for routing.

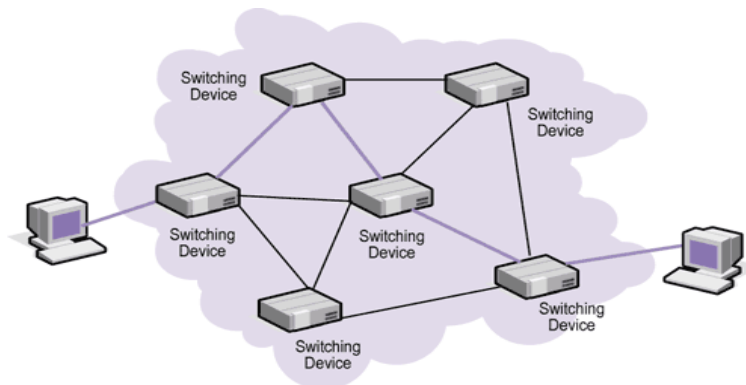


Figure 6.1: Switched Networks

Switched networks can be divided into three broad categories:

- circuit-switched networks
- packet-switched networks
- message-switched networks

Packet-switched networks can further be divided into two subcategories-

- virtual-circuit networks
- datagram networks

6.2 Circuit Switched Networks

- A circuit-switched network consists of a set of switches connected by physical links.
- A connection between two stations is a dedicated path made of one or more links. However, each connection uses only one dedicated channel on each link.
- Each link is normally divided into n channels by using FDM or TDM.
- Circuit switching takes place at the physical layer.
- Before starting communication, the stations must make a reservation for the resources to be used during the communication. These resources, such as channels (bandwidth in FDM and time slots

in TDM), switch buffers, switch processing time, and switch input/output ports, must remain dedicated during the entire duration of data transfer until the teardown phase.

- Data transferred between the two stations are not packetized. The data are a continuous flow sent by the source station and received by the destination station, although there may be periods of silence.
- There is no addressing involved during data transfer. The switches route the data based on their occupied band (FDM) or time slot (TDM).

Three Phases

The actual communication in a circuit-switched network requires three phases: connection setup, data transfer, and connection teardown.

1) Setup Phase

Before the two parties can communicate, a dedicated circuit needs to be established. The end systems are normally connected through dedicated lines to the switches, so connection setup means creating dedicated channels between the switches.

In the next step to making a connection, an acknowledgment from system M needs to be sent in the opposite direction to system A. Only after system A receives this acknowledgment is the connection established.

2) Data Transfer Phase

After the establishment of the dedicated circuit (channels), the two parties can transfer data.

3) Teardown Phase

When one of the parties needs to disconnect, a signal is sent to each switch to release the resources.

Efficiency

Circuit-switched networks are not as efficient as the other two types of networks because resources are allocated during the entire duration of the connection. These resources are unavailable to other connections.

Delay

Although a circuit-switched network normally has low efficiency, the delay in this type of network is minimal.

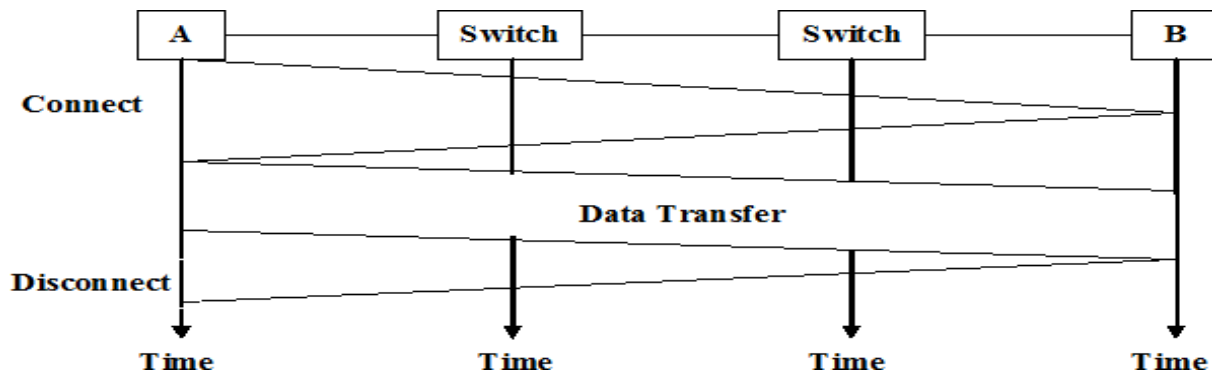


Figure 6.2: Delay in Circuit Switched Networks

- As Figure shows, there is no waiting time at each switch. The total delay is due to the time needed to create the connection, transfer data, and disconnect the circuit.
- The delay caused by the setup is the sum of four parts: the propagation time of the source computer request, the request signal transfer time, the propagation time of the acknowledgment from the destination computer, and the signal transfer time of the acknowledgment.
- The delay due to data transfer is the sum of two parts: the propagation time and data transfer time, which can be very long.
- The third box shows the time needed to tear down the circuit.

6.3 Packet Switched Networks

6.3.1 Datagram Networks

- In data communications, we need to send messages from one end system to another.
- If the message is going to pass through a packet-switched network, it needs to be divided into packets of fixed or variable size.
- The size of the packet is determined by the network and the governing protocol.
- In packet switching, there is no resource allocation for a packet. This means that there is no reserved bandwidth on the links, and there is no scheduled processing time for each packet.
- Resources are allocated on demand. The allocation is done on a first come, first-served basis. When a switch receives a packet, no matter what is the source or destination, the packet must wait if there are other packets being processed.
- In a datagram network, each packet is treated independently of all others. Even if a packet is part of a multipacket transmission, the network treats it as though it existed alone. Packets in this approach are referred to as datagrams.

- Datagram switching is normally done at the network layer.
- The datagram networks are sometimes referred to as connectionless networks. The term *connectionless* here means that the switch (packet switch) does not keep information about the connection state. There are no setup or teardown phases. Each packet is treated the same by a switch regardless of its source or destination.

Example: Below Figure shows how the datagram approach is used to deliver three packets from station A to station X. The switches in a datagram network are traditionally referred to as routers.

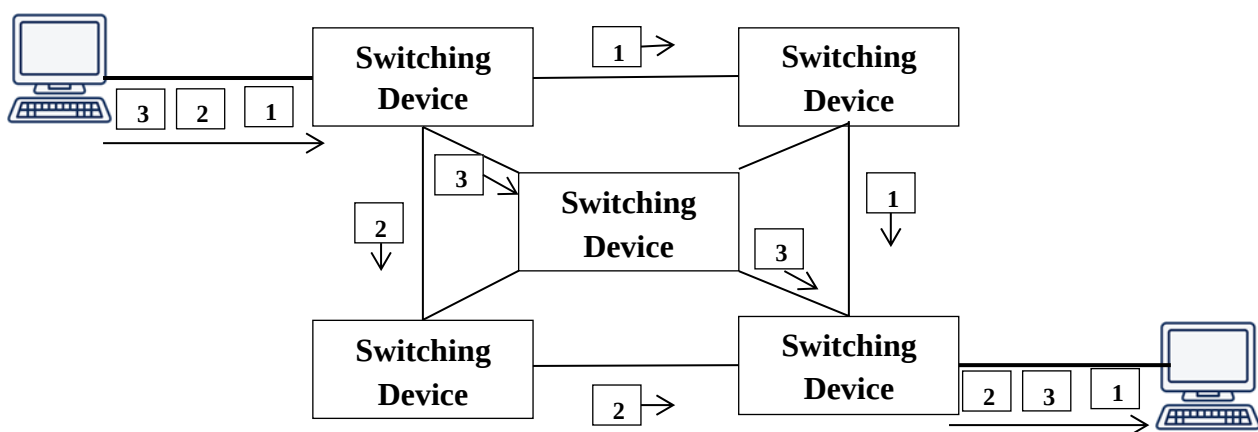


Figure 6.2: Packet Switched Networks

Routing Table

Each switch (or packet switch) has a routing table which is based on the destination address. The routing tables are dynamic and are updated periodically.

Destination Address	Output Port

Figure 6.3: Routing Table

Destination Address

- Every packet in a datagram network carries a header that contains the destination address of the packet.
- When the switch receives the packet, this destination address is examined; the routing table is consulted to find the corresponding port through which the packet should be forwarded.
- The destination address in the header of a packet in a datagram network remains the same during the entire journey of the packet.

Efficiency

The efficiency of a datagram network is better than that of a circuit-switched network; resources are

allocated only when there are packets to be transferred. If a source sends a packet and there is a delay of a few minutes before another packet can be sent, the resources can be reallocated during these minutes for other packets from other sources.

Delay

Below figure shows an example of delay in a datagram network for one single packet.

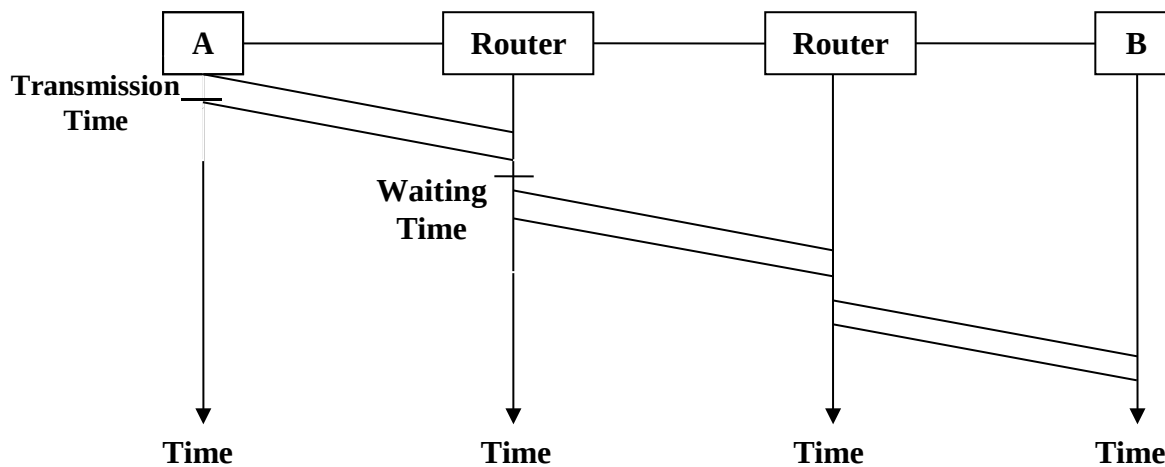


Figure 6.4: Delay in Datagram Networks

The packet travels through two switches. There are three transmission times ($3T$), three propagation delays (slopes 3τ of the lines), and two waiting times ($W1 + W2$). We ignore the processing time in each switch. The total delay is

$$\text{Total delay} = 3T + 3\tau + W1 + W2$$

6.3.2 Virtual-Circuit Networks

A virtual-circuit network is a cross between a circuit-switched network and a datagram network. It has some characteristics of both.

1. As in a circuit-switched network, there are setup and teardown phases in addition to the data transfer phase.
2. Resources can be allocated during the setup phase, as in a circuit-switched network, or on demand, as in a datagram network.
3. As in a datagram network, data are packetized and each packet carries an address in the header. However, the address in the header has local jurisdiction, not end-to-end jurisdiction.
4. As in a circuit-switched network, all packets follow the same path established during the connection.
5. A virtual-circuit network is normally implemented in the data link layer, while a circuit-switched

network is implemented in the physical layer and a datagram network in the network layer.

Addressing

In a virtual-circuit network, two types of addressing are involved: global and local (virtual-circuit identifier).

Global Addressing

A source or a destination needs to have a global address-an address that can be unique in the scope of the network or internationally if the network is part of an international network.

Virtual-Circuit Identifier

The identifier that is actually used for data transfer is called the virtual-circuit identifier (VCI). A VCI is a small number that has only switch scope; it is used by a frame between two switches. When a frame arrives at a switch, it has a VCI; when it leaves, it has a different VCI.

Three Phases

1) Data Transfer Phase

To transfer a frame from a source to its destination, all switches need to have a table entry for this virtual circuit. The table, in its simplest form, has four columns. This means that the switch holds four pieces of information for each virtual circuit that is already set up.

Below Figure shows such a switch and its corresponding table.

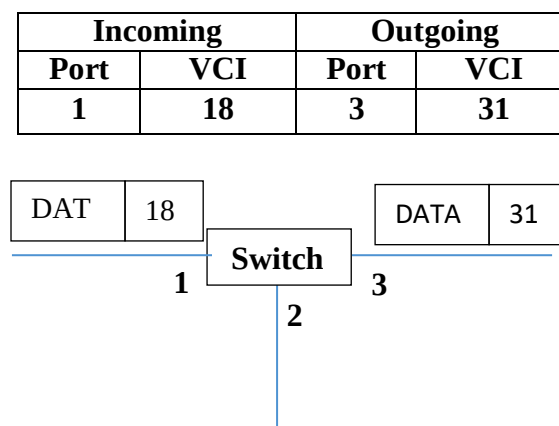


Figure 6.5: Routing Table

Figure shows a frame arriving at port 1 with a VCI of 18. When the frame arrives, the switch looks in its table to find port 1 and a VCI of 14. When it is found, the switch knows to change the VCI to 231 and send out the frame from port 3.

2) Setup Phase

In the setup phase, a switch creates an entry for a virtual circuit.

For example, suppose source A needs to create a virtual circuit to B. Two steps are required: the setup request and the acknowledgment.

Setup Request: A setup request frame is sent from the source to the destination.

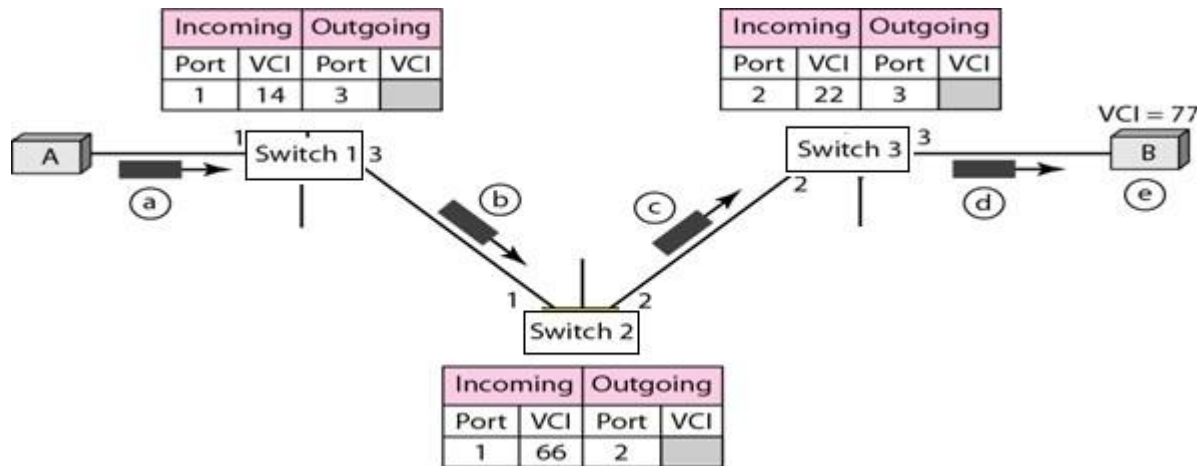


Figure 6.6: Setup request in a virtual-circuit network

- Source A sends a setup frame to switch 1.
- Switch 1 receives the setup request frame. It knows that a frame going from A to B goes out through port 3. The switch, in the setup phase, acts as a packet switch; it has a routing table which is different from the switching table. For the moment, assume that it knows the output port. The switch creates an entry in its table for this virtual circuit, but it is only able to fill three of the four columns. The switch assigns the incoming port (1) and chooses an available incoming VCI (14) and the outgoing port (3). It does not yet know the outgoing VCI, which will be found during the acknowledgment step. The switch then forwards the frame through port 3 to switch 2.
- Switch 2 receives the setup request frame. The same events happen here as at switch 1; three columns of the table are completed: in this case, incoming port (1), incoming VCI (66), and outgoing port (2).
- Switch 3 receives the setup request frame. Again, three columns are completed: incoming port (2), incoming VCI (22), and outgoing port (3).
- Destination B receives the setup frame, and if it is ready to receive frames from A, it assigns a VCI to the incoming frames that come from A, in this case 77. This VCI lets the destination know that the frames come from A, and not other sources.

Acknowledgment: A special frame, called the acknowledgment frame, completes the entries in the switching tables.

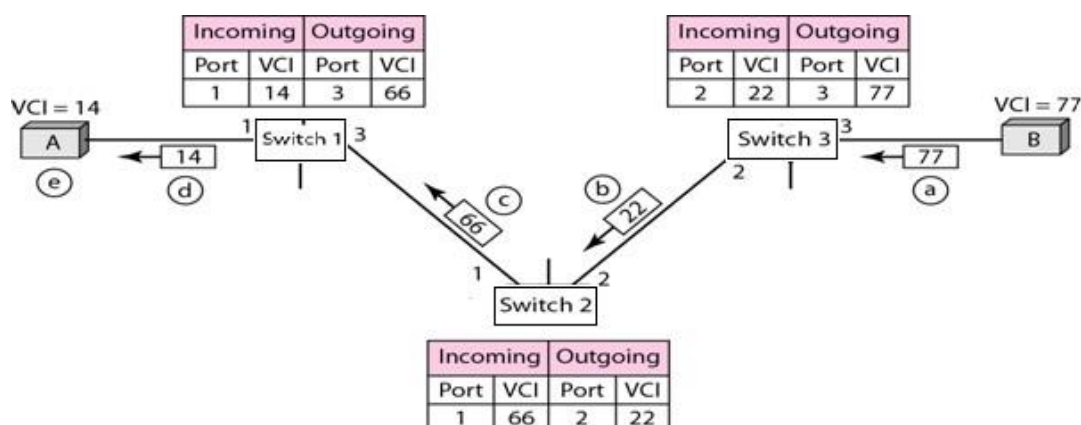


Figure 6.7: Setup acknowledgment in a virtual-circuit network

- The destination sends an acknowledgment to switch 3. The acknowledgment carries the global source and destination addresses so the switch knows which entry in the table is to be completed. The frame also carries VCI 77, chosen by the destination as the incoming VCI for frames from A. Switch 3 uses this VCI to complete the outgoing VCI column for this entry. Note that 77 is the incoming VCI for destination B, but the outgoing VCI for switch 3.
- Switch 3 sends an acknowledgment to switch 2 that contains its incoming VCI in the table, chosen in the previous step. Switch 2 uses this as the outgoing VCI in the table.
- Switch 2 sends an acknowledgment to switch 1 that contains its incoming VCI in the table, chosen in the previous step. Switch 1 uses this as the outgoing VCI in the table.
- Finally switch 1 sends an acknowledgment to source A that contains its incoming VCI in the table, chosen in the previous step.
- The source uses this as the outgoing VCI for the data frames to be sent to destination B.

3) Teardown Phase

In this phase, source A, after sending all frames to B, sends a special frame called a *teardown request*. Destination B responds with a teardown confirmation frame. All switches delete the corresponding entry from their tables.

Efficiency

Resource reservation in a virtual-circuit network can be made during the setup or can be on demand during the data transfer phase.

Delay in Virtual-Circuit Networks

In a virtual-circuit network, there is a one-time delay for setup and a one-time delay for teardown. If resources are allocated during the setup phase, there is no wait time for individual packets. Below Figure shows the delay for a packet traveling through two switches in a virtual-circuit network.

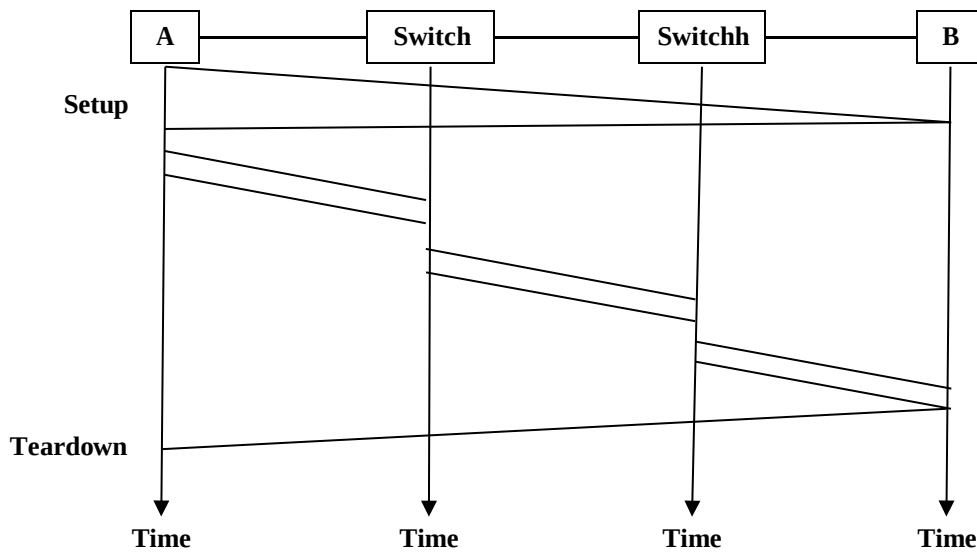


Figure 6.8: Delay in Virtual Circuit Networks

The packet is traveling through two switches (routers). There are three transmission times ($3T$), three propagation times (3τ), data transfer depicted by the sloping lines, a setup delay (which includes transmission and propagation in two directions), and a teardown delay (which includes transmission and propagation in one direction). We ignore the processing time in each switch. The total delay time is

$$\text{Total delay} = 3T + 3\tau + \text{setup delay} + \text{teardown delay}$$

Chapter 7: Error Detection and Correction

7.1 Introduction

Types of Errors

There are two types of error: Single bit error and Burst error.

1) Single-Bit Error

The term single-bit error means that only 1 bit of a given data unit (such as a byte, character, or packet)

is changed from 1 to 0 or from 0 to 1. Single-bit errors are the least likely type of error in serial data transmission.

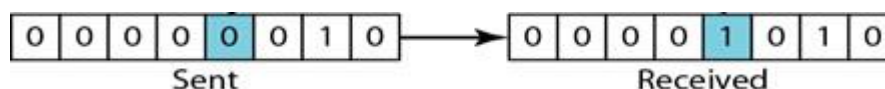


Figure 7.1: Single bit error

2) Burst Error

The term burst error means that 2 or more bits in the data unit have changed from 1 to 0 or from 0 to 1. In the below figure, 0100010001000011 was sent, but 0101110101100011 was received.

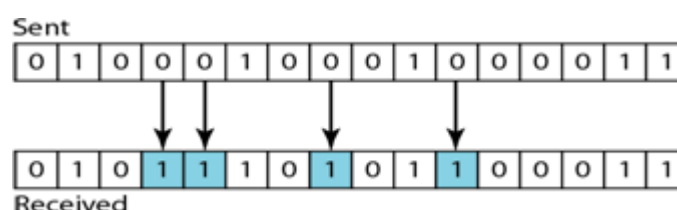


Figure 7.2: Burst Error

Redundancy

To detect or correct errors some extra bits are sent with data. These redundant bits are added by the sender and removed by the receiver. Their presence allows the receiver to detect or correct corrupted bits.

Error Detection versus Correction

- The correction of errors is more difficult than the detection.
- In error detection, we are looking only to see if any error has occurred.
- In error correction, we need to know the exact number of bits that are corrupted and more importantly, their location in the message.

Forward Error Correction versus Retransmission

There are two main methods of error correction.

- **Forward error correction** is the process in which the receiver tries to guess the message by using redundant bits. This is possible, if the number of errors is small.
- **Correction by retransmission** is a technique in which the receiver detects the occurrence of an error and asks the sender to resend the message. Resending is repeated until a message arrives that the receiver believes is error-free.

Coding

- Redundancy is achieved through various coding schemes.
- The sender adds redundant bits through a process that creates a relationship between the redundant bits and the actual data bits.
- The receiver checks the relationships between the two sets of bits to detect or correct the errors.
- Coding schemes can be divided into two broad categories: block coding and convolution coding.

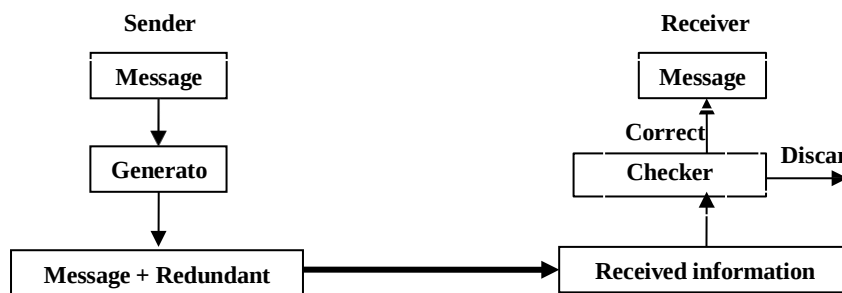


Figure 7.3: General Process of Coding

7.2 Block Coding

- In block coding message is divided into k bits blocks called **datawords**. Then r redundant bits are added to each block to make the length $n = k + r$. The resulting n -bit blocks are called **codewords**.

Error Detection

If the following two conditions are met, the receiver can detect a change in the original codeword.

1. The receiver has (or can find) a list of valid codewords.
2. The original codeword has changed to an invalid one.

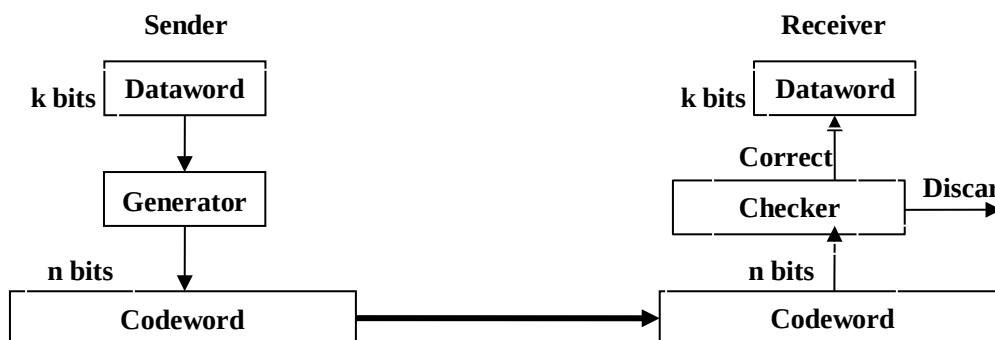


Figure 7.4: Process of error detection in block coding

- The sender creates codewords out of datawords by using a generator that applies the rules and procedures of encoding.

-
- Each codeword sent to the receiver may change during transmission.
 - If the received codeword is the same as one of the valid codewords, the word is accepted; the corresponding dataword is extracted for use. If the received codeword is not valid, it is discarded.
 - However, if the codeword is corrupted during transmission but the received word still matches a valid codeword, the error remains undetected.
 - This type of coding can detect only single errors. Two or more errors may remain undetected.

Example:

Let us assume that $k=2$ and $n=3$. Below Table shows the list of datawords and codewords.

Dataword	Codeword
00	000
01	011
10	101
11	110

Assume the sender encodes the dataword 01 as 011 and sends it to the receiver. Consider the following cases:

1. The receiver receives 011. It is a valid codeword. The receiver extracts the dataword 01 from it.
2. The codeword is corrupted during transmission, and 111 is received (the leftmost bit is corrupted). This is not a valid codeword and is discarded.
3. The codeword is corrupted during transmission, and 000 is received (the right two bits are corrupted). This is a valid codeword. The receiver incorrectly extracts the dataword 00. Two corrupted bits have made the error undetectable.

Error Correction

In error detection, the receiver needs to know only that the received codeword is invalid; in error correction the receiver needs to find (or guess) the original codeword sent.

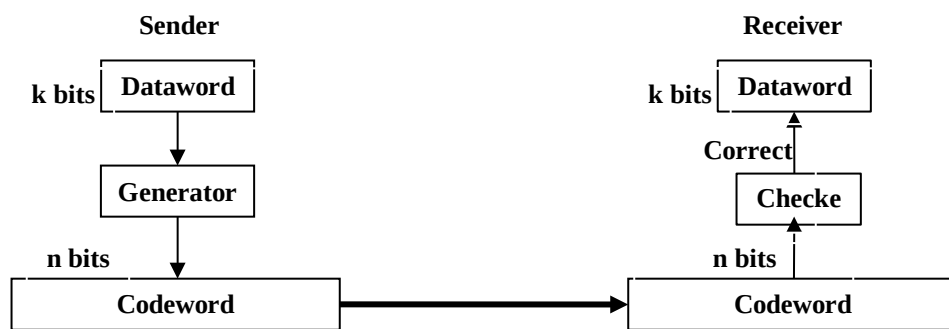


Figure 7.5: Process of error Correction in block coding

Below Table shows the datawords and codewords.

Dataword	Codeword
00	00000
01	01011
10	10101
11	11110

Assume the dataword is 01. The sender consults the table (or uses an algorithm) to create the codeword 01011.

The codeword is corrupted during transmission, and 01001 is received (error in the second bit from the right).

First, the receiver finds that the received codeword is not in the table. This means an error has occurred. (Detection must come before correction.)

The receiver, assuming that there is only 1 bit corrupted, uses the following strategy to guess the correct dataword.

1. Comparing the received codeword with the first codeword in the table (01001 versus 00000), the receiver decides that the first codeword is not the one that was sent because there are two different bits.
2. By the same reasoning, the original codeword cannot be the third or fourth one in the table.
3. The original codeword must be the second one in the table because this is the only one that differs from the received codeword by 1 bit. The receiver replaces 01001 with 01011 and consults the table to find the dataword 01.

Hamming Distance

The Hamming distance between two words (of the same size) is the number of differences between the corresponding bits. Hamming distance between two words x and y is represented as $d(x, y)$. The Hamming distance can be found by applying the XOR operation on the two words and counting the number of 1s in the result.

Example:

1. The Hamming distance $d(000, 011)$ is 2 because $000 \oplus 011$ is 011 (two 1s).
2. The Hamming distance $d(10101, 11110)$ is 3 because $10101 \oplus 11110$ is 01011 (three 1s).

Minimum Hamming Distance: The minimum Hamming distance is the smallest Hamming distance between all possible pairs in a set of words. It is represented as $d_{\min}$.

Example 7.1

Find the minimum Hamming distance of the coding scheme in below table:

Dataword	Codeword
00	000
01	011
10	101
11	110

Solution

$d(000, 011) = 2$, $d(000, 101) = 2$, $d(000, 110) = 2$, $d(011, 101) = 2$, $d(011, 110) = 2$, $d(101, 110) = 2$

The $d_{\min}$ in this case is 2.

Example 7.2

Find the minimum Hamming distance of the coding scheme in below table:

Dataword	Codeword
00	00000
01	01011
10	10101
11	11110

Solution

$d(00000, 01011) = 3$, $d(00000, 10101) = 3$, $d(00000, 11110) = 4$

$d(01011, 10101) = 4$, $d(01011, 11110) = 3$, $d(10101, 11110) = 3$

The $d_{\min}$ in this case is 3.

Linear Block Codes

Linear block code is a code in which the exclusive OR of two valid codewords creates another valid

codeword.

Minimum Distance for Linear Block Codes: The minimum Hamming distance is the number of 1s in the nonzero valid codeword with the smallest number of 1s.

Simple Parity-Check Code

- In this code, a k -bit dataword is changed to an n -bit codeword where $n = k + 1$. The extra bit, called the parity bit, is selected to make the total number of 1s in the codeword even.
- A simple parity-check code is a single-bit error-detecting code in which $n = k + 1$ with $d_{\min} = 2$.
- The encoder uses a generator that takes a copy of a 4-bit dataword (a_0, a_1, a_2 , and a_3) and generates a parity bit r_0 .
- The dataword bits and the parity bit create the 5-bit codeword. The parity bit that is added makes the number of 1s in the codeword even.

Example: Simple parity-check code $C(5, 4)$

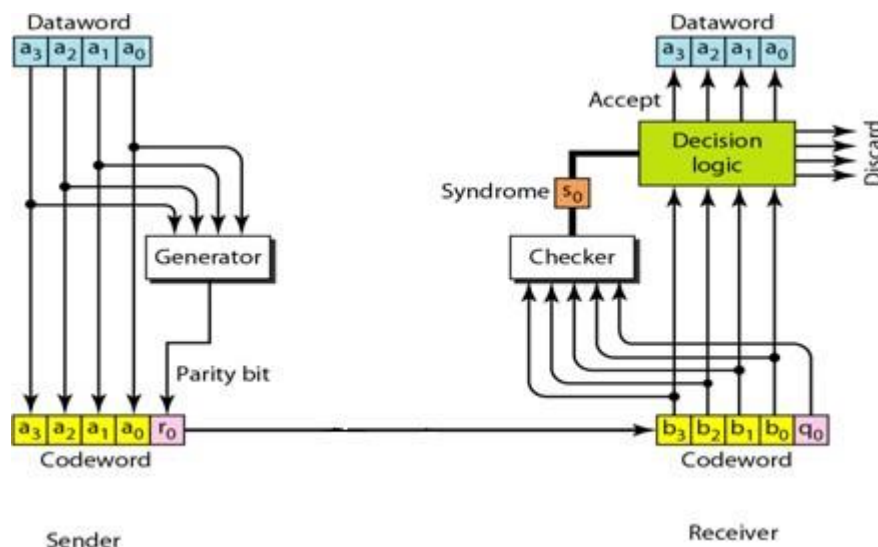


Figure 7.6: Process of Error detection in Simple parity check code

- This is normally done by adding the 4 bits of the dataword (modulo-2); the result is the parity bit. In other words,

$$r_0 = a_3 + a_2 + a_1 + a_0 \text{ (modulo - 2)}$$

- If the number of 1s is even, the result is 0; if the number of 1s is odd, the result is 1. In both cases, the total number of 1s in the codeword is even.
- The sender sends the codeword which may be corrupted during transmission.
- The receiver receives a 5-bit word.
- The checker at the receiver does the same thing as the generator in the sender with one exception:

The addition is done over all 5 bits.

- The result, which is called the syndrome, is just 1 bit. The syndrome is 0 when the number of 1s in the received codeword is even; otherwise, it is 1.

$$s_0 = b_3 + b_2 + b_1 + b_0 + q_0 \text{ (modulo } - 2)$$

- The syndrome is passed to the decision logic analyzer.
- If the syndrome is 0, there is no error in the received codeword; the data portion of the received codeword is accepted as the dataword.
- If the syndrome is 1, the data portion of the received codeword is discarded. The dataword is not created.

Example:

Assume the sender sends the dataword 1011. The codeword created from this dataword is 10111, which is sent to the receiver.

1. No error occurs; the received codeword is 10111. The syndrome is 0. The dataword 1011 is created.
2. One single-bit error changes a_1 . The received codeword is 10011. The syndrome is 1. No dataword is created.
3. One single-bit error changes r_0 . The received codeword is 10110. The syndrome is 1. No dataword is created. Note that although none of the dataword bits are corrupted, no dataword is created because the code is not sophisticated enough to show the position of the corrupted bit.
4. An error changes r_0 and a second error changes a_3 . The received codeword is 00110. The syndrome is 0. The dataword 0011 is created at the receiver. Note that here the dataword is wrongly created due to the syndrome value. The simple parity-check decoder cannot detect an even number of errors. The errors cancel each other out and give the syndrome a value of 0.
5. Three bits— a_3 , a_2 , and a_1 —are changed by errors. The received codeword is 01011. The syndrome is 1. The dataword is not created. This shows that the simple parity check, guaranteed to detect one single error, can also find any odd number of errors.

Limitation: A simple parity-check code can detect an odd number of errors.

7.3 Cyclic Codes

Cyclic codes are special linear block codes in which, if a codeword is cyclically shifted (rotated), the result is another codeword.

For example, if 1011000 is a codeword and we cyclically left-shift, then 0110001 is also a codeword.

Cyclic Redundancy Check

- In the encoder, the dataword has k bits (4 here); the codeword has n bits (7 here). The size of the dataword is augmented by adding $n - k$ (3 here) 0s to the right-hand side of the word. The n -bit result is fed into the generator.
- The generator uses a divisor of size $n - k + 1$ (4 here), predefined and agreed upon. The generator divides the augmented dataword by the divisor (modulo-2 division).
- The quotient of the division is discarded; the remainder is appended to the dataword to create the codeword.
- The decoder receives the possibly corrupted codeword. A copy of all n bits is fed to the checker which is a replica of the generator.
- The remainder produced by the checker is a syndrome of $n - k$ (3 here) bits, which is fed to the decision logic analyzer.

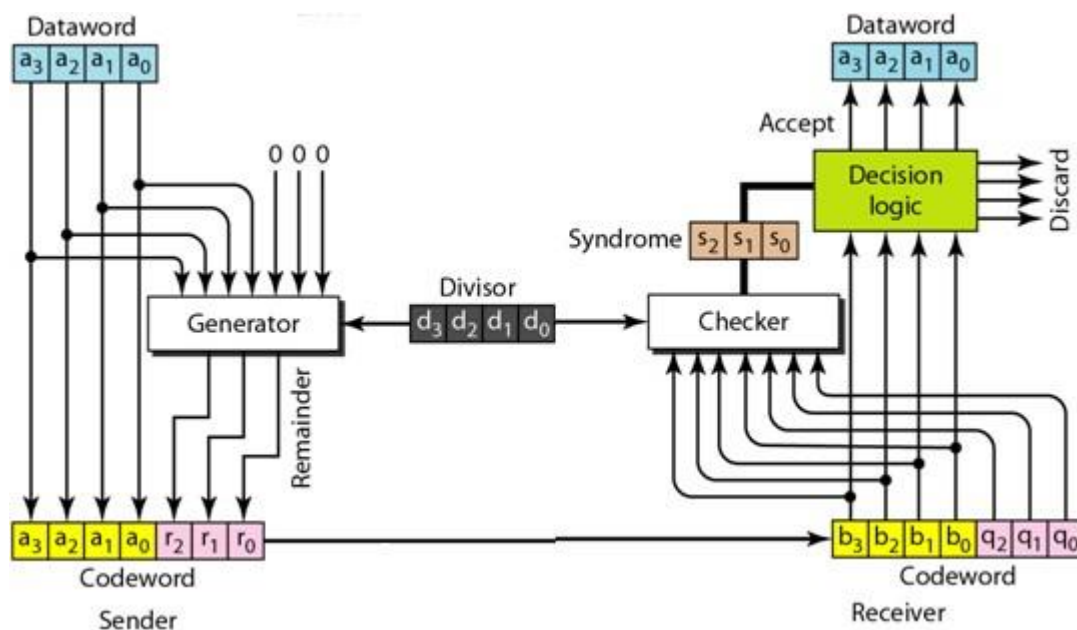
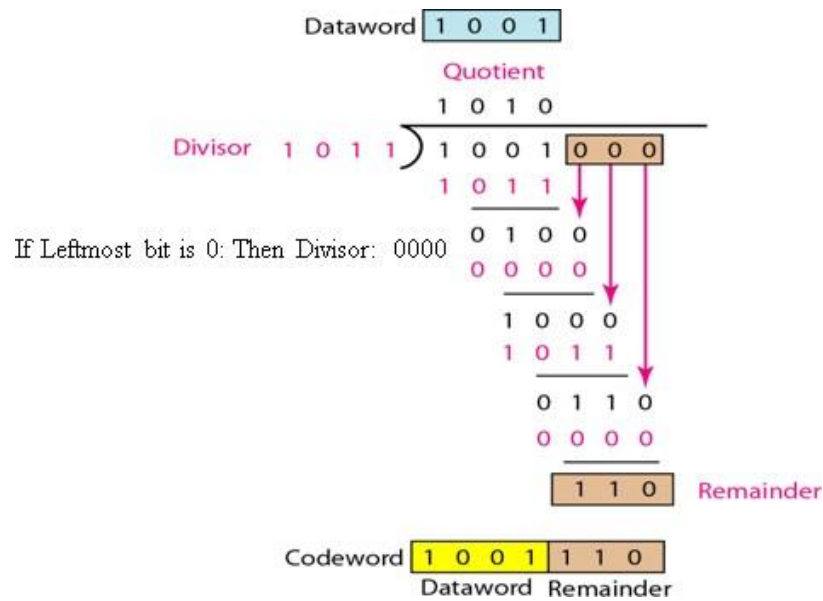


Figure 7.7: Cyclic redundancy code

- The analyzer has a simple function. If the syndrome bits are all as, the 4 leftmost bits of the codeword are accepted as the dataword (interpreted as no error); otherwise, the 4 bits are discarded (error).

Encoder

The encoder takes the dataword and augments it with $n - k$ number of 0s. It then divides the augmented dataword by the divisor.

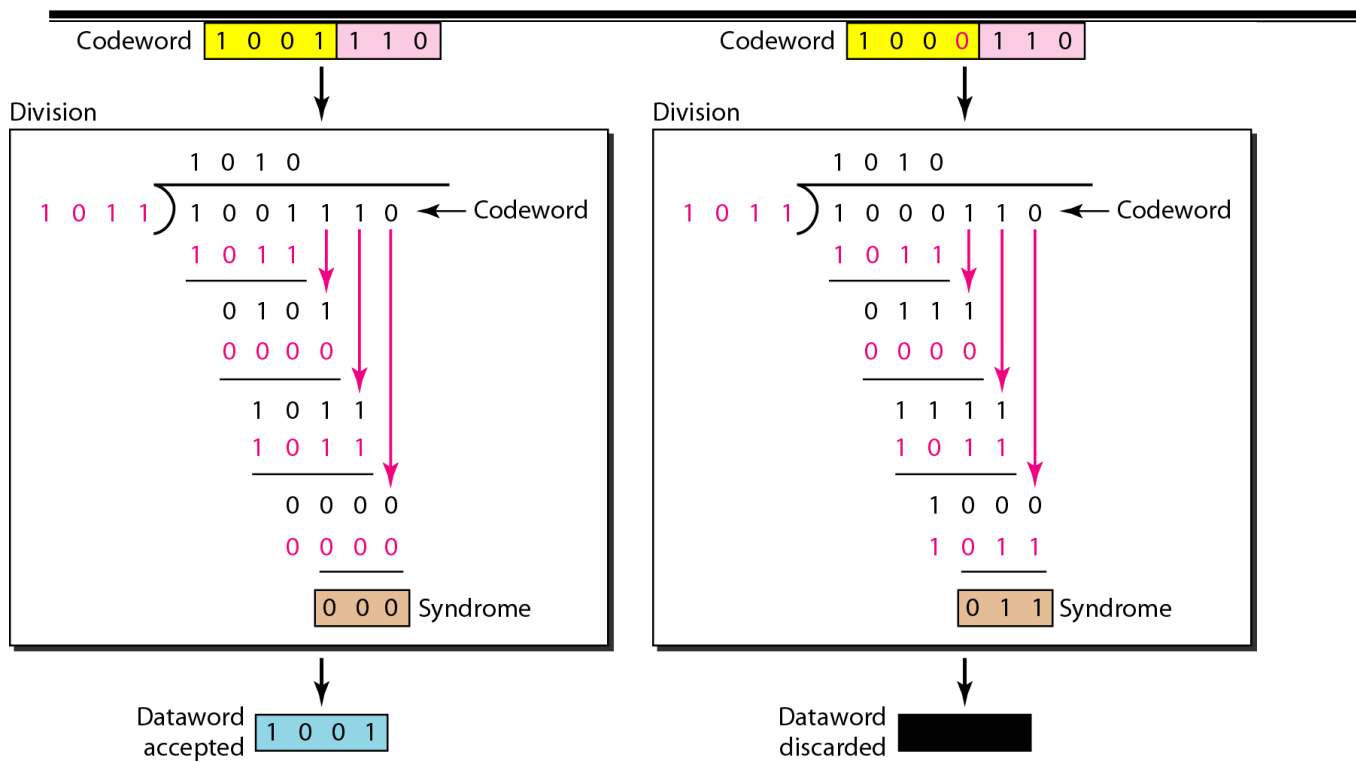


Decoder

The codeword can change during transmission. The decoder does the same division process as the encoder.

The remainder of the division is the syndrome. If the syndrome is all 0s, there is no error; the dataword is separated from the received codeword and accepted. Otherwise, everything is discarded.

The left hand figure shows the value of syndrome when no error has occurred; the syndrome is 000. The right-hand part of the figure shows the case in which there is one single error. The syndrome is not all 0s



Hardware Implementation

One of the advantages of a cyclic code is that the encoder and decoder can easily and cheaply be implemented in hardware by using a handful of electronic devices. Also, a hardware implementation increases the rate of check bit and syndrome bit calculation.

Divisor:

1. The divisor is repeatedly XORed with part of the dividend.
2. The divisor has $n - k + 1$ bits which either are predefined or are all 0s. In other words, the bits do not change from one dataword to another. In previous example, the divisor bits were either 1011 or 0000. The choice was based on the leftmost bit of the part of the augmented data bits that are active in the XOR operation.
3. A close look shows that only $n - k$ bits of the divisor is needed in the XOR operation. The leftmost bit is not needed because the result of the operation is always 0, no matter what the value of this bit. The reason is that the inputs to this XOR operation are either both 0s or both 1s.

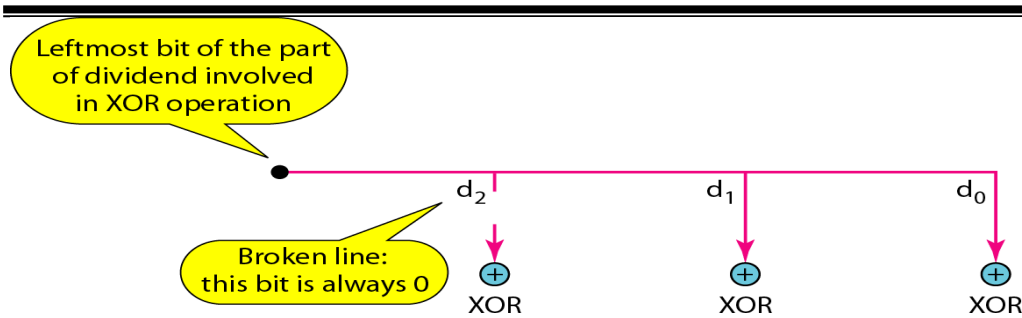


Figure 7.8: Hardware implementation of cyclic code

Steps:

1. Assume that the remainder is originally all 0s (000 in our example).
2. At each time click (arrival of 1 bit from an augmented dataword), repeat the following two actions:
 - a. Use the leftmost bit to make a decision about the divisor (011 or 000).
 - b. The other 2 bits of the remainder and the next bit from the augmented dataword (total of 3 bits) are XORed with the 3-bit divisor to create the next remainder.

Below Figure shows this simulator, but note that this is not the final design; there will be more i

Polynomials

A pattern of 0s and 1s can be represented as a **polynomial** with coefficients of 0 and 1. The power of each term shows the position of the bit; the coefficient shows the value of the bit. Figure shows a binary pattern and its polynomial representation.

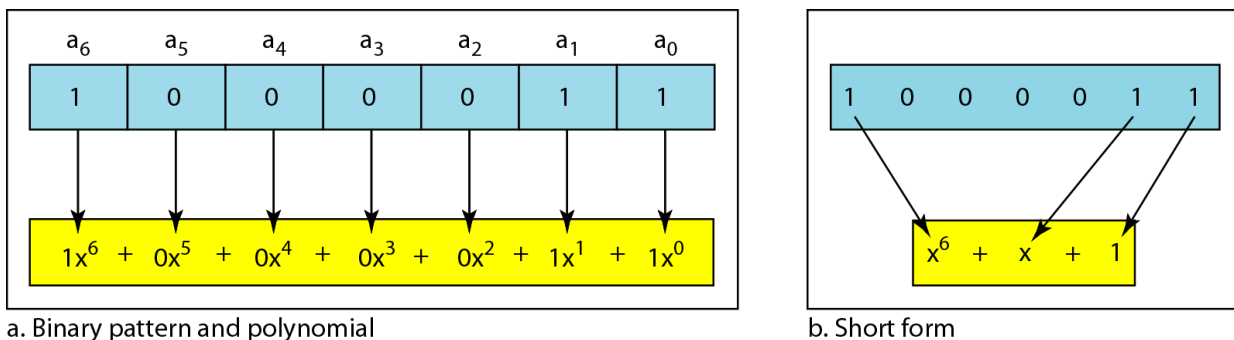


Figure 7.9: Binary to polynomial

1) Degree of a Polynomial

The degree of a polynomial is the highest power in the polynomial.

Example: the degree of the polynomial $x^6 + x + 1$ is 6.

2) Adding and Subtracting Polynomials

Adding or subtracting is done by combining terms and deleting pairs of identical terms.

Example: $(x^5 + x^4 + x^2) + (x^6 + x^4 + x^2) = x^6 + x^5$.

3) Multiplying or Dividing Terms

Multiplying a term by another term is just adding the powers.

Example: $x^3 \times x^4 = x^7$

For dividing, just subtract the power of the second term from the power of the first.

4) Multiplying Two Polynomials

Multiplying a polynomial by another is done term by term. Each term of the first polynomial must be multiplied by all terms of the second. The result, of course, is then simplified, and pairs of equal terms are deleted. The following is an example:

$$\begin{aligned}(x^5 + x^3 + x^2 + x)(x^2 + x + 1) \\= x^7 + x^6 + x^5 + x^5 + x^4 + x^3 + x^4 + x^3 + x^2 + x^3 + x^2 + x \\= x^7 + x^6 + x^3 + x\end{aligned}$$

5) Dividing One Polynomial by Another

We divide the first term of the dividend by the first term of the divisor to get the first term of the quotient.

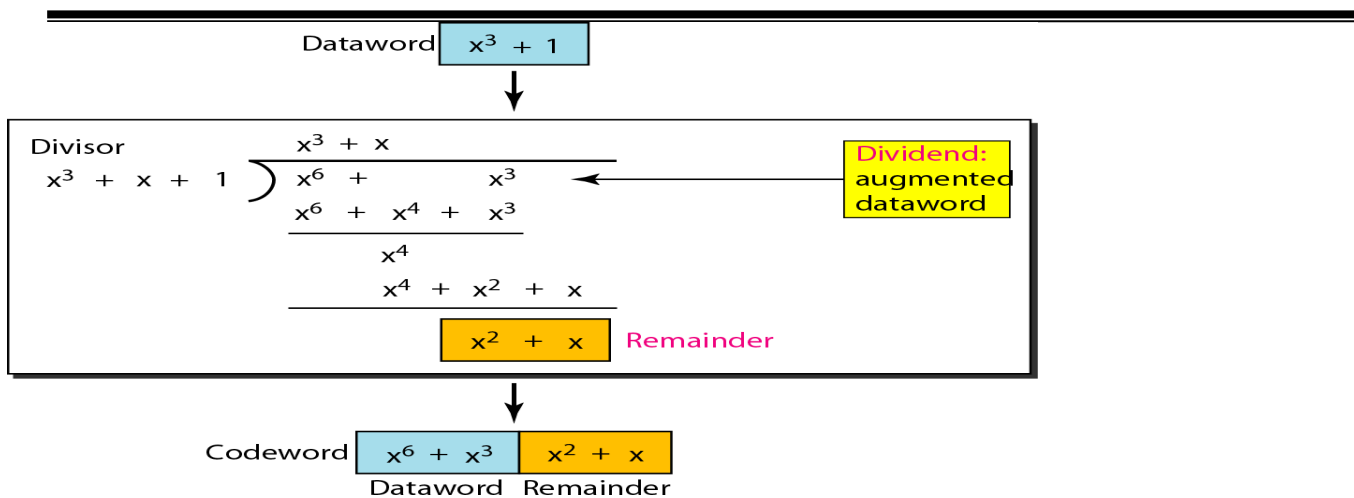
We multiply the term in the quotient by the divisor and subtract the result from the dividend. We repeat the process until the dividend degree is less than the divisor degree.

6) Shifting

A binary pattern is often shifted a number of bits to the right or left. Shifting to the left means adding extra

Os as rightmost bits; shifting to the right means deleting some rightmost bits.

Shifting left 3 bits:	10011 becomes 10011000	$x^4 + x + 1$ becomes $x^7 + x^4 + x^3$
Shifting right 3 bits:	10011 becomes 10	$x^4 + x + 1$ becomes x



Cyclic Code Analysis

Following notations can be used in the cyclic codes:

Dataword: $d(x)$ Syndrome: $s(x)$ Codeword: $c(x)$
 Error: $e(x)$ Generator: $g(x)$

In a cyclic code,

1. If $s(x) \neq 0$, one or more bits is corrupted.
2. If $s(x) = 0$, either
 - a. No bit is corrupted. or
 - b. Some bits are corrupted, but the decoder failed to detect them.

The received codeword is the sum of the sent codeword and the error.

Received codeword $= c(x) + e(x)$

The receiver divides the received codeword by $g(x)$ to get the syndrome.

$$\frac{\text{Received codeword}}{g(x)} = \frac{c(x)}{g(x)} + \frac{e(x)}{g(x)}$$

The Right hand side of above equation is called as syndrome.

If Syndrome does not have a remainder (syndrome $= 0$), either $e(x)$ is 0 or $e(x)$ is divisible by $g(x)$.

In a cyclic code, those $e(x)$ errors that are divisible by $g(x)$ are not caught.

Single-Bit Error

A single-bit error is $e(x) = x^i$, where i is the position of the bit. If a single-bit error is caught, then x^i is not divisible by $g(x)$.

If the generator has more than one term and the coefficient of x^0 is 1, all single errors can be caught.

Example 8.3:

Which of the following $g(x)$ values guarantees that a single-bit error is caught? For each case, what is the error that cannot be caught?

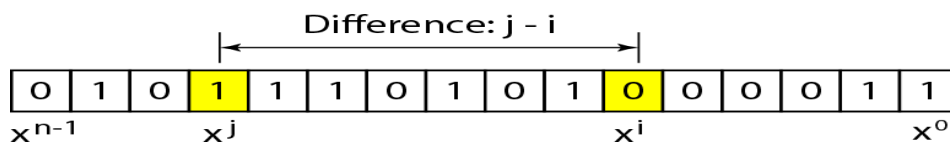
- a. $x + 1$
- b. x^3
- c. 1

Solution:

- a. No x^i can be divisible by $x + 1$. In other words, $x^i/(x + 1)$ always has a remainder. So the syndrome is nonzero. Any single-bit error can be caught.
- b. If i is equal to or greater than 3, x^i is divisible by $g(x)$. The remainder of x^i/x^3 is zero, and the receiver is fooled into believing that there is no error, although there might be one. All single-bit errors in positions 1 to 3 are caught.
- c. All values of i make j divisible by $g(x)$. No single-bit error can be caught

Two Isolated Single-Bit Errors

Two isolated single bit errors can be represented as $e(x) = x^j + x^i$. The values of i and j define the positions of the errors, and the difference $j - i$ defines the distance between the two errors.



We can write $e(x) = x^i (x^{j-i} + 1) = x^i (x^t + 1)$.

If a generator cannot divide $x^t + 1$ (t between 0 and $n - 1$), then all isolated double errors can be detected.

Example 8.4:

Find the status of the following generators related to two isolated, single-bit errors.

- a. $x + 1$
- b. $x^4 + 1$
- c. $x^7 + x^6 + 1$
- d. $x^{15} + x^{14} + 1$

Solution:

- a. This is a very poor choice for a generator. Any two errors next to each other cannot be detected.
- b. This generator cannot detect two errors that are four positions apart. The two errors can be anywhere, but if their distance is 4, they remain undetected.

-
- c. This is a good choice for this purpose.
 - d. This polynomial cannot divide any error of type $x^t + 1$ if t is less than 32,768. This means that a codeword with two isolated errors that are next to each other or up to 32,768 bits apart can be detected by this generator.

Odd Numbers of Errors

A generator that contains a factor of $x + 1$ can detect all odd-numbered errors.

Burst Errors

If L is the burst size and r is the degree of generator polynomial.

- All burst errors with $L \leq r$ will be detected.
- All burst errors with $L = r + 1$ will be detected with probability $1 - (1/2)^{r-1}$
- All burst errors with $L > r + 1$ will be detected with probability $1 - (1/2)^r$

Example 8.5:

Find the suitability of the following generators in relation to burst errors of different lengths.

- a. $x^6 + 1$
- b. $x^{18} + x^7 + x + 1$
- c. $x^{32} + x^{23} + x^7 + 1$

Solution:

- a. This generator can detect all burst errors with a length less than or equal to 6 bits; 3 out of 100 burst errors with length 7 will slip by; 16 out of 1000 burst errors of length 8 or more will slip by.
- b. This generator can detect all burst errors with a length less than or equal to 18 bits; 8 out of 1 million burst errors with length 19 will slip by; 4 out of 1 million burst errors of length 20 or more will slip by.
- c. This generator can detect all burst errors with a length less than or equal to 32 bits; 5 out of 10 billion burst errors with length 33 will slip by; 3 out of 10 billion burst errors of length 34 or more will slip by.

A good polynomial generator needs to have the following characteristics:

1. It should have at least two terms.
2. The coefficient of the term x^0 should be 1.

3. It should not divide $x^t + 1$, for t between 2 and $n - 1$.
4. It should have the factor $x + 1$.

Advantages of Cyclic Codes

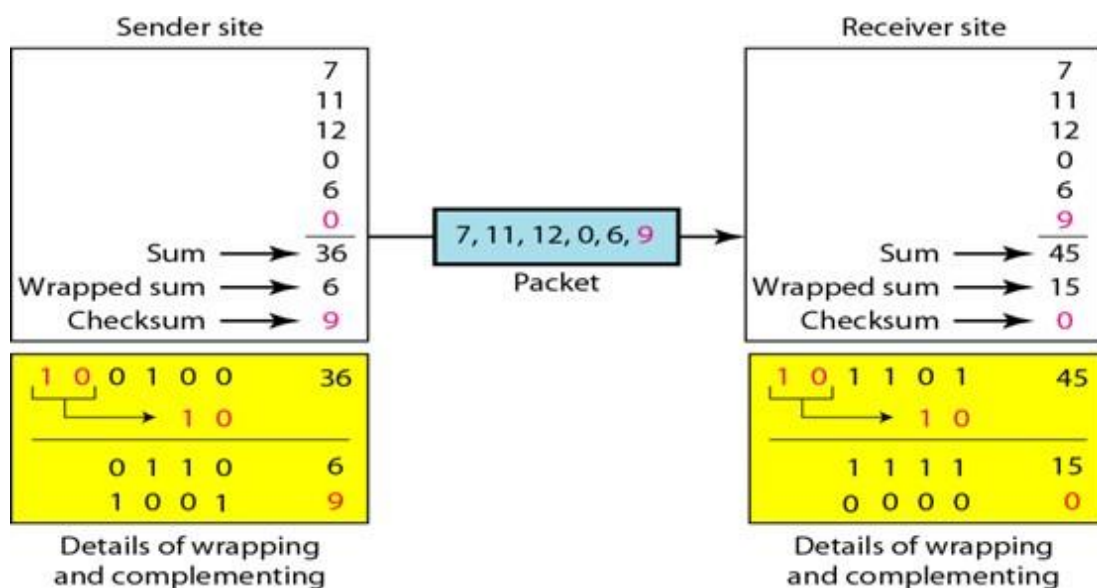
- Cyclic codes have a very good performance in detecting single-bit errors, double errors, an odd number of errors, and burst errors.
- They can easily be implemented in hardware and software.
- They are especially fast when implemented in hardware.

7.4 Checksum

The checksum is used in the Internet by several protocols. The checksum is based on the concept of redundancy.

The sender initializes the checksum to 0 and adds all data items and the checksum. The result is 36.

However, 36 cannot be expressed in 4 bits. The extra two bits are wrapped and added with the sum to create the wrapped sum value 6. The sum is then complemented, resulting in the checksum value 9 ($15 - 6 = 9$). The sender now sends six data items to the receiver including the checksum 9. The receiver follows the same procedure as the sender. It adds all data items (including the checksum); the result is 45. The sum is wrapped and becomes 15. The wrapped sum is complemented and becomes 0. Since the value of the checksum is 0, this means that the data is not corrupted. The receiver drops the checksum and keeps the other data items. If the checksum is not zero, the entire packet is dropped.



7.4.1 Internet Checksum

Traditionally, the Internet has been using a 16-bit checksum.

Sender site:

1. The message is divided into 16-bit words.
2. The value of the checksum word is set to 0.
3. All words including the checksum are added using one's complement addition.
4. The sum is complemented and becomes the checksum.
5. The checksum is sent with the data.

Receiver site:

1. The message (including checksum) is divided into 16-bit words.
2. All words are added using one's complement addition.
3. The sum is complemented and becomes the new checksum.
4. If the value of checksum is 0, the message is accepted; otherwise, it is rejected.

Example:

1	0	1	2	Carries
4	6	6	F	
7	2	6	7	
7	5	7	A	
6	1	6	E	
0	0	0	0	Checksum (initial)
8	F	B	E	Sum (partial)
8	F	B	F	Sum
7	0	4	0	Checksum (to send)

a. Checksum at the sender site

1	0	1	2	Carries
4	6	6	F	
7	2	6	7	
7	5	7	A	
6	1	6	E	
7	0	4	0	Checksum (received)
F	F	F	E	Sum (partial)
F	F	F	F	Sum
0	0	0	0	Checksum (new)

a. Checksum at the receiver site

Data Link Control

- The two main functions of the data link layer are data link control and media access control.
- The first, data link control, deals with the design and procedures for communication between two adjacent nodes: node-to-node communication.
- The second function of the data link layer is media access control, or how to share the link.
- **Data link control** functions include framing, flow and error control, and software implemented protocols that provide smooth and reliable transmission of frames between nodes.

8.1 DLC Services

8.1.1 Framing

- Data transmission in the physical layer means moving bits in the form of a signal from the source to the destination.
- The physical layer provides bit synchronization to ensure that the sender and receiver use the same bit durations and timing.
- The data link layer, on the other hand, needs to pack bits into frames, so that each frame is distinguishable from another.
- Framing in the data link layer separates a message from one source to a destination, or from other messages to other destinations, by adding a sender address and a destination address.
- The destination address defines where the packet is to go; the sender address helps the recipient acknowledge the receipt.
- Frames can be of fixed or variable size.

1) Fixed-Size Framing

In fixed-size framing, there is no need for defining the boundaries of the frames; the size itself can be used as a delimiter.

2) Variable-Size Framing

- In variable-size framing, we need a way to define the end of the frame and the beginning of the next.
- Historically, two approaches were used for this purpose: a character-oriented approach and a bit-oriented approach.

Character-Oriented Protocols

- In a character-oriented protocol, data to be carried are 8-bit characters from a coding system such as ASCII.
- The header, which normally carries the source and destination addresses and other control information, and the trailer, which carries error detection or error correction redundant bits, are also multiples of 8 bits.
- To separate one frame from the next, an 8-bit (1-byte) flag is added at the beginning and the end of a frame. The flag, composed of protocol-dependent special characters, signals the start or end of a frame.



Figure 8.1: Frame in a character oriented protocol

- Flag should be different from the text pattern used in the data section. Otherwise, when the receiver encounters this pattern in the middle of the data, it thinks that it has reached the end of the frame. To fix this problem, a **byte-stuffing** strategy was added to character-oriented framing.
- **Byte stuffing** is the process of adding 1 extra byte whenever there is a flag or escape character in the text.
- The data section is stuffed with an extra byte. This byte is usually called the escape character (ESC), which has a predefined bit pattern. Whenever the receiver encounters the ESC character, it removes it from the data section and treats the next character as data, not a delimiting flag.
- If even the escape character is also part of the text, an extra escape character is added to show that the second one is part of the text.
- Character-oriented protocols present another problem in data communications. The universal coding systems in use today, such as Unicode, have 16-bit and 32-bit characters that conflict with 8-bit characters.

Bit-Oriented Protocols:

- In a bit-oriented protocol, the data section of a frame is a sequence of bits to be interpreted by the upper layer as text, graphic, audio, video, and so on.
- In addition to headers a special 8-bit pattern flag 01111110 is used as the delimiter to define the beginning and the end of the frame.

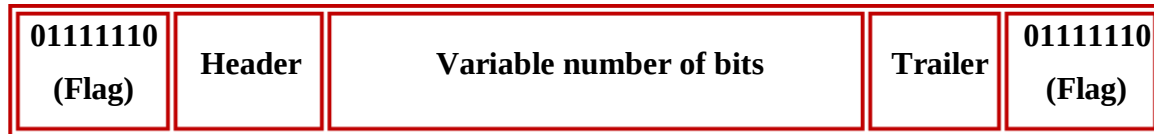


Figure 8.2: Frame in a Bit oriented protocol

- If the flag pattern appears in the data, **bit stuffing** is used to differentiate the flag from information.
- Bit stuffing is the process of adding one extra 0 whenever five consecutive 1s follow a 0 in the data, so that the receiver does not mistake the pattern 0111110 for a flag.
- This extra stuffed bit is eventually removed from the data by the receiver.

8.1.2 Flow and Error Control

Flow Control

- Flow control coordinates the amount of data that can be sent before receiving an acknowledgment.
- Flow control refers to a set of procedures used to restrict the amount of data that the sender can send before waiting for acknowledgment.
- The flow of data must not be allowed to overwhelm the receiver.
- Any receiving device has a limited speed at which it can process incoming data and a limited amount of memory in which to store incoming data.
- The receiving device must be able to inform the sending device before those limits are reached and to request that the transmitting device send fewer frames or stop temporarily.
- Incoming data must be checked and processed before they can be used. The rate of such processing is often slower than the rate of transmission. For this reason, each receiving device has a block of memory, called a *buffer*, reserved for storing incoming data until they are processed. If the buffer begins to fill up, the receiver must be able to tell the sender to halt transmission until it is once again able to receive.

Error Control

- Error control is both error detection and error correction.
- It allows the receiver to inform the sender of any frames lost or damaged in transmission and coordinates the retransmission of those frames by the sender.

-
- In the data link layer, the term *error control* refers primarily to methods of error detection and retransmission.
 - Error control in the data link layer is based on automatic repeat request, which is the retransmission of data.

8.1.3 Connectionless and Connection-Oriented

A DLC protocol can be either connectionless or connection-oriented.

Connectionless Protocol

In a connectionless protocol, frames are sent from one node to the next without any relationship between the frames; each frame is independent. The frames are not numbered and there is no sense of ordering.

Connection-Oriented Protocol

In a connection-oriented protocol, a logical connection should first be established between the two nodes (setup phase). After all frames that are somehow related to each other are transmitted (transfer phase), the logical connection is terminated (teardown phase). In this type of communication, the frames are numbered and sent in order. If they are not received in order, the receiver needs to wait until all frames belonging to the same set are received and then deliver them in order to the network layer.

8.2 Data Link Layer Protocols

The protocols are normally implemented in software by using one of the common programming languages.

In a real-life network, the data link protocols are implemented as bidirectional; data flow in both directions.

In these protocols the flow and error control information such as ACKs and NAKs is included in the data frames in a technique called **piggybacking**.

8.2.1 Simple Protocol

Simplest protocol is one that has no flow or error control.

It is a unidirectional protocol in which data frames are traveling in only one direction—from the sender to receiver.

Working

- The data link layer at the sender site gets data from its network layer, makes a frame out of the

data, and sends it.

- The data link layer at the receiver site receives a frame from its physical layer, extracts data from the frame, and delivers the data to its network layer.
- The data link layers of the sender and receiver provide transmission services for their network layers.
- The data link layers use the services provided by their physical layers (such as signaling, multiplexing, and so on) for the physical transmission of bits.
- The sender site cannot send a frame until its network layer has a data packet to send. The receiver site cannot deliver a data packet to its network layer until a frame arrives.



Figure 8.3: Simple protocol

Example:

Figure shows an example of communication using this protocol. It is very simple. The sender sends a sequence of frames without even thinking about the receiver. To send three frames, three events occur at the sender site and three events at the receiver site.

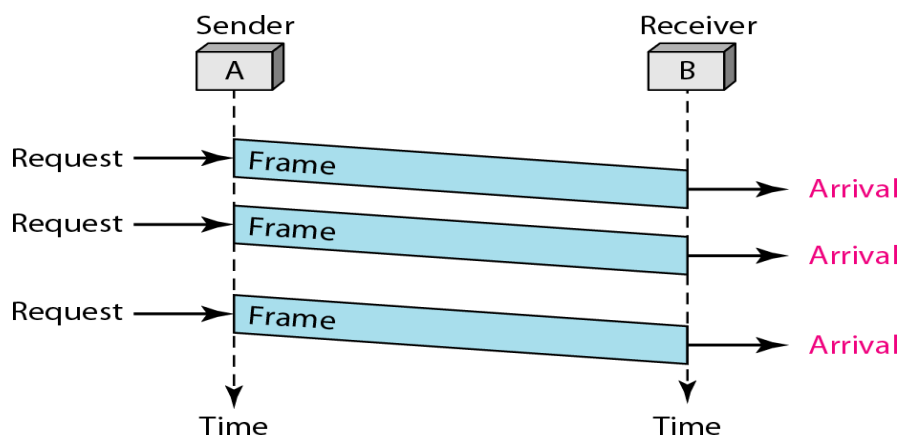


Figure 8.4: Simple protocol Example

8.2.2 Stop-and-Wait Protocol

- If data frames arrive at the receiver site faster than they can be processed, the frames must be stored until their use.
- Normally, the receiver does not have enough storage space, especially if it is receiving data from many sources. This may result in either the discarding of frames or denial of service.
- To prevent the receiver from becoming overwhelmed with frames, we somehow need to tell the sender to slow down. There must be feedback from the receiver to the sender.
- The protocol is called the Stop-and-Wait Protocol because the sender sends one frame, stops until it receives confirmation from the receiver, and then sends the next frame.
- At any time, there is either one data frame on the forward channel or one ACK frame on the reverse channel. We therefore need a half-duplex link.

Working:

- The data link layer of sending device waits for a packet from the network layer. If a packet comes from the network layer, the sender creates a frame, saves a copy of the frame, starts the timer and sends the frame.
- If a time-out occurs, the sender resends the saved copy of the frame and restarts the timer.
- If a corrupted ACK arrives, it is discarded.
- If an error-free ACK arrives, the sender stops the timer and discards the saved copy of the frame. Then it sends next frame
- The data link layer at the receiver site receives a frame from its physical layer, extracts data from the frame, and delivers the data to its network layer. Then it sends ACK frame to sender site. If a corrupted frame arrives, the frame is discarded.

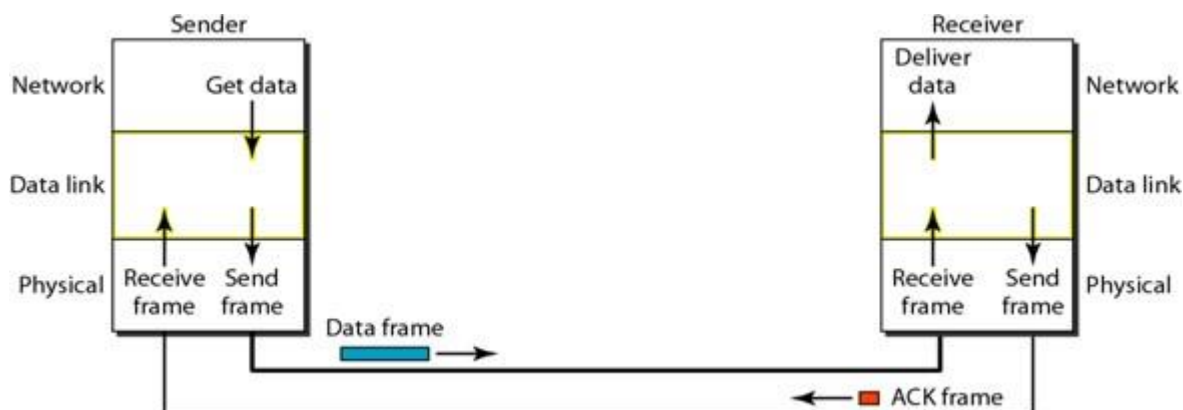


Figure 8.5: Stop and Wait Protocol

Sequence Numbers

- The protocol specifies that frames need to be numbered. This is done by using sequence numbers. A field is added to the data frame to hold the sequence number of that frame.

Acknowledgment Numbers

- Since the sequence numbers must be suitable for both data frames and ACK frames the acknowledgment numbers always announce the sequence number of the next frame expected by the receiver.

Example:

Frame 0 is sent and acknowledged. Frame 1 is lost and resent after the time-out. The resent frame 1 is acknowledged and the timer stops. Frame 0 is sent and acknowledged, but the acknowledgment is lost. The sender has no idea if the frame or the acknowledgment is lost, so after the time-out, it resends frame 0, which is acknowledged.

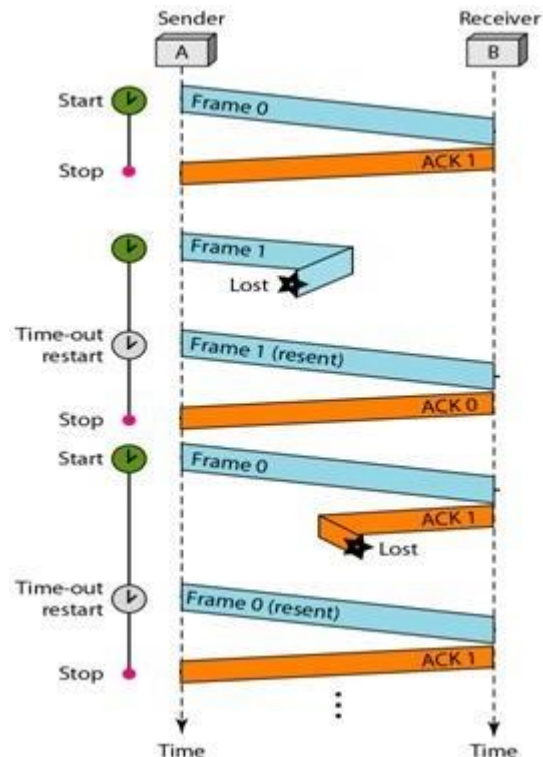


Figure 8.6: Stop and Wait Protocol Example

8.2.3 Piggybacking

A technique called piggybacking is used to improve the efficiency of the bidirectional protocols. When a frame is carrying data from A to B, it can also carry control information about arrived (or lost) frames from B; when a frame is carrying data from B to A, it can also carry control information about the arrived (or lost) frames from A.

8.3 HDLC

High-level Data Link Control (HDLC) is a bit-oriented protocol for communication over point-to-point and multipoint links. It implements the ARQ mechanisms.

8.3.1 Configurations and Transfer Modes

HDLC provides two common transfer modes that can be used in different configurations:

1. Normal response mode (NRM)
2. Asynchronous balanced mode (ABM)

Normal Response Mode:

- In normal response mode (NRM), the station configuration is unbalanced. We have one primary station and multiple secondary stations.
- A primary station can send commands; a secondary station can only respond.
- The NRM is used for both point-to-point and multiple-point links

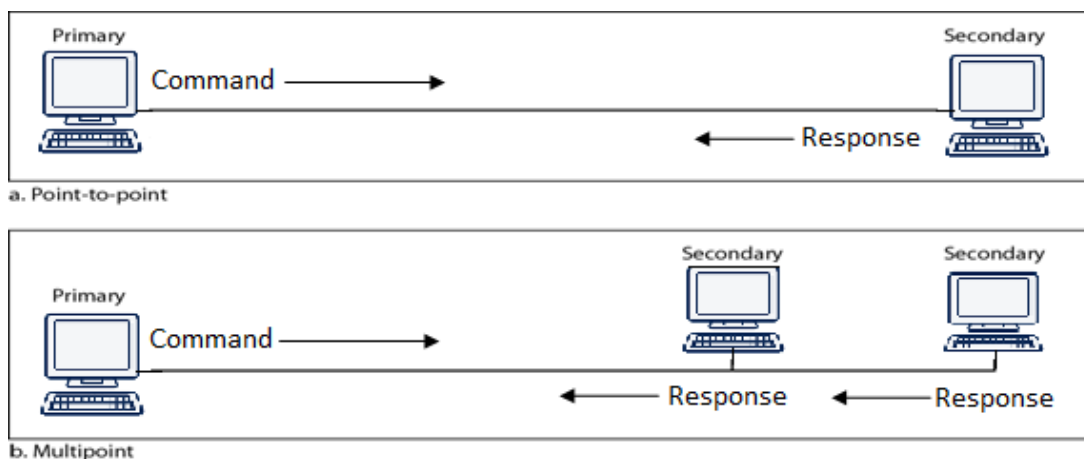


Figure 8.7: Normal Response Mode

Asynchronous Balanced Mode

In asynchronous balanced mode (ABM), the configuration is balanced. The link is point-to-point, and each

station can function as a primary and a secondary.

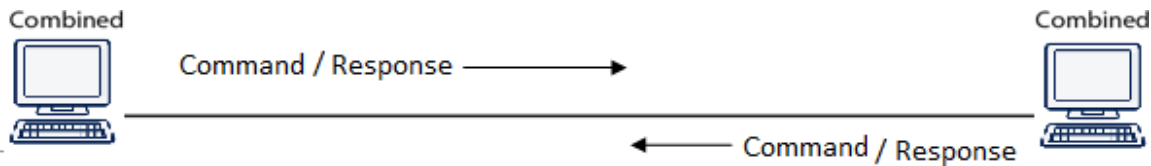


Figure 8.8: Asynchronous Balanced Mode

8.3.2 Frames

- HDLC defines three types of frames:
 1. Information frames (I-frames)
 2. Supervisory frames (S-frames)
 3. Unnumbered frames (U-frames)
- Each type of frame serves as an envelope for the transmission of a different type of message.
- I-frames are used to transport user data and control information relating to user data (piggybacking).
- S-frames are used only to transport control information.
- U-frames are reserved for system management. Information carried by U-frames is intended for managing the link itself.

Frame Format

Flag	Address	Control	User information		FCS	Flag	I-Frame
Flag	Address	Control	FCS	Flag	S-Frame		
Flag	Address	Control	Management information		FCS	Flag	U-Frame

Figure 8.9: HDLC Frames

Fields

- **Flag field:** The flag field of an HDLC frame is an 8-bit sequence with the bit pattern 01111110 that identifies both the beginning and the end of a frame and serves as a synchronization pattern

for the receiver.

- **Address field:** The second field of an HDLC frame contains the address of the secondary station. If a primary station created the frame, it contains to address. If a secondary creates the frame, it contains from address.
- **Control field:** The control field is a 1- or 2-byte segment of the frame used for flow and error control. The interpretation of bits in this field depends on the frame type.
- **Information field:** The information field contains the user's data from the network layer or management information. Its length can vary from one network to another.
- **FCS field:** The frame check sequence (FCS) is the HDLC error detection field. It can contain either a 2- or 4-byte ITU-T CRC.

Control Field

The control field determines the type of frame and defines its functionality.

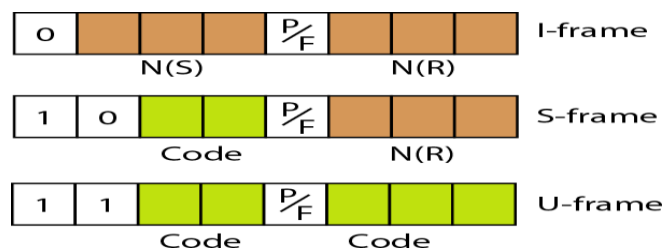


Figure 8.10: Control Field of HDLC frames

Control Field for I-Frames

- I-frames are designed to carry user data from the network layer. In addition, they can include flow and error control information (piggybacking).
- If the first bit of the control field is 0, this means the frame is an I-frame.
- The next 3 bits, called N(S), define the sequence number of the frame. Note that with 3 bits, we can define a sequence number between 0 and 7; but in the extension format, in which the control field is 2 bytes, this field is larger.
- The last 3 bits, called N(R), correspond to the acknowledgment number when piggybacking is used.
- The single bit between N(S) and N(R) is called the P/F bit. The P/F field is a single bit with a dual purpose. It has meaning only when it is set (bit = 1) and can mean poll or final. It means poll when the frame is sent by a primary station to a secondary. It means final when the frame is sent by a

secondary to a primary.

Control Field for S-Frames

- Supervisory frames are used for flow and error control whenever piggybacking is either impossible or inappropriate.
- S-frames do not have information fields.
- If the first 2 bits of the control field is 10, this means the frame is an S-frame.
- The last 3 bits, called N(R), corresponds to the acknowledgment number (ACK) or negative acknowledgment number (NAK) depending on the type of S-frame.
- The 2 bits called code is used to define the type of S-frame itself. With 2 bits, we can have four types of S-frames, as described below:
 - **Receive ready (RR):** If the value of the code subfield is 00, it is an RR S-frame. This kind of frame acknowledges the receipt of a safe and sound frame or group of frames. In this case, the value N(R) field defines the acknowledgment number.
 - **Receive not ready (RNR):** If the value of the code subfield is 10, it is an RNR S-frame. This kind of frame is an RR frame with additional functions. It acknowledges the receipt of a frame or group of frames, and it announces that the receiver is busy and cannot receive more frames. It acts as a kind of congestion control mechanism by asking the sender to slow down. The value of NCR) is the acknowledgment number.
 - **Reject (REJ):** If the value of the code subfield is 01, it is a REJ S-frame. This is a NAK frame. The value of N(R) is the negative acknowledgment number.
 - **Selective reject (SREJ):** If the value of the code subfield is 11, it is an SREJ S-frame. The value of N(R) is the negative acknowledgment number.

Control Field for U-Frames

- Unnumbered frames are used to exchange session management and control information between connected devices. Unlike S-frames, U-frames contain an information field, but one used for system management information, not user data.
- As with S-frames, however, much of the information carried by U-frames is contained in codes included in the control field.
- U-frame codes are divided into two sections: a 2-bit prefix before the P/F bit and a 3-bit suffix after the P/F bit. Together, these two segments (5 bits) can be used to create up to 32 different types of U-frames. Some of the more common types are shown

Code	Command	Response	Meaning
001	RM		normal response mode
011	RME		normal response mode, extended
100	BM	M	asynchronous balanced mode
110	BME		asynchronous balanced mode, extended
000			numbered information
110		A	numbered acknowledgement
010	SC		disconnect or request disconnect
000	M	M	initialization mode or request information mode
100			numbered poll
001	ET		set
101	D	D	change ID
001	MR	MR	name reject

Figure 8.11: U-Frame code

8.4 Point-To-Point Protocol

Today, millions of Internet users who need to connect their home computers to the server of an Internet service provider use PPP. The majority of these users have a traditional modem; they are connected to the Internet through a telephone line, which provides the services of the physical layer. But to control and manage the transfer of data, there is a need for a point-to-point protocol at the data link layer.

PPP provides several services:

1. PPP defines the format of the frame to be exchanged between devices.
2. PPP defines how two devices can negotiate the establishment of link and the exchange of data.
3. PPP defines how network layer data are encapsulated in the data link frame.
4. PPP defines how two devices can authenticate each other.
5. PPP provides multiple network layer services supporting a variety of network layer protocols.
6. PPP provides connections over multiple links.
7. PPP provides network address configuration. This is particularly useful when a home user needs a temporary network address to connect to the Internet.

Limitations of PPP:

1. PPP does not provide flow control.
2. PPP has a very simple mechanism for error control. A CRC field is used to detect errors. If the frame is corrupted, it is silently discarded; the upper-layer protocol needs to take care of the problem. Lack of error control and sequence numbering may cause a packet to be received out of order.
3. PPP does not provide a sophisticated addressing mechanism to handle frames in a multipoint configuration.

8.4.1 Framing

Frame Format

Flag 11111111	Address	Control 11000000	Protocol	Payload	FCS	Flag
1 byte	1 byte	1 byte	1 or 2 bytes	Variable	2 or 4 bytes	1 byte

Figure 8.12: Point – point protocol Frame format

- **Flag:** A PPP frame starts and ends with 1-byte flag with the bit pattern 01111110.
- **Address:** The address field in this protocol is a constant value and set to 11111111 (broadcast address).
- **Control:** This field is set to the constant value 11000000
- **Protocol:** The protocol field defines what is being carried in the data field: either user data or other information.
- **Payload field:** This field carries either the user data or other information. The data field is a sequence of bytes with the default of a maximum of 1500 bytes; but this can be changed during negotiation. Because there is no field defining the size of the data field, padding is needed if the size is less than the maximum default value or the maximum negotiated value.
- **FCS:** The frame check sequence (FCS) is simply a 2-byte or 4-byte standard CRC. **PPP** is a byte-oriented protocol using byte stuffing with the escape byte 01111101.

8.4.2 Transition Phases

- **Dead:** In the dead phase the link is not being used. There is no active carrier (at the physical layer) and the line is quiet.
- **Establish:** When one of the nodes starts the communication, the connection goes into this phase.

In this phase, options are negotiated between the two parties. If the negotiation is successful, the system goes to the authentication phase (if authentication is required) or directly to the networking phase.

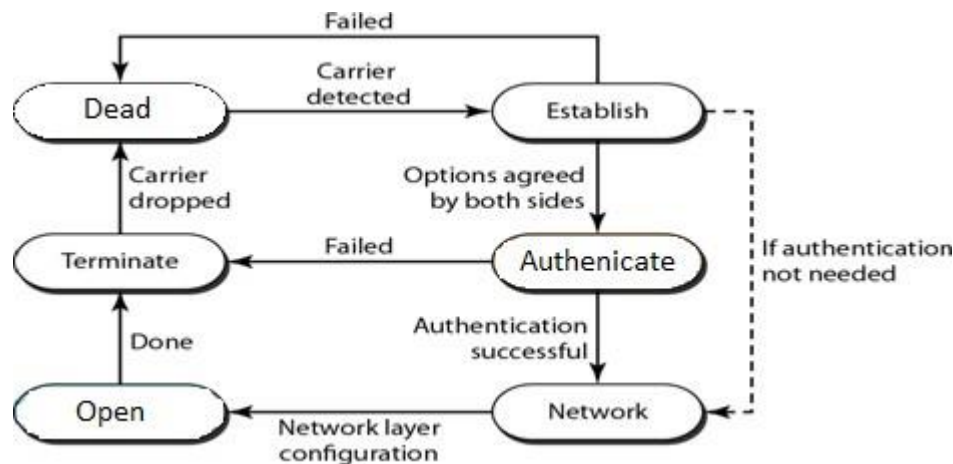


Figure 8.13: PPP Transition Phases

- **Authenticate:** The authentication phase is optional; the two nodes may decide, during the establishment phase, not to skip this phase. However, if they decide to proceed with authentication, they send several authentication packets. If the result is successful, the connection goes to the networking phase; otherwise, it goes to the termination phase.
- **Network:** In the network phase, negotiation for the network layer protocols takes place. PPP specifies that two nodes establish a network layer agreement before data at the network layer can be exchanged. The reason is that PPP supports multiple protocols at the network layer. If a node is running multiple protocols simultaneously at the network layer, the receiving node needs to know which protocol will receive the data.
- **Open:** In the open phase, data transfer takes place. When a connection reaches this phase, the exchange of data packets can be started. The connection remains in this phase until one of the endpoints wants to terminate the connection.
- **Terminate:** In the termination phase the connection is terminated. Several packets are exchanged between the two ends for house cleaning and closing the link.

Media Access Control

9.1 Random Access

- In random access or contention methods, no station is superior to another station and none is assigned the control over another. No station permits, or does not permit, another station to send.
- At each instance, a station that has data to send uses a procedure defined by the protocol to make a decision on whether or not to send. This decision depends on the state of the medium (idle or busy).
- There is no scheduled time for a station to transmit. Transmission is random among the stations. That is why these methods are called random access.
- No rules specify which station should send next. Stations compete with one another to access the medium. That is why these methods are also called *contention* methods.

9.1.1 ALOHA

ALOHA, the earliest random access method, was developed at the University of Hawaii in early 1970. It was designed for a radio (wireless) LAN, but it can be used on any shared medium.

Pure ALOHA

- The original ALOHA protocol is called pure ALOHA. The idea is that each station sends a frame whenever it has a frame to send. However, since there is only one channel to share, there is the possibility of collision between frames from different stations.

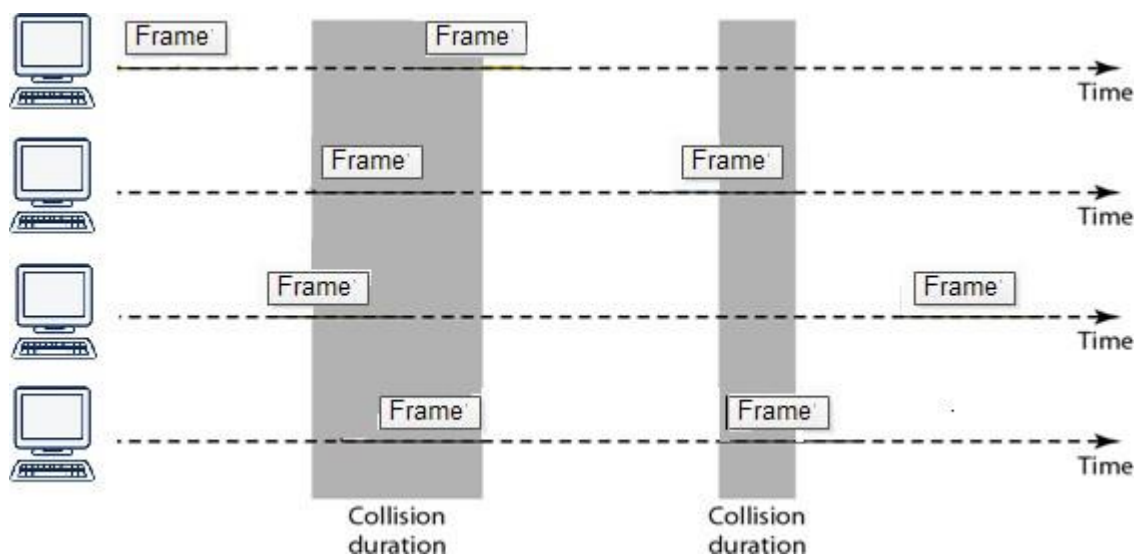
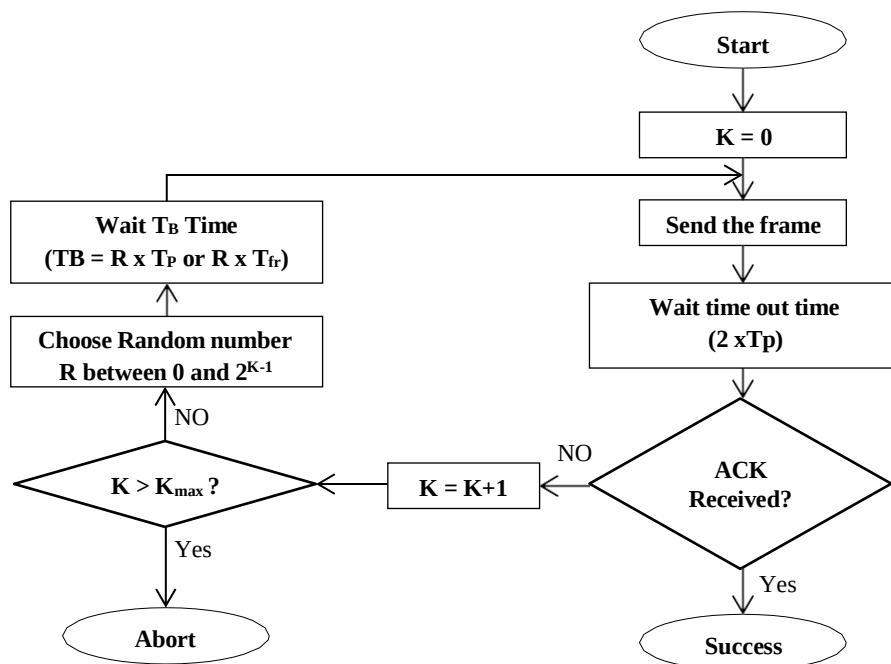


Figure 9.1: Pure ALOHA

Here there are four stations that contend with one another for access to the shared channel. The figure shows that each station sends two frames. Some of these frames collide because multiple frames are in contention for the shared channel. Figure shows that only two frames survive: frame from station 1

and frame from station 3.

- The pure ALOHA protocol relies on acknowledgments from the receiver. When a station sends a frame, it expects the receiver to send an acknowledgment. If the acknowledgment does not arrive after a time-out period, the station assumes that the frame (or the acknowledgment) has been destroyed and resends the frame.
- If two or more stations try to resend their frames after the time-out, the frames will collide again. Pure ALOHA dictates that when the time-out period passes, each station waits a random amount of time before resending its frame. The randomness will help avoid more collisions.
- After a maximum number of retransmission attempts $K_{\max}$ a station must give up and try later.



Where K is number of attempts, T_p is maximum propagation time T_{fr} is average transmission time for a frame and T_B is back off time

Vulnerable time: is the length of time in which there is a possibility of collision.

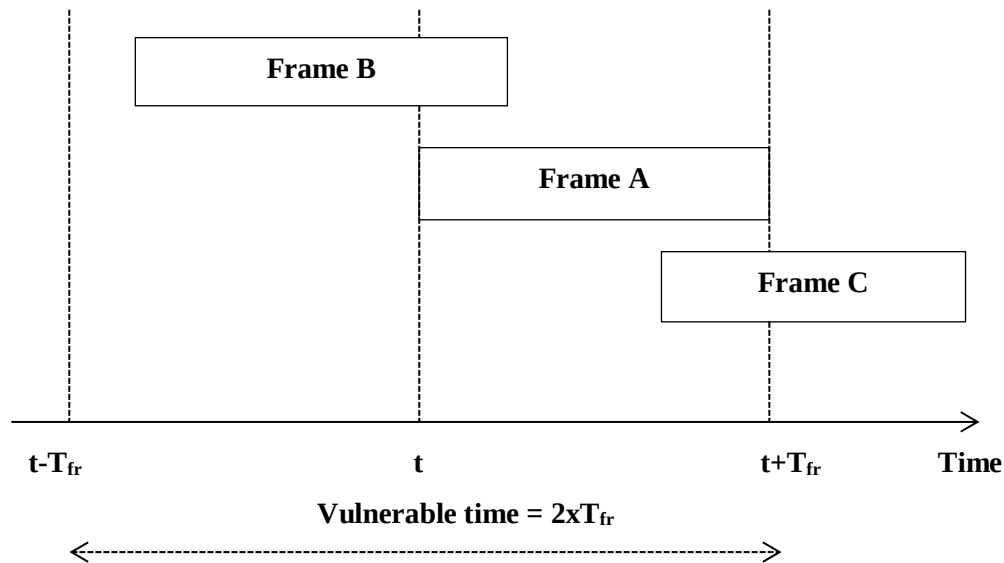


Figure 9.3: Vulnerable time in Pure ALOHA

The vulnerable time, during which a collision may occur in pure ALOHA, is 2 times the frame transmission time.

$$\text{Pure ALOHA vulnerable time} = 2 \times T_{fr}$$

Throughput: The throughput for pure ALOHA is $S = Ge^{-2G}$

Where G is the average number of frames generated by the system during one frame transmission time. The maximum throughput $S_{max} = 0.184$ when $G = (1/2)$.

Slotted ALOHA

- Pure ALOHA has a vulnerable time of $2 \times T_{fr}$. This is so because there is no rule that defines when the station can send. A station may send soon after another station has started or soon before another station has finished.
- Slotted ALOHA was invented to improve the efficiency of pure ALOHA.
- In slotted ALOHA the time is divided into slots of T_{fr} s and force the station to send only at the beginning of the time slot.

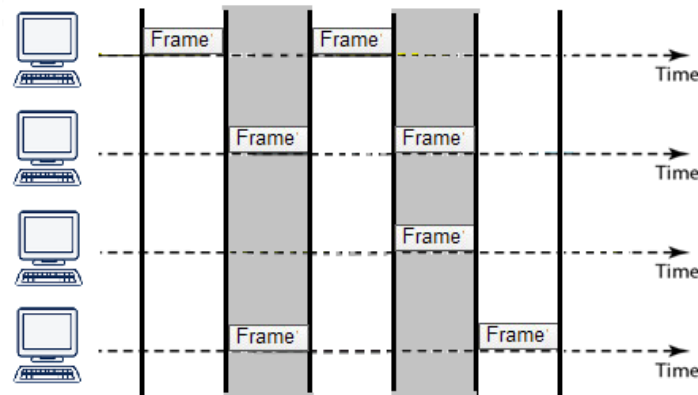


Figure 9.3: Slotted ALOHA

Because a station is allowed to send only at the beginning of the synchronized time slot, if a station misses this moment, it must wait until the beginning of the next time slot. This means that the station which started at the beginning of this slot has already finished sending its frame. There is still the possibility of collision if two stations try to send at the beginning of the same time slot.

The vulnerable time for slotted ALOHA is one-half that of pure ALOHA.

Slotted ALOHA vulnerable time = T_{fr}

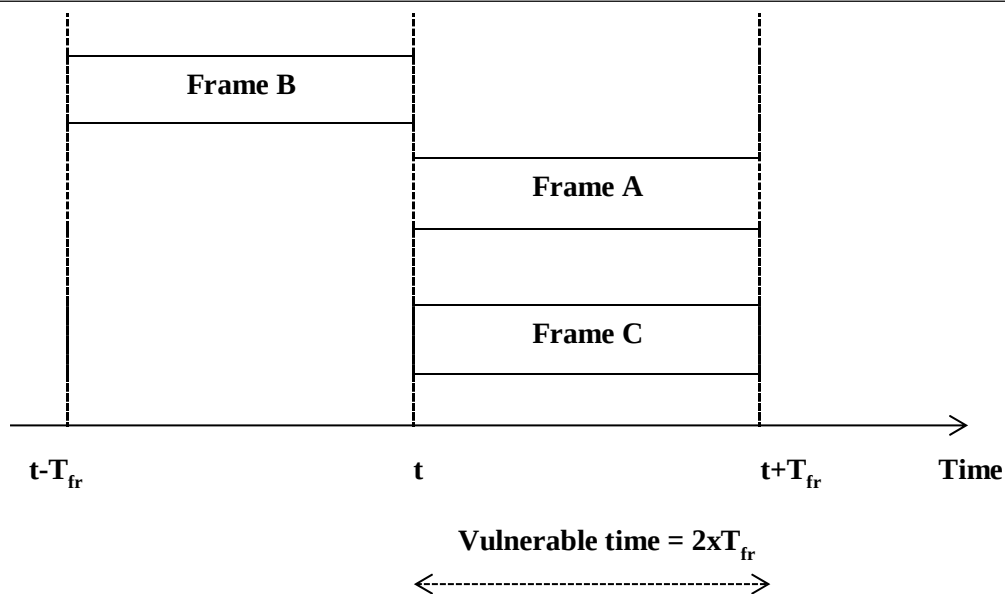


Figure 9.4: Vulnerable time in Slotted ALOHA

The throughput for slotted ALOHA is $S = G e^{-G}$

The maximum throughput $S_{max} = 0.368$ when $G=1$.

9.1.2 Carrier Sense Multiple Access (CSMA)

Carrier sense multiple access (CSMA) requires that each station first listen to the medium (or check the state of the medium) before sending. In other words, CSMA is based on the principle "**sense before**

transmit" or "listen before talk."

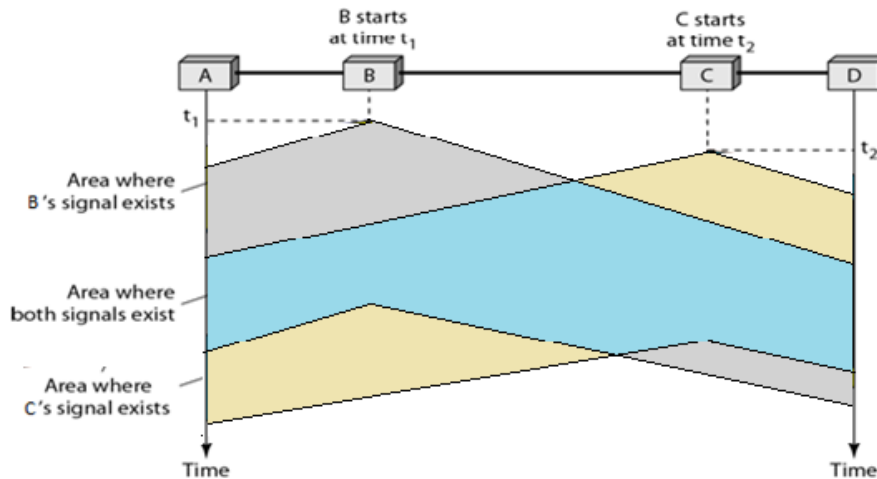


Figure 9.5: Timing diagram of collision in CSMA

The possibility of collision still exists because of propagation delay; when a station sends a frame, it still takes time (although very short) for the first bit to reach every station and for every station to sense it. In other words, a station may sense the medium and find it idle, only because the first bit sent by another station has not yet been received.

At time t_1 station B senses the medium and finds it idle, so it sends a frame. At time t_2 ($t_2 > t_1$) station C senses the medium and finds it idle because, at this time, the first bits from station B have not reached station C. Station C also sends a frame. The two signals collide and both frames are destroyed.

Vulnerable Time

The vulnerable time for CSMA is the propagation time T_p . This is the time needed for a signal to propagate from one end of the medium to the other. When a station sends a frame, and any other station tries to send a frame during this time, a collision will result.

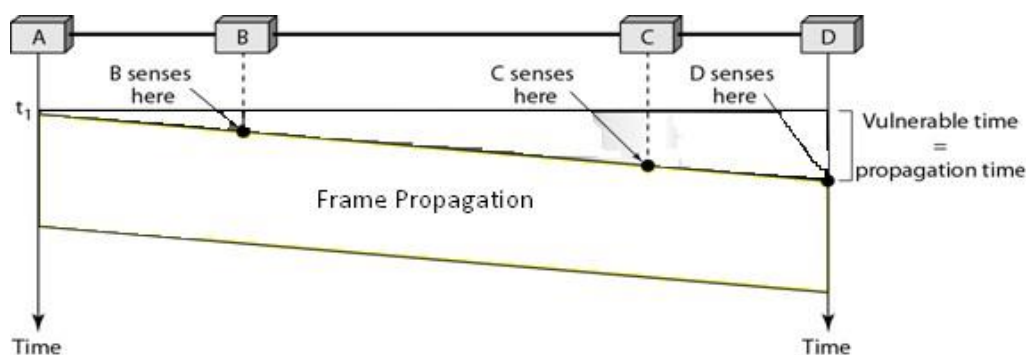


Figure 9.6: Vulnerable time in CSMA

Persistence Methods

There are three persistence methods which decide the action that has to be performed by station when the channel is busy or idle.

1) 1-Persistent:

The 1-persistent method is simple and straightforward. In this method, after the station finds the line idle, it sends its frame immediately (with probability 1).

This method has the highest chance of collision because two or more stations may find the line idle and send their frames immediately.

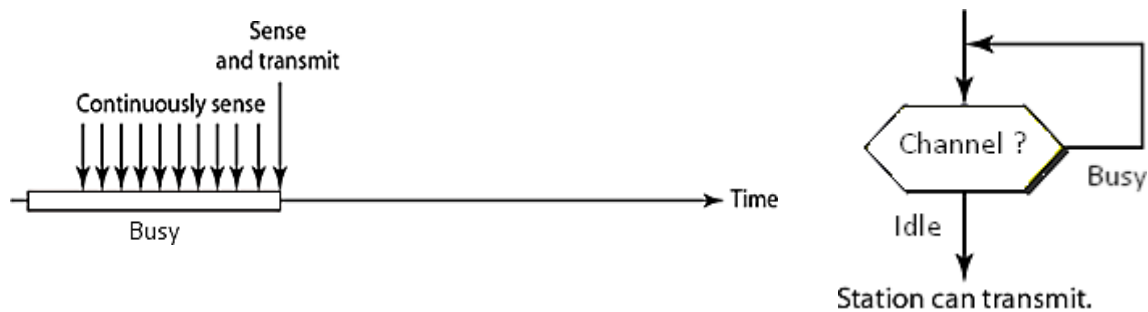


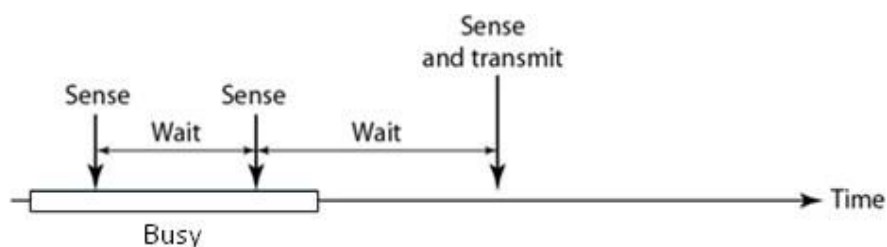
Figure 9.7: 1-Persistent Method

2) Nonpersistent:

In the nonpersistent method, a station that has a frame to send senses the line. If the line is idle, it sends immediately. If the line is not idle, it waits a random amount of time and then senses the line again.

The nonpersistent approach reduces the chance of collision because it is unlikely that two or more stations will wait the same amount of time and retry to send simultaneously.

However, this method reduces the efficiency of the network because the medium remains idle when there may be stations with frames to send.



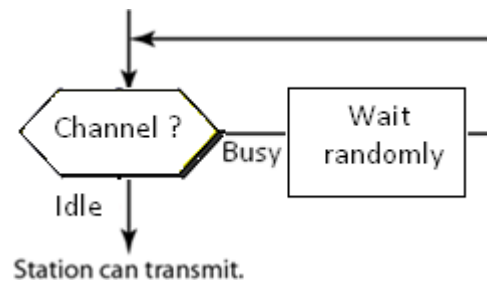


Figure 9.8: Non Persistent Method

3) p-Persistent:

The p-persistent method is used if the channel has time slots with a slot duration equal to or greater than the maximum propagation time.

It reduces the chance of collision and improves efficiency. In this method, after the station finds the line idle it follows these steps:

1. With probability p , the station sends its frame.
2. With probability $q = 1-p$, the station waits for the beginning of the next time slot and checks the line again.
 - a. If the line is idle, it goes to step 1.
 - b. If the line is busy, it acts as though a collision has occurred and uses the back off procedure.

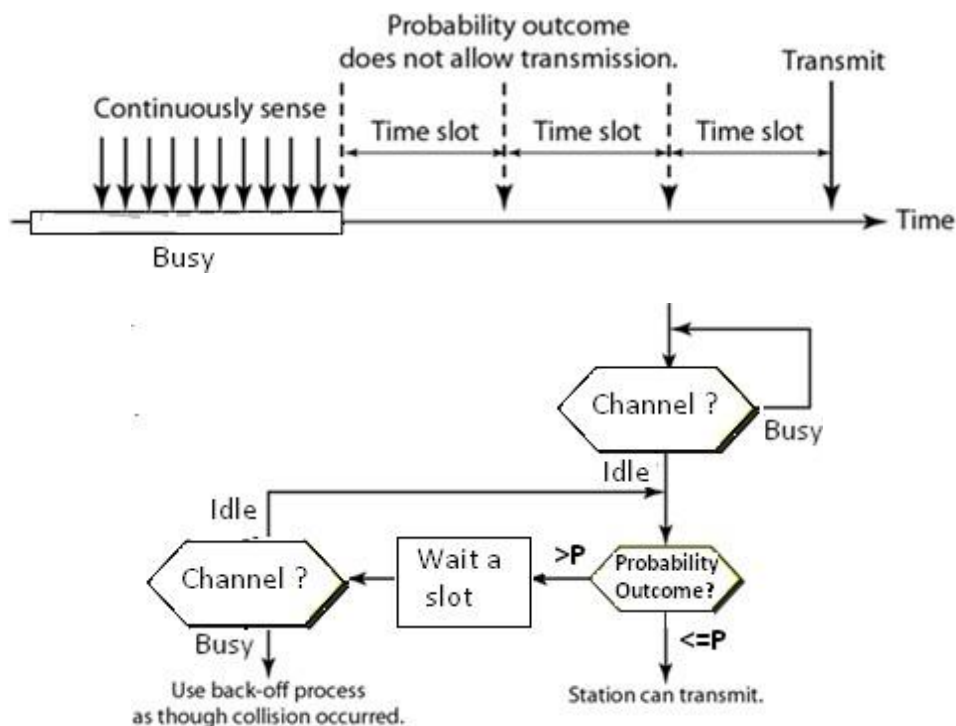


Figure 9.9: P-Persistent Method

9.1.3 Carrier Sense Multiple Access with Collision Detection (CSMA/CD)

In Carrier sense multiple access with collision detection (CSMA/CD) a station monitors the medium after it sends a frame to see if the transmission was successful. If so, the station is finished. If there is a collision, the frame is sent again.

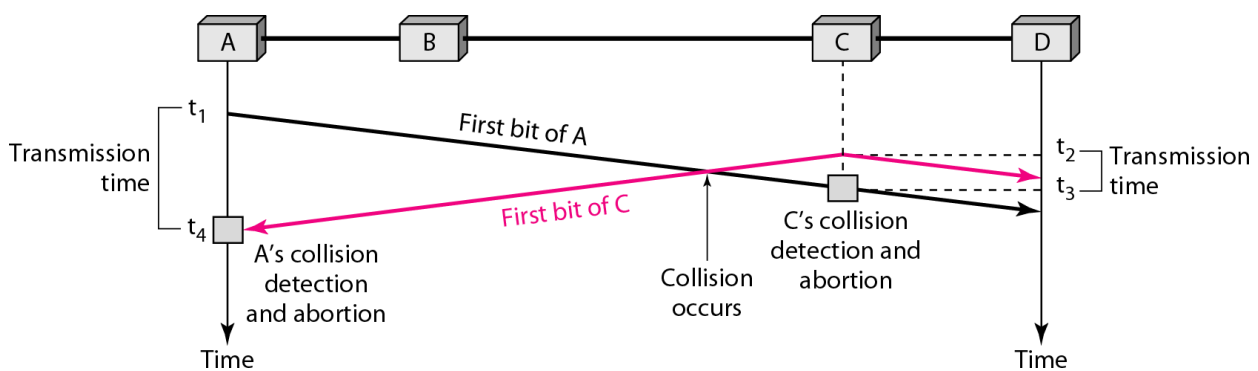


Figure 9.10: CSMA/CD

At time t_1 , station A has executed its persistence procedure and starts sending the bits of its frame. At time t_2 , station C has not yet sensed the first bit sent by A. Station C executes its persistence procedure and starts sending the bits in its frame, which propagate both to the left and to the right. The collision occurs sometime after time t_2 . Station C detects a collision at time t_3 when it receives the first bit of A's frame. Station C immediately aborts transmission. Station A detects collision at time t_4 when it receives the first bit of C's frame; it also immediately aborts transmission.

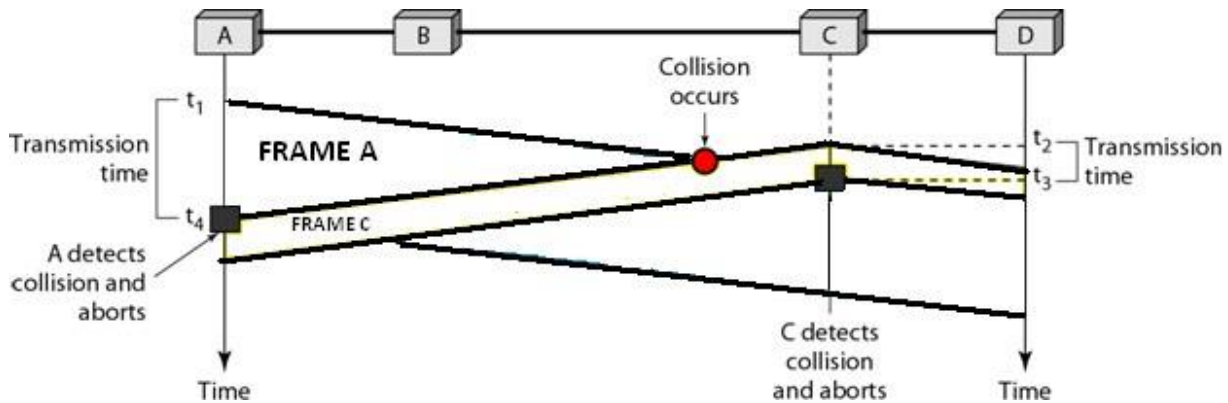


Figure 9.11: Collision and abortion in CSMA/CD

Minimum Frame Size

Before sending the last bit of the frame, the sending station must detect a collision, if any, abort the transmission. This is so because the station, once the entire frame is sent, does not keep a copy of the frame and does not monitor the line for collision detection. Therefore, the frame transmission time T_{fr} must be at least two times the maximum propagation time T_p .

If the two stations involved in a collision are the maximum distance apart, the signal from the first takes time T_p to reach the second, and the effect of the collision takes another time T_p to reach the first. So the requirement is that the first station must still be transmitting after $2T_p$.

Difference between ALOHA and CSMA/CD

- The first difference is the addition of the persistence process.
- The second difference is the frame transmission. In ALOHA, we first transmit the entire frame and then wait for an acknowledgment. In *CSMA/CD*, transmission and collision detection is a continuous process.
- The third difference is the sending of a short jamming signal that enforces the collision in case other stations have not yet sensed the collision.

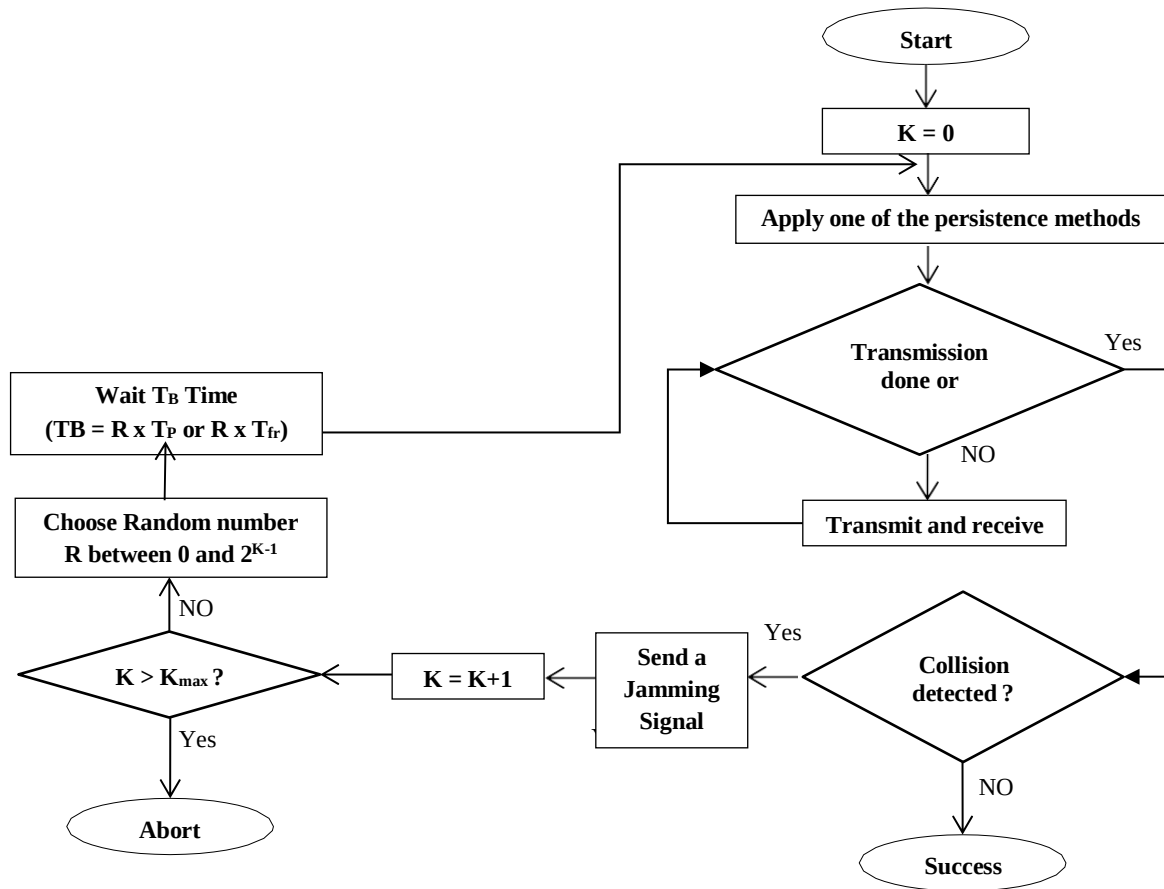


Figure 9.12: Flow Diagram for CSMA/CD

Energy Level

The level of energy in a channel can have three values: zero, normal, and abnormal. At the zero level, the channel is idle. At the normal level, a station has successfully captured the channel and is sending its frame. At the abnormal level, there is a collision and the level of the energy is twice the normal level. A station that has a frame to send or is sending a frame needs to monitor the energy level to determine if the channel is idle, busy, or in collision mode.

Throughput

The maximum throughput occurs at a different value of G and is based on the persistence method and the value of p in the p -persistent approach. For 1-persistent method the maximum throughput is around 50 percent when $G = 1$. For nonpersistent method, the maximum throughput can go up to 90 percent when G is between 3 and 8.

9.1.4 Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA)

Collisions are avoided through the use of CSMA/CA's three strategies: the interframe space, the contention window, and acknowledgments.

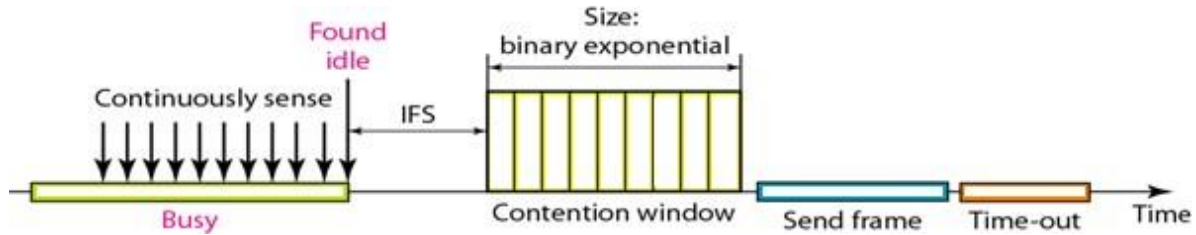


Figure 9.13: CSMA/CA

Interframe Space (IFS)

Collisions are avoided by deferring transmission even if the channel is found idle. When an idle channel is found, the station does not send immediately. It waits for a period of time called the interframe space or IFS. Even though the channel may appear idle when it is sensed, a distant station may have already started transmitting. The distant station's signal has not yet reached this station. The IFS time allows the front of the transmitted signal by the distant station to reach this station. If after the IFS time the channel is still idle, the station can send, but it still needs to wait a time equal to the contention time. The IFS variable can also be used to prioritize stations or frame types. For example, a station that is assigned a shorter IFS has a higher priority.

Contention Window

The contention window is an amount of time divided into slots. A station that is ready to send chooses a random number of slots as its wait time. The number of slots in the window changes according to the binary exponential back-off strategy. This means that it is set to one slot the first time and then doubles each time the station cannot detect an idle channel after the IFS time. The station needs to sense the channel after each time slot. However, if the station finds the channel busy, it does not restart the process; it just stops the timer and restarts it when the channel is sensed as idle. This gives priority to the station with the longest waiting time.

Acknowledgment

With all these precautions, there still may be a collision resulting in destroyed data. In addition, the data may be corrupted during the transmission. The positive acknowledgment and the time-out timer can help guarantee that the receiver has received the frame.

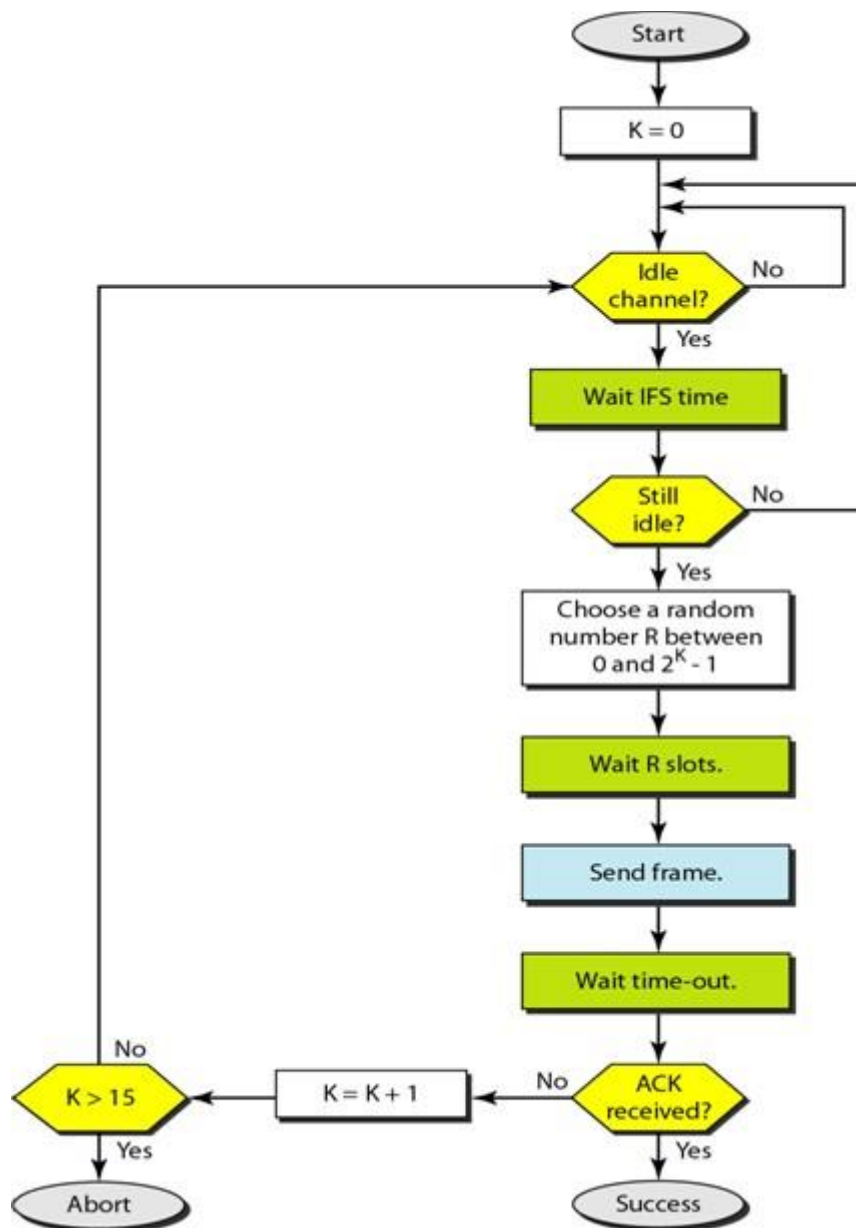


Figure 9.14: Collision and abortion in CSMA/CD

9.2 Controlled Access

In controlled access, the stations consult one another to find which station has the right to send. A station cannot send unless it has been authorized by other stations.

9.2.1 Reservation

- In the reservation method, a station needs to make a reservation before sending data.
- Time is divided into intervals. In each interval, a reservation frame precedes the data frames sent in that

interval.

- If there are N stations in the system, there are exactly N reservation minislots in the reservation frame. Each minislot belongs to a station. When a station needs to send a data frame, it makes a reservation in its own minislot. The stations that have made reservations can send their data frames after the reservation frame.

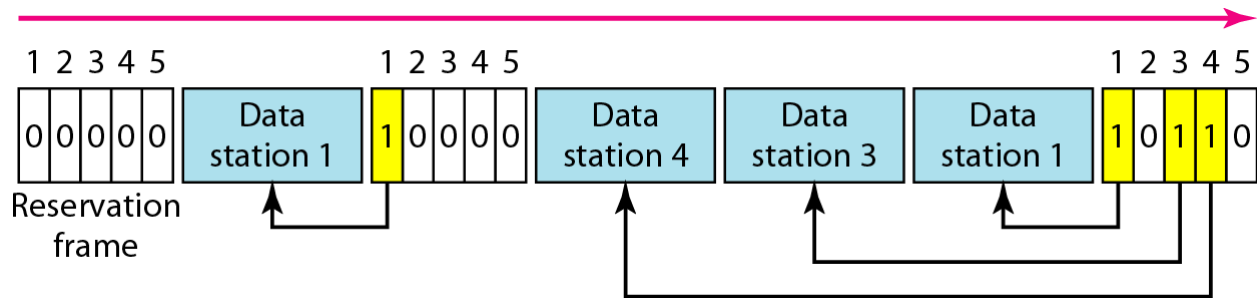


Figure 9.15: Reservation access method

- Here in the first interval, only stations 1, 3, and 4 have made reservations. In the second interval, only station 1 has made a reservation.

9.2.2 Polling

- Polling works with topologies in which one device is designated as a primary station and the other devices are secondary stations.
- All data exchanges must be made through the primary device even when the ultimate destination is a secondary device.
- The primary device controls the link; the secondary devices follow its instructions. It is up to the primary device to determine which device is allowed to use the channel at a given time. The primary device, therefore, is always the initiator of a session

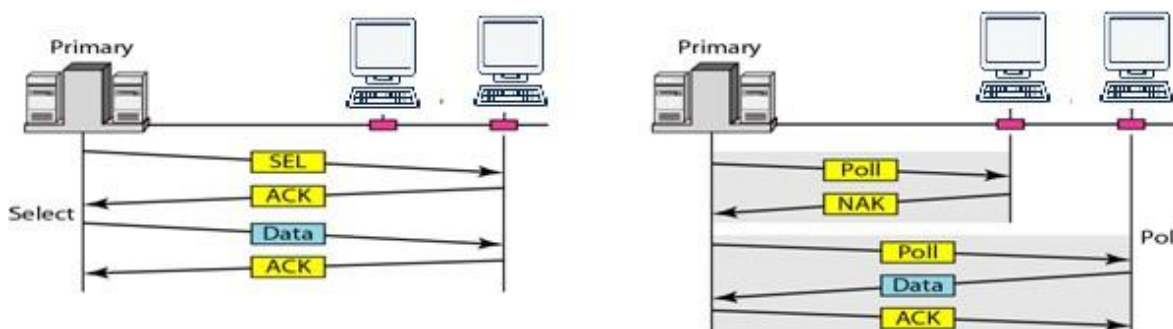


Figure 9.16: Polling access method

- If the primary wants to receive data, it asks the secondaries if they have anything to send; this is called poll

function.

- If the primary wants to send data, it tells the secondary to get ready to receive; this is called select function.

Select

- The *select* function is used whenever the primary device has something to send.
- If it has something to send, the primary device sends it.
- The primary must alert the secondary to the upcoming transmission and wait for an acknowledgment of the secondary's ready status.
- Before sending data, the primary creates and transmits a select (SEL) frame, one field of which includes the address of the intended secondary.

Poll

- The *poll* function is used by the primary device to solicit transmissions from the secondary devices.
- When the primary is ready to receive data, it must ask (poll) each device in turn if it has anything to send.
- When the first secondary is approached, it responds either with a NAK frame if it has nothing to send or with data if it does.
- If the response is negative (a NAK frame), then the primary polls the next secondary in the same manner until it finds one with data to send. When the response is positive (a data frame), the primary reads the frame and returns an acknowledgment (ACK frame), verifying its receipt.

9.2.3 Token Passing

- In the token-passing method, the stations in a network are organized in a logical ring.
- For each station, there is a *predecessor* and a *successor*. The predecessor is the station which is logically before the station in the ring; the successor is the station which is after the station in the ring. The current station is the one that is accessing the channel now.
- The right to this access has been passed from the predecessor to the current station. The right will be passed to the successor when the current station has no more data to send.
- In this method, a special packet called a token circulates through the ring. The possession of the token gives the station the right to access the channel and send its data. When a station has some data to send, it waits until it receives the token from its predecessor. It then holds the token and sends its data. When the station has no more data to send, it releases the token, passing it to the next logical station in the ring.
- The station cannot send data until it receives the token again in the next round.
- In this process, when a station receives the token and has no data to send, it just passes the data to the next station.

- Token management is needed for this access method.
 - Stations must be limited in the time they can have possession of the token.
 - The token must be monitored to ensure it has not been lost or destroyed.
 - Another function of token management is to assign priorities to the stations and to the types of data being transmitted. And finally, token management is needed to make low-priority stations release the token to high priority stations.
- In a token-passing network, stations do not have to be physically connected in a ring; the ring can be a logical one.
- In the **physical ring topology**, when a station sends the token to its successor, the token cannot be seen by other stations; the successor is the next one in line. This means that the token does not have to have the address of the next successor. The problem with this topology is that if one of the links-the medium between two adjacent stations fails, the whole system fails.
- The **dual ring topology** uses a second (auxiliary) ring which operates in the reverse direction compared with the main ring. The second ring is for emergencies only. If one of the links in the main ring fails, the system automatically combines the two rings to form a temporary ring. After the failed link is restored, the auxiliary ring becomes idle again.
- In the **bus ring topology**, also called a token bus, the stations are connected to a single cable called a bus. They, however, make a logical ring, because each station knows the address of its successor. When a station has finished sending its data, it releases the token and inserts the address of its successor in the token. Only the station with the address matching the destination address of the token gets the token to access the shared media.
- In a **star ring topology**, the physical topology is a star. There is a hub, however, that acts as the connector. The wiring inside the hub makes the ring; the stations are connected to this ring through the two wire connections. This topology makes the network less prone to failure because if a link goes down, it will be bypassed by the hub and the rest of the stations can operate. Also adding and removing stations from the ring is easier.

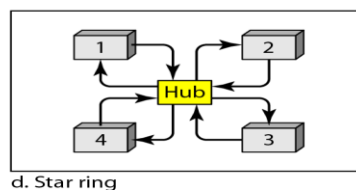
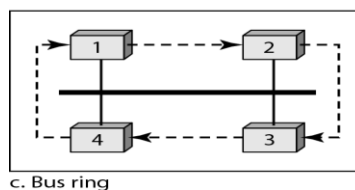
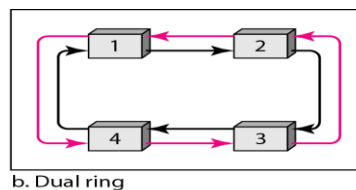
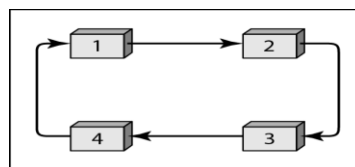


Figure 9.17: Token Passing

9.3 Channelization

Channelization is a multiple-access method in which the available bandwidth of a link is shared in time, frequency, or through code, between different stations.

9.3.1 Frequency-Division Multiple Access (FDMA)

- In frequency-division multiple access (FDMA), the available bandwidth is divided into frequency bands.
- Each station is allocated a band to send its data. In other words, each band is reserved for a specific station, and it belongs to the station all the time.
- Each station also uses a bandpass filter to confine the transmitter frequencies.
- To prevent station interferences, the allocated bands are separated from one another by small **guard bands**.

9.3.2 Time-Division Multiple Access (TDMA)

- In time-division multiple access (TDMA), the stations share the bandwidth of the channel in time.
- Each station is allocated a time slot during which it can send data. Each station transmits its data in its assigned time slot.
- The main problem with TDMA lies in achieving synchronization between the different stations.
- Each station needs to know the beginning of its slot and the location of its slot. This may be difficult because of propagation delays introduced in the system if the stations are spread over a large area.
- To compensate for the delays, we can insert **guard times**. Synchronization is normally accomplished by having some synchronization bits at the beginning of each slot.

9.3.3 Code-Division Multiple Access (CDMA)

CDMA differs from FDMA because only one channel occupies the entire bandwidth of the link. It differs from TDMA because all stations can send data simultaneously; there is no timesharing.

Let us assume we have four stations 1, 2, 3, and 4 connected to the same channel. The data from station 1 are d_1 , from station 2 are d_2 , and so on. The code assigned to the first station is c_1 , to the second is c_2 , and so on. We assume that the assigned codes have two properties.

1. If we multiply each code by another, we get 0.
2. If we multiply each code by itself, we get 4 (the number of stations).

Station 1 multiplies its data by its code to get $d_1.c_1$ Station 2 multiplies its data by its code to get $d_2.c_2$

and so on. The data that go on the channel are the sum of all these terms.

Any station that wants to receive data from one of the other three multiplies the data on the channel by the code of the sender. For example, suppose stations 1 and 2 are talking to each other. Station 2 wants to hear what station 1 is saying. It multiplies the data on the channel by c_1 the code of station 1.

Because $(c_1 \cdot c_1)$ is 4, but $(c_2 \cdot c_1)$, $(c_3 \cdot c_1)$, and $(c_4 \cdot c_1)$ are all 0s, station 2 divides the result by 4 to get the data from station 1.

$$\text{data} = (d_1.C_1 + d_2.C_2 + d_3.C_3 + d_4.C_4) \cdot C_1 = 4 \times d_1$$

CDMA is based on coding theory. Each station is assigned a code, which is a sequence of numbers called chips.

C1 : [+1 +1 +1 +1]

C2 : [+1 -1 +1 -1]

C3 : [+1 +1 -1 -1]

C4 : [+1 -1 -1 +1]

They are called orthogonal sequences and have the following properties:

1. Each sequence is made of N elements, where N is the number of stations.
2. If we multiply a sequence by a number, every element in the sequence is multiplied by that element.

This is called multiplication of a sequence by a scalar. For example,

$$2. [+1 +1 -1 -1] = [+2 +2 -2 -2]$$

3. If we multiply two equal sequences, element by element, and add the results, we get N , where N is the number of elements in the each sequence. This is called the inner product of two equal sequences.

For example,

$$[+1 +1 -1 -1] \cdot [+1 +1 -1 -1] = 1 + 1 + 1 + 1 = 4$$

4. If we multiply two different sequences, element by element, and add the results, we get 0. This is called inner product of two different sequences. For example,

$$[+1 +1 -1 -1] \cdot [+1 +1 +1 +1] = 1 + 1 - 1 - 1 = 0$$

5. Adding two sequences means adding the corresponding elements. The result is another sequence.

For example,

$$[+1 +1 -1 -1] + [+1 +1 +1 +1] = [+2 +2 0 0]$$

Data Representation

Data bit 0 $\rightarrow$ -1

Data bit 1 -> +1

Silence -> 0

Sequence Generation

To generate chip sequences, Walsh table is used, which is a two-dimensional table with an equal number of rows and columns.

$$W_1 = \begin{bmatrix} +1 \end{bmatrix} \qquad W_{2N} = \begin{bmatrix} W_N & W_N \\ W_N & \overline{W_N} \end{bmatrix}$$

a. Two basic rules

$$W_1 = \begin{bmatrix} +1 \end{bmatrix}$$

$$W_2 = \begin{bmatrix} +1 & +1 \\ +1 & -1 \end{bmatrix}$$

$$W_4 = \begin{bmatrix} +1 & +1 & +1 & +1 \\ +1 & -1 & +1 & -1 \\ +1 & +1 & -1 & -1 \\ +1 & -1 & -1 & +1 \end{bmatrix}$$

b. Generation of W_1 , W_2 , and W_4

Figure 9.18: Sequence Generation

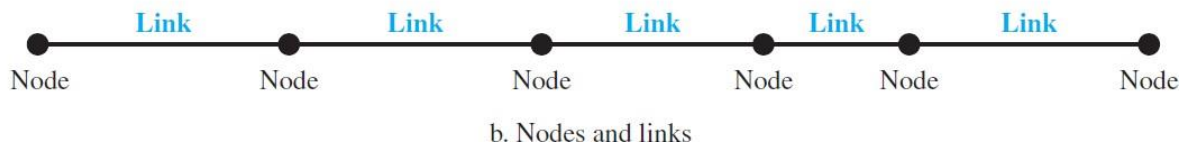
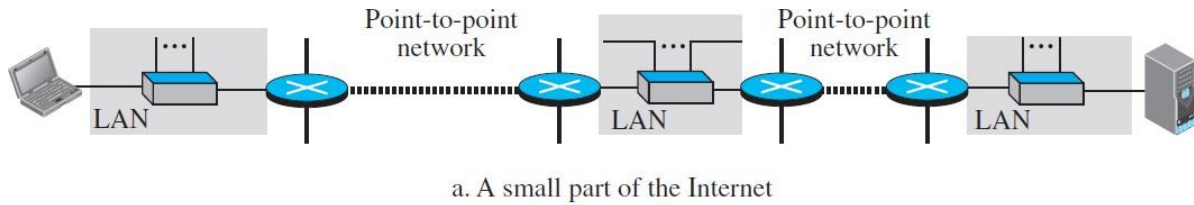
In the Walsh table, each row is a sequence of chips.

The number of sequences in a Walsh table needs to be $N = 2m$.

INTRODUCTION TO DATA LINK LAYER

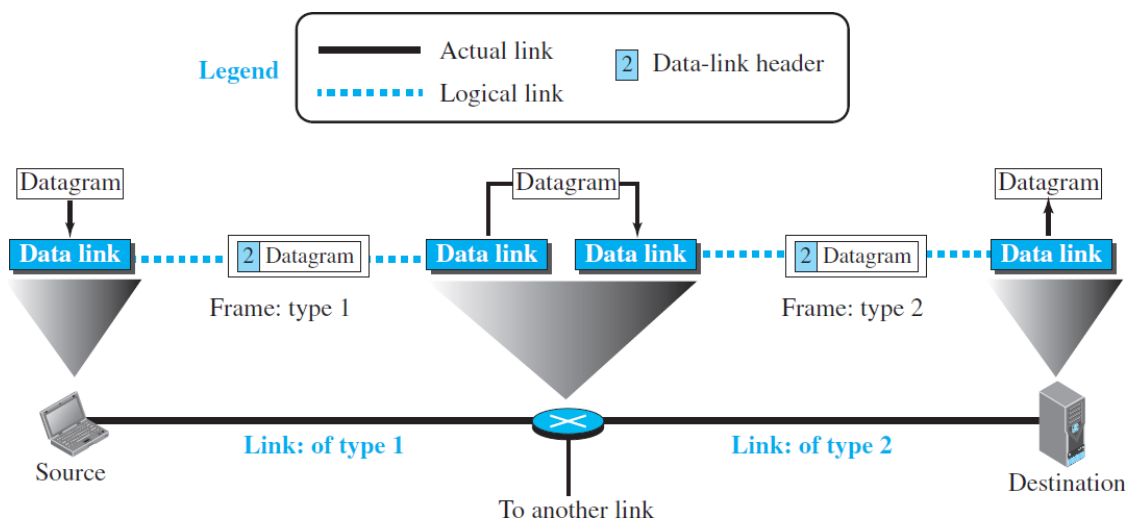
Nodes and Links

- Communication at the data-link layer is node-to-node.
- A data unit from one point in the Internet needs to pass through many networks (LANs and WANs) to reach another point.
- These LANs and WANs are connected by routers. It is customary to refer to the two end hosts and the routers as nodes and the networks in between as links.



Services

- The data-link layer is located between the physical and the network layers. The datalink layer provides services to the network layer; it receives services from the physical layer.
- The duty scope of the data-link layer is node-to-node. When a packet is travelling in the Internet, the data-link layer of a node (host or router) is responsible for delivering a datagram to the next node in the path.
- For this purpose, the data-link layer of the sending node needs to encapsulate the datagram received from the network in a frame, and the data-link layer of the receiving node needs to decapsulate the datagram from the frame.
- The data-link layer of the source host needs only to encapsulate, the data-link layer of the destination host needs to decapsulate, but each intermediate node needs to both encapsulate and decapsulate.



1) Framing

- The first service provided by the data-link layer is framing. The data-link layer at each node needs to encapsulate the datagram (packet received from the network layer) in a frame before sending it to the next node.
- The node also needs to decapsulate the datagram from the frame received on the logical channel.
- A packet at the data-link layer is normally called a frame.

2) Flow Control

- Flow control is the process of controlling the flow of data between sender and receiver according to the receiver capacity.

3) Error Control

- At the sending node, a frame in a data-link layer needs to be changed to bits, transformed to electromagnetic signals, and transmitted through the transmission media.
- At the receiving node, electromagnetic signals are received, transformed to bits, and put together to create a frame.
- Since electromagnetic signals are susceptible to error, a frame is susceptible to error.
- The error needs first to be detected.
- After detection, it needs to be either corrected at the receiver node or discarded and retransmitted by the sending node.

4) Congestion Control

- Although a link may be congested with frames, which may result in frame loss, most data-link-layer protocols do not directly use a congestion control to alleviate congestion, although some wide-area networks do.
- In general, congestion control is considered an issue in the network layer or the transport layer because of its end-to-end nature.

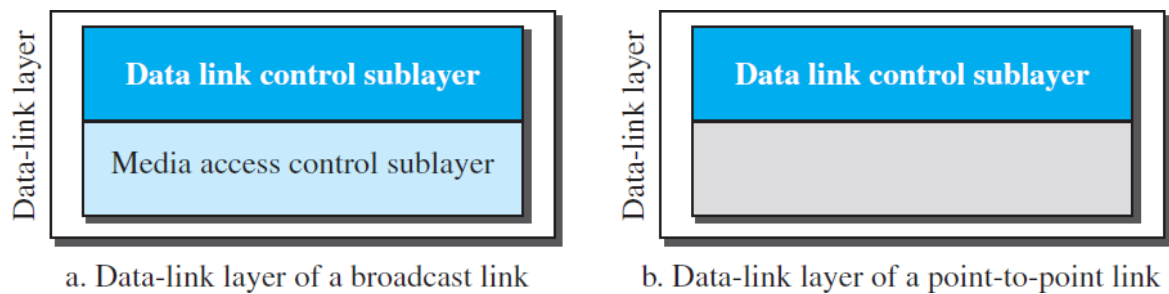
Two Categories of Links

- In a point-to-point link, the link is dedicated to the two devices; in a broadcast link, the link is shared between several pairs of devices.
- For example, when two friends use the traditional home phones to chat, they are using a point-to-point link; when the same two friends use their cellular phones, they are using a broadcast

link

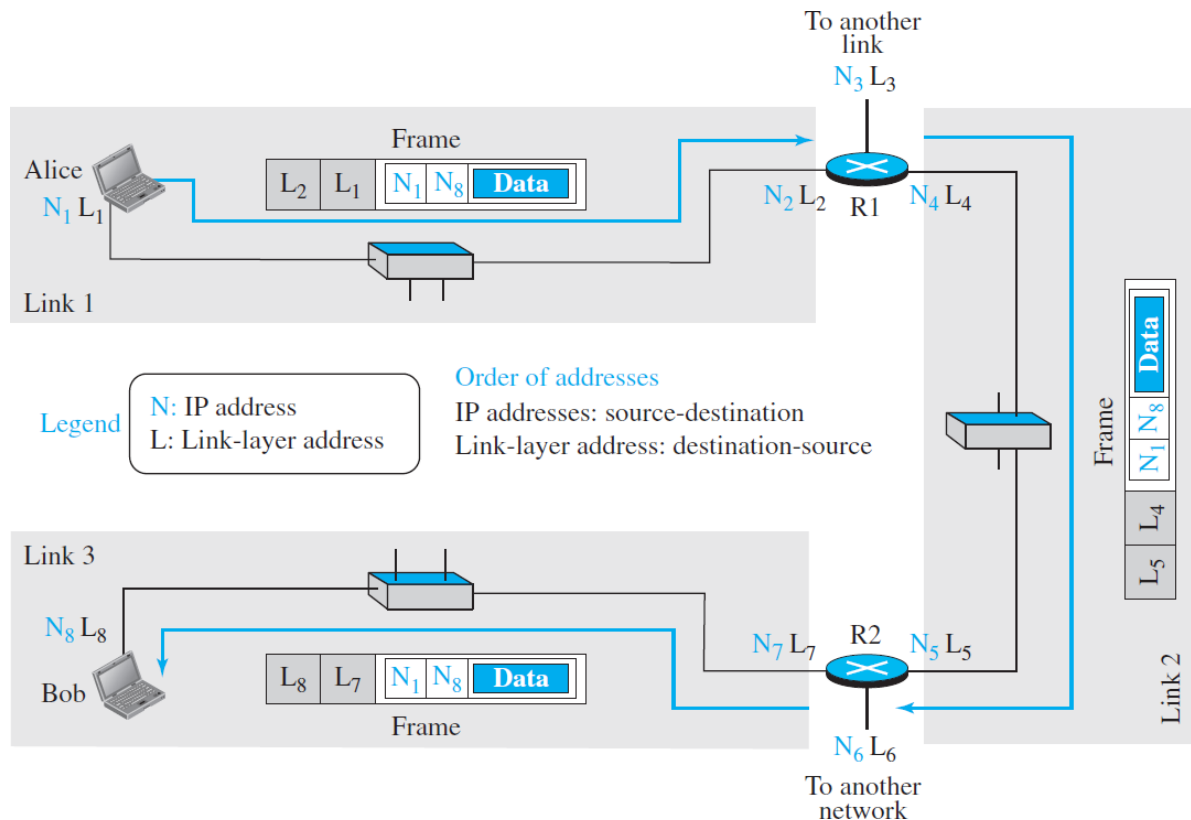
Two Sublayers

- The data-link layer can be divided into two sublayers: data link control (DLC) and media access control (MAC).
- The data link control sublayer deals with all issues common to both point-to-point and broadcast links; the media access control sublayer deals only with issues specific to broadcast links.



LINK-LAYER ADDRESSING

- The source and destination IP addresses define the two ends but cannot define which links the datagram should pass through.
- The IP addresses
- in a datagram should not be changed. If the destination IP address in a datagram changes, the packet never reaches its destination; if the source IP address in a datagram changes, the destination host or a router can never communicate with the source if a response needs to be sent back or an error needs to be reported back to the source.
- Hence we need another addressing mechanism in a connectionless internetwork: the link-layer addresses of the two nodes.
- A link-layer address is sometimes called a link address, sometimes a physical address, and sometimes a MAC address.
- When a datagram passes from the network layer to the data-link layer, the datagram will be encapsulated in a frame and two data-link addresses are added to the frame header. These two addresses are changed every time the frame moves from one link to another.



- MAC address is a 48bit Address

Three Types of addresses

Unicast Address

Each host or each interface of a router is assigned a unicast address. Unicasting means one-to-one communication. A frame with a unicast address destination is destined only for one entity in the link.

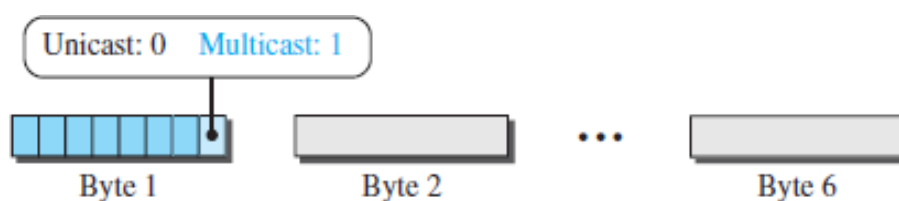
Multicast Address

Some link-layer protocols define multicast addresses. Multicasting means one-to-many communication. However, the jurisdiction is local

Broadcast Address

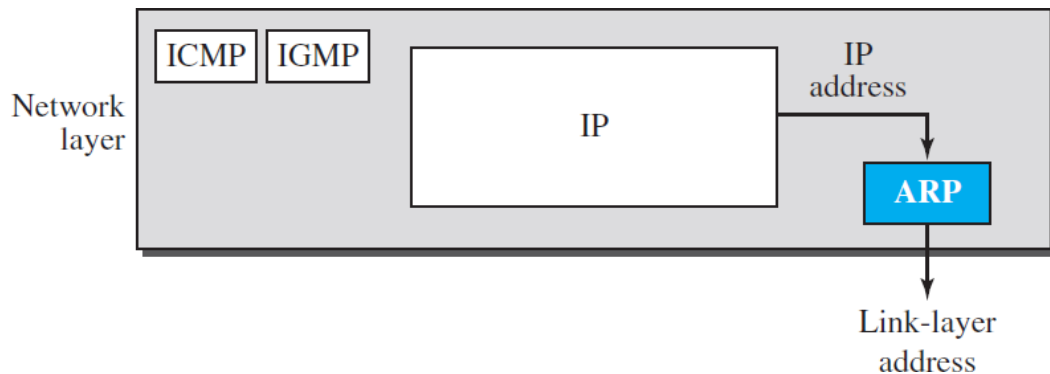
Some link-layer protocols define a broadcast address. Broadcasting means one-to-all communication. A frame with a destination broadcast address is sent to all entities in the link.

Ex: FF:FF:FF:FF:FF:FF

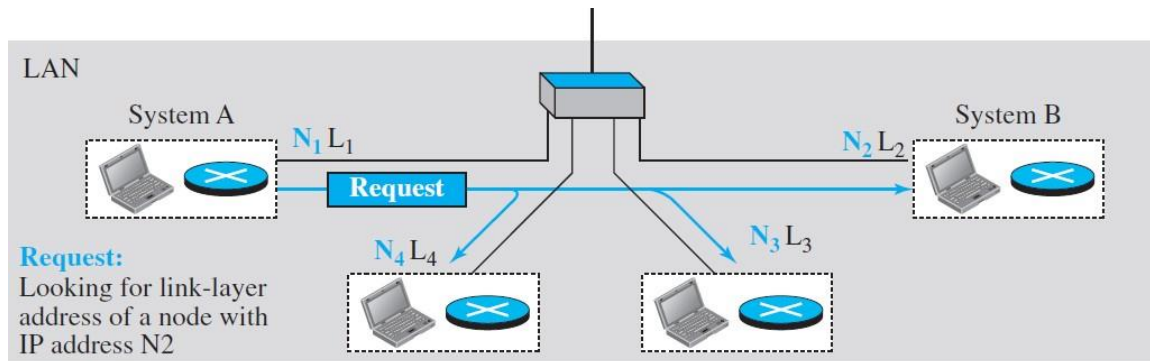


ADDRESS RESOLUTION PROTOCOL (ARP)

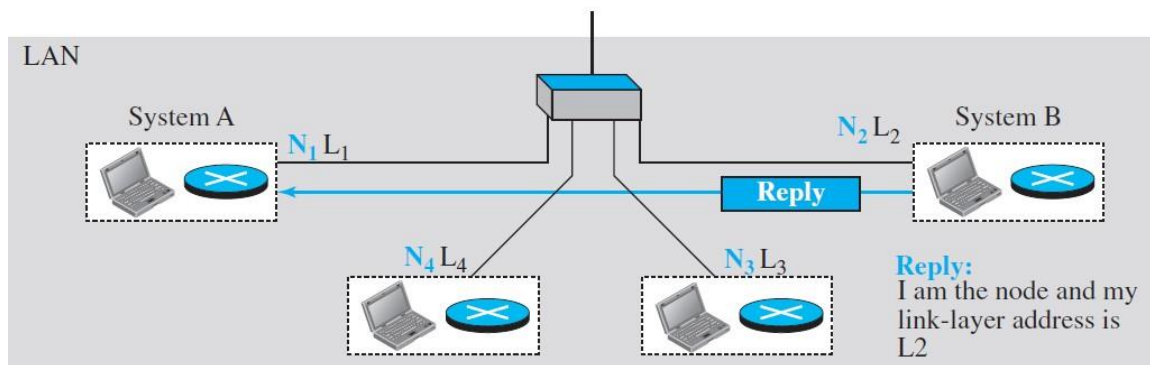
ARP accepts an IP address from the IP protocol, maps the address to the corresponding link-layer address, and passes it to the data-link layer.



- Anytime a host or a router needs to find the link-layer address of another host or router in its network, it sends an ARP request packet.
- The packet includes the link-layer and IP addresses of the sender and the IP address of the receiver. Because the sender does not know the link-layer address of the receiver, the query is broadcast over the link using the link-layer broadcast address.
- Every host or router on the network receives and processes the ARP request packet, but only the intended recipient recognizes its IP address and sends back an ARP response packet.
- The response packet contains the recipient's IP and link-layer addresses. The packet is unicast directly to the node that sent the request packet.



a. ARP request is broadcast



b. ARP reply is unicast

Packet Format

0	8	16	31
Hardware Type		Protocol Type	
Hardware length	Protocol length	Operation Request:1, Reply:2	
Source hardware address			
Source protocol address			
Destination hardware address (Empty in request)			
Destination protocol address			

Hardware: LAN or WAN protocol

Protocol: Network-layer protocol

- The hardware type field defines the type of the link-layer protocol; Ethernet is given the type 1.
- The protocol type field defines the network-layer protocol: IPv4 protocol is (0800)16.

- The source hardware and source protocol addresses are variable-length fields defining the link-layer and network-layer addresses of the sender.
- The destination hardware address and destination protocol address fields define the receiver link-layer and network-layer addresses.
- An ARP packet is encapsulated directly into a data-link frame. The frame needs to have a field to show that the payload belongs to the ARP and not to the network-layer datagram.

IPv4 Addressing and Subnetting

1.1 IPv4 ADDRESSES

An IPv4 address is a 32-bit address that uniquely and universally defines the connection of a device (for example, a computer or a router) to the Internet.

Address Space

- An address space is the total number of addresses used by the protocol. If a protocol uses N bits to define an address, the address space is 2^N because each bit can have two different values (0 or 1) and N bits can have 2^N values.
- IPv4 uses 32-bit addresses, which means that the address space is 2^{32} or 4,294,967,296 (more than 4 billion).

Notations

There are two prevalent notations to show an IPv4 address: binary notation and dotted decimal notation.

1) Binary Notation:

In binary notation, the IPv4 address is displayed as 32 bits. Each octet is often referred to as a byte. The following is an example of an IPv4 address in binary notation:

01110101 10010101 00011101 00000010

2) Dotted-Decimal Notation:

To make the IPv4 address more compact and easier to read, Internet addresses are usually written in decimal form with a decimal point (dot) separating the bytes. The following is the dotted decimal notation of the above address:

117.149.29.2

Example:

Change the following IPv4 addresses from binary notation to dotted-decimal notation. a.

10000001 00001011 00001011 11101111

b. 11000001 10000011 00011011 11111111

Solution:

a. 129.11.11.239

b. 193.131.27.255

Classful Addressing

IPv4 addressing, at its inception, used the concept of classes. This architecture is called classful

addressing.

In classful addressing, the address space is divided into five classes: A, B, C, D, and E. Each class occupies some part of the address space.

	First byte	Second byte	Third byte	Fourth byte
Class A	0			
Class B	10			
Class C	110			
Class D	1110			
Class E	1111			

a. Binary notation

	First byte	Second byte	Third byte	Fourth byte
Class A	0–127			
Class B	128–191			
Class C	192–223			
Class D	224–239			
Class E	240–255			

b. Dotted-decimal notation

Classes and Blocks

One problem with classful addressing is that each class is divided into a fixed number of blocks with each block having a fixed size.

Class	Number of Blocks	Block Size	Application
A	128	16,777,216	Unicast
B	16,384	65,536	Unicast
C	2,097,152	256	Unicast
D	1	268,435,456	Multicast
E	1	268,435,456	Reserved

- Class A addresses were designed for large organizations with a large number of attached hosts or routers
- Class B addresses were designed for midsize organizations with tens of thousands of attached hosts or routers.
- Class C addresses were designed for small organizations with a small number of attached hosts or routers.
- Class D addresses were designed for multicasting.
- Class E addresses were reserved for future use.
- In classful addressing, a large part of the available addresses were wasted.

Netid and Hostid

- In classful addressing, an IP address in class A, B, or C is divided into netid and hostid. These parts are of varying lengths, depending on the class of the address.
- In class A, one byte defines the netid and three bytes define the hostid.
- In class B, two bytes define the netid and two bytes define the hostid.
- In class C, three bytes define the netid and one byte defines the hostid.

Mask:

Although the length of the netid and hostid (in bits) is predetermined in classful addressing, we can also use a mask (also called the default mask), a 32-bit number made of contiguous 1s followed by contiguous as.

The mask can help us to find the netid and the hostid. For example, the mask for a class A address has eight 1s, which means the first 8 bits of any address in class A define the netid; the next 24 bits define the hostid.

Subnetting:

If an organization was granted a large block in class A or B, it could divide the addresses into several contiguous groups and assign each group to smaller networks (called subnets) or, in rare cases, share part of the addresses with neighbors.

Supernetting:

In supernetting, an organization can combine several class C blocks to create a larger range of addresses. In other words, several networks are combined to create a supernet or a supernet.

Address Depletion

The flaws in classful addressing scheme combined with the fast growth of the Internet led to the near depletion of the available addresses. We have run out of class A and B addresses, and a class C block is too small for most midsize organizations. One solution that has alleviated the problem is the idea of classless addressing

Classless Addressing

To overcome address depletion and give more organizations access to the Internet, classless addressing was designed and implemented. In this scheme, there are no classes, but the addresses are

still granted in blocks.

Address Blocks

In classless addressing, when an entity, small or large, needs to be connected to the Internet, it is granted a block (range) of addresses. The size of the block (the number of addresses) varies based on the nature and size of the entity. To simplify the handling of addresses, the Internet authorities impose three restrictions on classless address blocks:

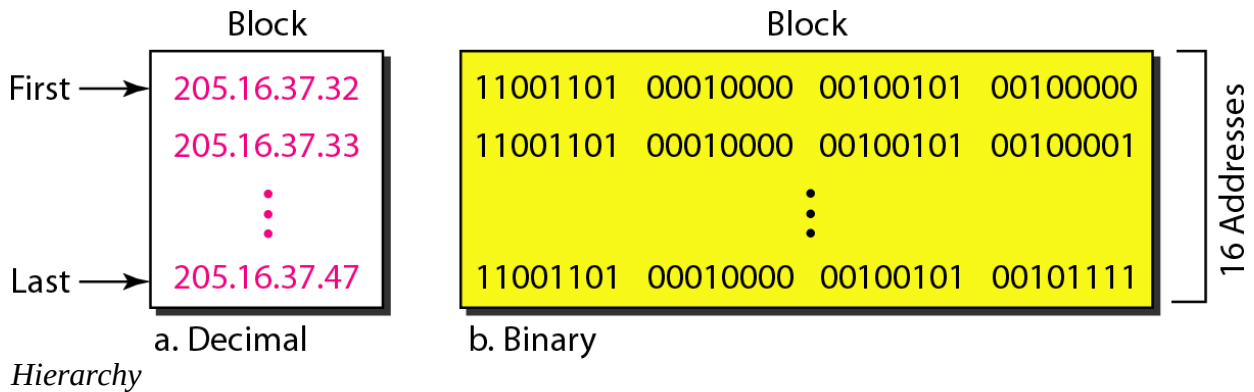
1. The addresses in a block must be contiguous, one after another.
2. The number of addresses in a block must be a power of 2 (1, 2, 4, 8, ...).
3. The first address must be evenly divisible by the number of addresses.

Mask

- In classless addressing the mask for a block can take any value from 0 to 32. It is very convenient to give just the value of n preceded by a slash (CIDR notation).
- In IPv4 addressing, a block of addresses can be defined as $x.y.z.t/n$ in which $x.y.z.t$ defines one of the addresses and the $/n$ defines the mask.
- The first address in the block can be found by setting the rightmost $32 - n$ bits to 0s
- The last address in the block can be found by setting the rightmost $32 - n$ bits to 1s.
- The number of addresses in the block can be found by using the formula 2^{32-n}

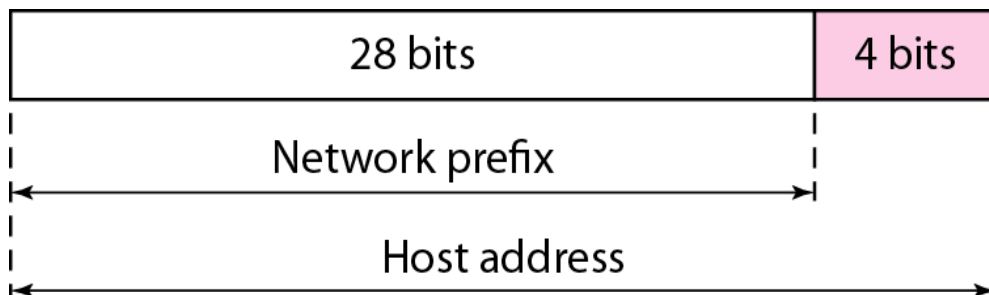
Network Addresses

When an organization is given a block of addresses, the organization is free to allocate the addresses to the devices that need to be connected to the Internet. The first address in the class, however, is normally (not always) treated as a special address. The first address is called the network address and defines the organization network. It defines the organization itself to the rest of the world.



Two-Level Hierarchy: No Subnetting

- An IP address can define only two levels of hierarchy when not subnetted.
- The n leftmost bits of the address $x.y.z.t/n$ define the network (organization network); the 32
 - n rightmost bits define the particular host (computer or router) to the network.
- The part of the address that defines the network is called the prefix; the part that defines the host is called the suffix.



Three-Levels of Hierarchy: Subnetting

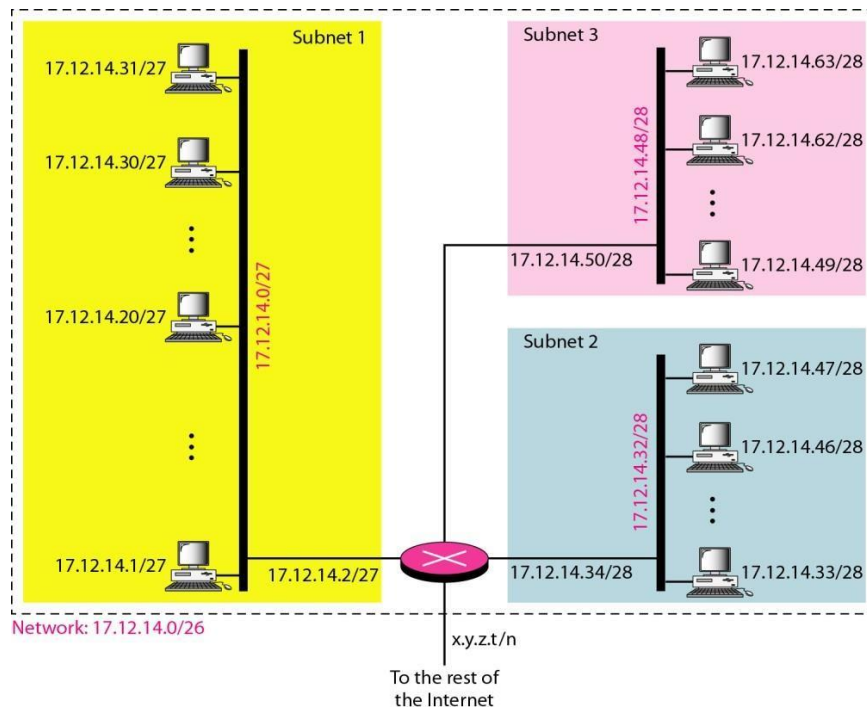
- An organization that is granted a large block of addresses may want to create clusters of networks (called subnets) and divide the addresses between the different subnets. The rest of the world still sees the organization as one entity; however, internally there are several subnets.
- All messages are sent to the router address that connects the organization to the rest of the Internet; the router routes the message to the appropriate subnets.
- The organization, however, needs to create small sub blocks of addresses, each assigned to specific subnets. The organization has its own mask; each subnet must also have its own.

Example:

Suppose an organization is given the block 17.12.40.0/26, which contains 64 addresses. The organization has three offices and needs to divide the addresses into three sub blocks of 32, 16, and

16 addresses. We can find the new masks by using the following arguments:

1. Suppose the mask for the first subnet is n_1 , then 2^{32-n_1} must be 32, which means that $n_1 = 27$.
2. Suppose the mask for the second subnet is n_2 , then 2^{32-n_2} must be 16, which means that $n_2 = 28$.
3. Suppose the mask for the third subnet is n_3 , then 2^{32-n_3} must be 16, which means that $n_3 = 28$.



a. In subnet 1, the address 17.12.14.29/27 can give us the subnet address if we use the mask /27 because

Host: 00010001 00001100 00001110 00011101

Mask: /27

Subnet: 00010001 00001100 00001110 00000000.....(17.12.14.0)

b. In subnet 2, the address 17.12.14.45/28 can give us the subnet address if we use the mask /28 because

Host: 00010001 00001100 00001110 00101101

Mask: /28

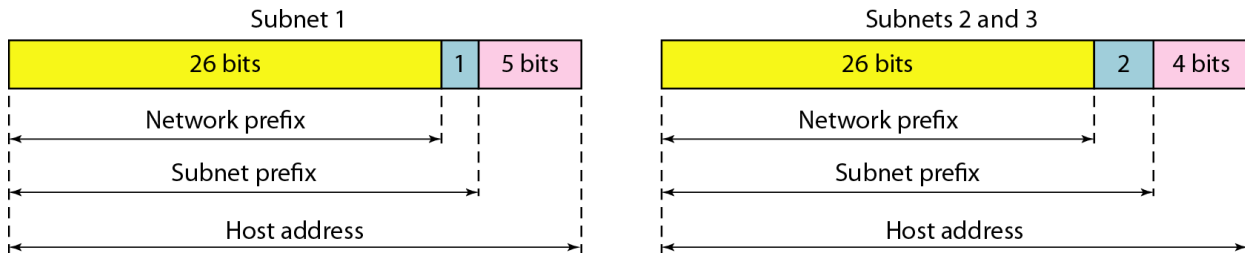
Subnet: 00010001 00001100 00001110 00100000 (17.12.14.32)

c. In subnet 3, the address 17.12.14.50/28 can give us the subnet address if we use the mask /28 because

Host: 00010001 00001100 00001110 00110010

Mask: /28

Subnet: 00010001 00001100 00001110 00110000.....(17.12.14.48)



Example:

An organization is granted a block of addresses with the beginning address 14.24.74.0/24. The organization needs to have 3 subblocks of addresses to use in its three subnets: one subblock of 10 addresses, one subblock of 60 addresses, and one subblock of 120 addresses. Design the subblocks.

Solution

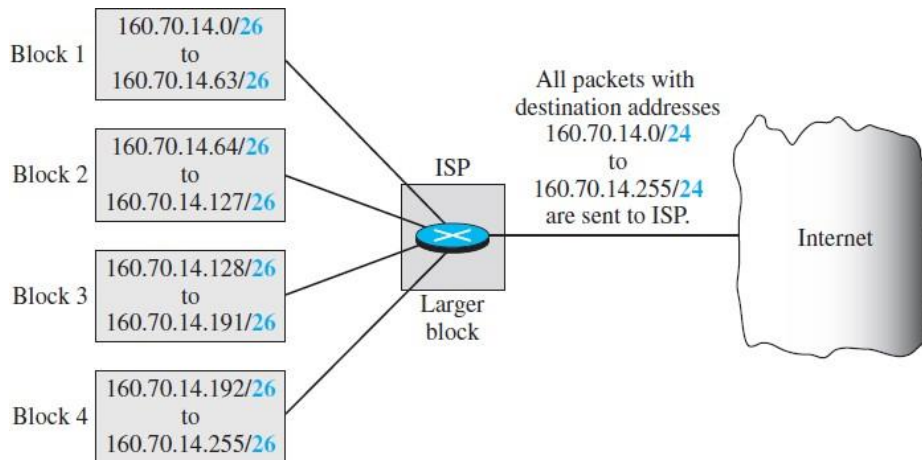
There are $2^{32-24} = 256$ addresses in this block. The first address is 14.24.74.0/24; the last address is 14.24.74.255/24. To satisfy the third requirement, we assign addresses to subblocks, starting with the largest and ending with the smallest one.

- The number of addresses in the largest subblock, which requires 120 addresses, is not a power of 2. We allocate 128 addresses. The subnet mask for this subnet can be found as $n_1 = 32 - \log_2 128 = 25$. The first address in this block is 14.24.74.0/25; the last address is 14.24.74.127/25.
- The number of addresses in the second largest subblock, which requires 60 addresses, is not a power of 2 either. We allocate 64 addresses. The subnet mask for this subnet can be found as $n_2 = 32 - \log_2 64 = 26$. The first address in this block is 14.24.74.128/26; the last address is 14.24.74.191/26.
- The number of addresses in the smallest subblock, which requires 10 addresses, is not a power of 2 either. We allocate 16 addresses. The subnet mask for this subnet can be found as $n_3 = 32 - \log_2 16 = 28$. The first address in this block is 14.24.74.192/28; the last address is 14.24.74.207/28.

If we add all addresses in the previous subblocks, the result is 208 addresses, which means 48 addresses are left in reserve. The first address in this range is 14.24.74.208. The last address is 14.24.74.255.

Address Aggregation

One of the advantages of the CIDR strategy is address aggregation (sometimes called address summarization or route summarization). When blocks of addresses are combined to create a larger block, routing can be done based on the prefix of the larger block.



More Levels of Hierarchy

The structure of classless addressing does not restrict the number of hierarchical levels. An organization can divide the granted block of addresses into sub blocks. Each sub block can in turn be divided into smaller sub blocks.

Special Addresses

1) This-host Address

The only address in the block 0.0.0.0/32 is called the this-host address. It is used whenever a host needs to send an IP datagram but it does not know its own address to use as the source address.

2) Limited-broadcast Address

The only address in the block 255.255.255.255/32 is called the limited-broadcast address. It is used whenever a router or a host needs to send a datagram to all devices in a network.

3) Loopback Address

The block 127.0.0.0/8 is called the loopback address. A packet with one of the addresses in this block as the destination address never leaves the host; it will remain in the host.

4) Private Addresses

Four blocks are assigned as private addresses: 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16, and 169.254.0.0/16.

5) Multicast Addresses

The block 224.0.0.0/4 is reserved for multicast addresses.

Dynamic Host Configuration Protocol

- Once an organization has obtained a block of addresses, it can assign individual IP addresses to

the host and router interfaces in its organization.

- A system administrator will typically manually configure the IP addresses into the.
- Host addresses can also be configured manually, but more often this task is now done using the Dynamic Host Configuration Protocol (DHCP).
- DHCP allows a host to obtain (be allocated) an IP address automatically.
- DHCP works over UDP with port number 67.
- DHCP involves four steps:

1) DHCP server discovery

- Host broadcast DHCP discovery message with source address 0.0.0.0 and destination address 255.255.255.255

2) DHCP server offer(s)

- A DHCP server receiving a DHCP discover message responds to the client with a DHCP offer message that is broadcast to all nodes on the subnet, again using the IP broadcast address of 255.255.255.255.

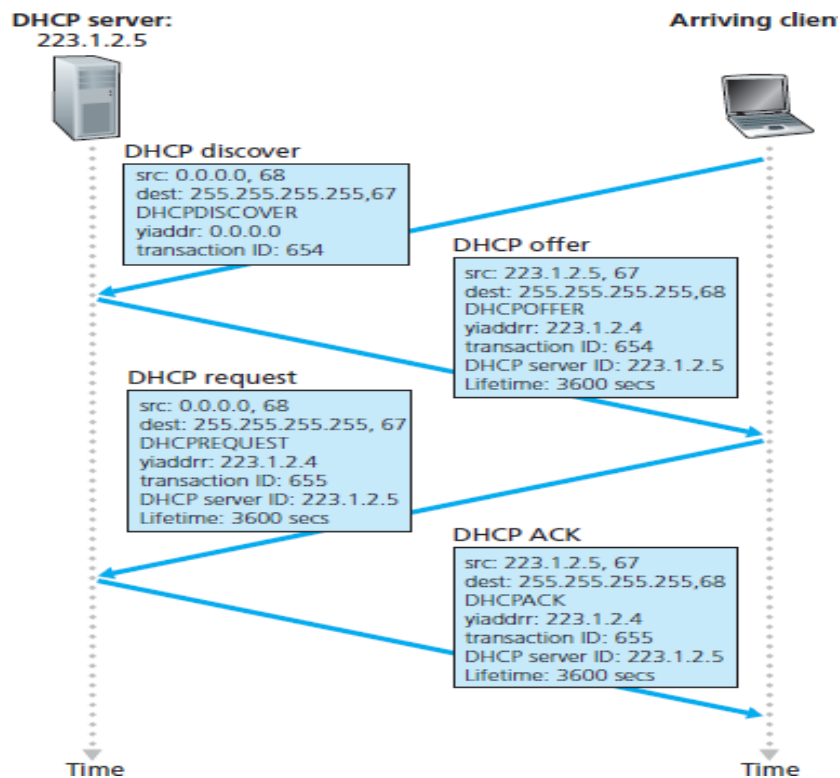
3) DHCP request

- The newly arriving client will choose from among one or more server offers and respond to its selected offer with a DHCP request message

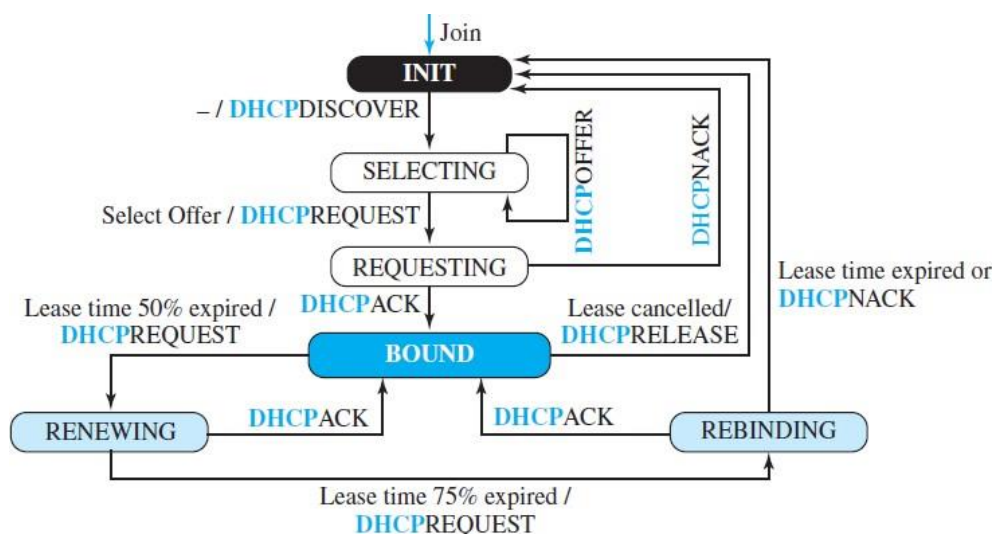
4) DHCP ACK

- The server responds to the DHCP request message with a DHCP ACK message, confirming the requested parameters.

0	8	16	24	31	
Opcode	Htype	HLen	HCount		Fields:
Transaction ID					Opcode: Operation code, request (1) or reply (2)
Time elapsed		Flags			Htype: Hardware type (Ethernet, ...)
Client IP address					HLen: Length of hardware address
Your IP address					HCount: Maximum number of hops the packet can travel
Server IP address					Transaction ID: An integer set by the client and repeated by the server
Gateway IP address					Time elapsed: The number of seconds since the client started to boot
Client hardware address					Flags: First bit defines unicast (0) or multicast (1); other 15 bits not used
Server name					Client IP address: Set to 0 if the client does not know it
Boot file name					Your IP address: The client IP address sent by the server
Options					Server IP address: A broadcast IP address if client does not know it
					Gateway IP address: The address of default router
					Server name: A 64-byte domain name of the server
					Boot file name: A 128-byte file name holding extra information
					Options: A 64-byte field with dual purpose described in text



Transition States:



- When the DHCP client first starts, it is in the INIT state (initializing state). The client broadcasts a discover message.
- When it receives an offer, the client goes to the SELECTING state. While it is there, it may

receive more offers.

- After it selects an offer, it sends a request message and goes to the REQUESTING state. If an ACK arrives while the client is in this state, it goes to the BOUND state and uses the IP address.
- When the lease is 50 percent expired, the client tries to renew it by moving to the RENEWING state.
- If the server renews the lease, the client moves to the BOUND state again.
- If the lease is not renewed and the lease time is 75 percent expired, the client moves to the REBINDING state.
- If the server agrees with the lease (ACK message arrives), the client moves to the BOUND state and continues using the IP address; otherwise, the client moves to the INIT state and requests another IP address.

Network Address Translation (NAT)

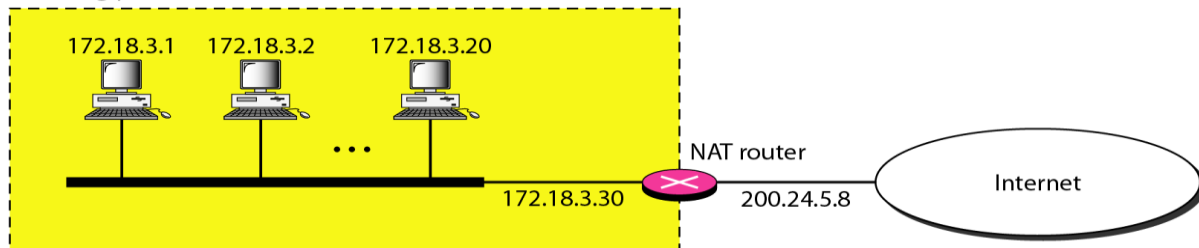
- The number of home users and small businesses that want to use the Internet is ever increasing.
- In the beginning, a user was connected to the Internet with a dial-up line. An ISP with a block of addresses could dynamically assign an address to this user. An address was given to a user when it was needed.
- But the situation is different today. Home users and small businesses can be connected by an ADSL line or cable modem. In addition, many are not happy with one address; many have created small networks with several hosts and need an IP address for each host. With the shortage of addresses, this is a serious problem.
- A quick solution to this problem is called network address translation (NAT).
- NAT enables a user to have a large set of addresses internally and one address, or a small set of addresses, externally. The traffic inside can use the large set; the traffic outside, the small set.

- To separate the addresses used inside the home or business and the ones used for the Internet, the Internet authorities have reserved three sets of addresses as private addresses.

Range			Total
10.0.0.0	to	10.255.255.255	2^{24}
172.16.0.0	to	172.31.255.255	2^{20}
192.168.0.0	to	192.168.255.255	2^{16}

- Any organization can use an address out of this set without permission from the Internet authorities. Everyone knows that these reserved addresses are for private networks. They are unique inside the organization, but they are not unique globally. No router will forward a packet that has one of these addresses as the destination address.
- The site must have only one single connection to the global Internet through a router that runs the NAT software

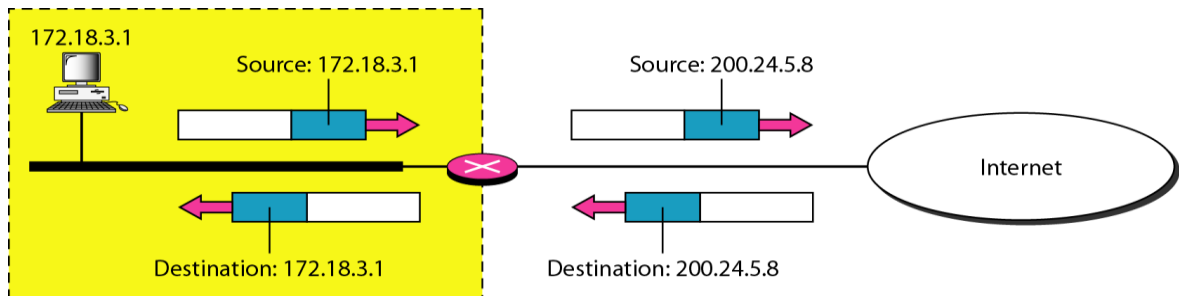
Site using private addresses



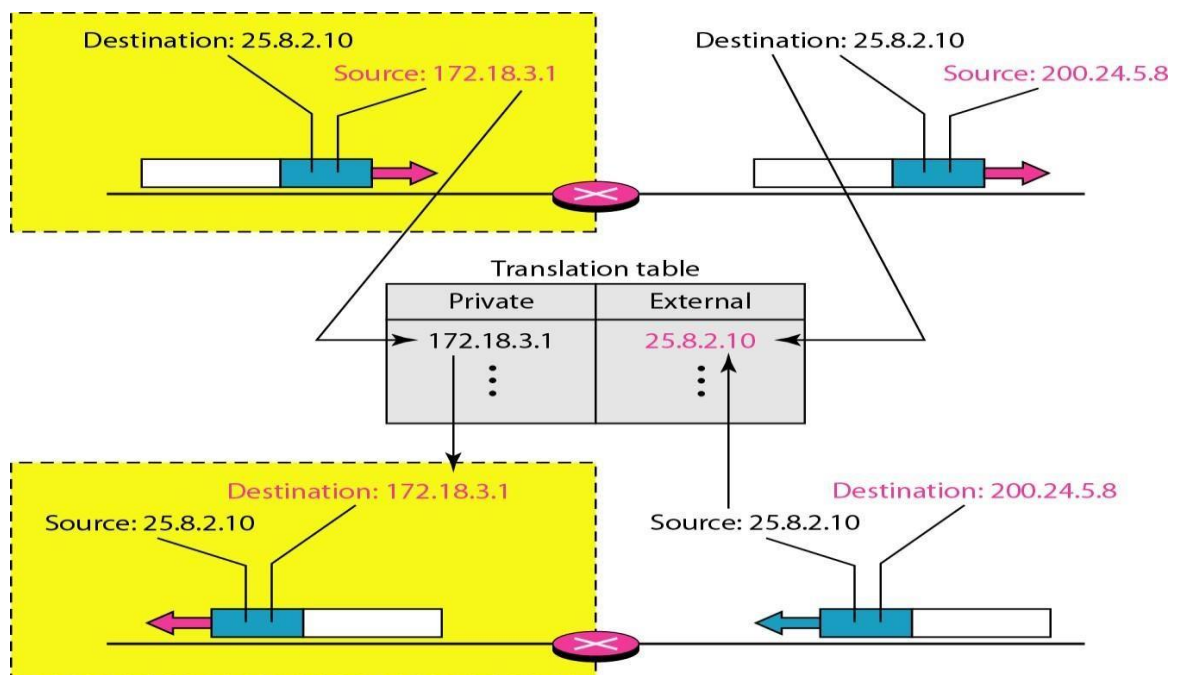
In the above figure, the router that connects the network to the global address uses one private address and one global address. The private network is transparent to the rest of the Internet; the rest of the Internet sees only the NAT router with the address 200.24.5.8.

Address Translation

- All the outgoing packets go through the NAT router, which replaces the *source address* in the packet with the global NAT address.
- All incoming packets also pass through the NAT router, which replaces the *destination address* in the packet (the NAT router global address) with the appropriate private address.



A **translation table** has only two columns: the private' address and the external address (destination address of the packet). When the router translates the source address of the outgoing packet, it also makes note of the destination address-where the packet is going. When the response comes back from the destination, the router uses the source address of the packet (as the external address) to find the private address of the packet.



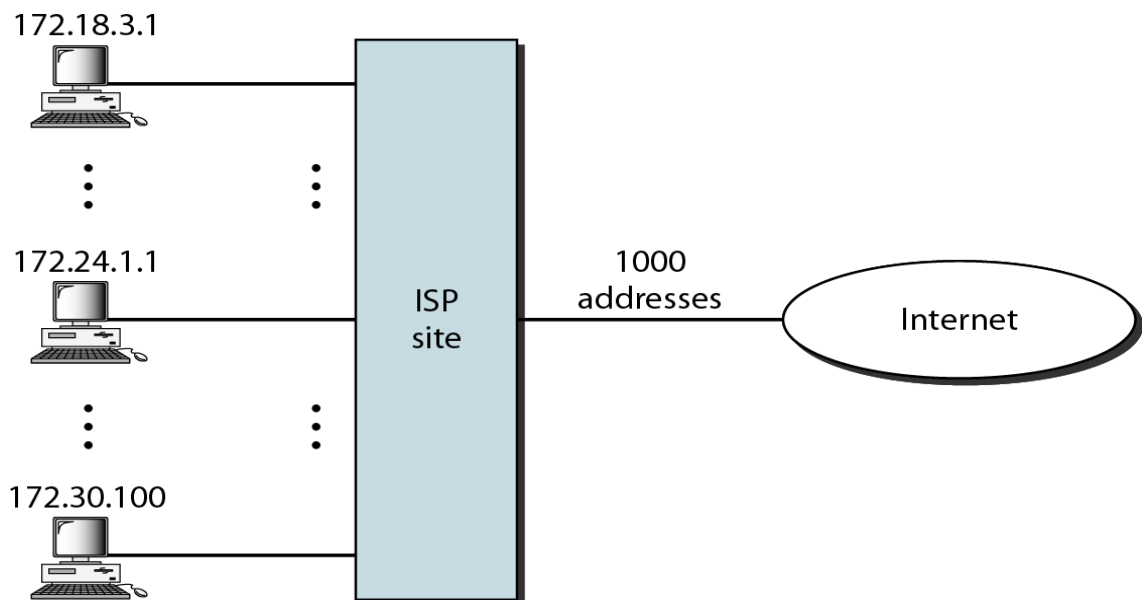
- **Using a Pool of IP Addresses** Since the NAT router has only one global address, only one private network host can access the same external host. To remove this restriction, the NAT router uses a pool of global addresses.

- Using Both **IP** Addresses and Port Numbers To allow a many-to-many relationship between private-network hosts and external server programs, we need more information in the translation table.

<i>Private Address</i>	<i>Private Port</i>	<i>External Address</i>	<i>External Port</i>	<i>Transport Protocol</i>
172.18.3.1	1400	25.8.3.2	80	TCP
172.18.3.2	1401	25.8.3.2	80	TCP
...	...	...	...	...

NAT and ISP

An ISP that serves dial-up customers can use NAT technology to conserve addresses. For example, suppose an ISP is granted 1000 addresses, but has 100,000 customers. Each of the customers is assigned a private network address. The ISP translates each of the 100,000 source addresses in outgoing packets to one of the 1000 global addresses; it translates the global destination address in incoming packets to the corresponding private address



Wired LANs Ethernet

10.1 Ethernet Protocol

10.1.1 IEEE 802

The IEEE has subdivided the data link layer into two sublayers: logical link control (LLC) and media access control (MAC).

IEEE has also created several physical layer standards for different LAN protocols.

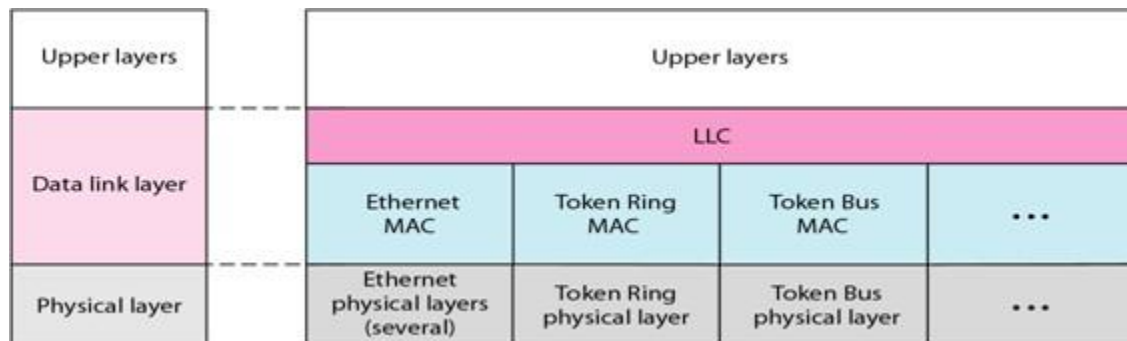


Figure 10.1: IEEE Standard for LANs

Data Link Layer

The data link layer in the IEEE standard is divided into two sub layers: LLC and MAC.

Logical Link Control (LLC) - **flow control, error control, part of framing.** Provides one single data link control for all IEEE LANs

Framing: LLC defines a protocol data unit (PDU) that is somewhat similar to that of HDLC.

- The header contains a control field like the one in HDLC; this field is used for flow and error control.
- The two other header fields define the upper-layer protocol at the source and destination that uses LLC. These fields are called the destination service access point (DSAP) and the source service access point (SSAP).
- In other words, a frame defined in HDLC is divided into a PDU at the LLC sublayer and a frame at the MAC sublayer, as shown in figure.

Need for LLC The purpose of the LLC is to provide flow and error control for the upper-layer protocols that actually demand these services.

For example, if a LAN or several LANs are used in an isolated system, LLC may be needed to provide flow and error control for the application layer protocols.

Media Access Control (MAC) - Defines the specific access method for each type of LAN (Ethernet–CSMA/CD, Token Ring and Token Bus-Token Passing). Provides part of framing function.

Physical Layer

- The physical layer is dependent on the implementation and type of physical media used.
- IEEE defines detailed specifications for each LAN implementation.
- For example, although there is only one MAC sublayer for Standard Ethernet, there is a different physical layer specification for each Ethernet implementations.

10.2 Standard Ethernet

Ethernet has gone through four generations: Standard Ethernet (10 t Mbps), Fast Ethernet (100 Mbps), Gigabit Ethernet (1 Gbps), and Ten-Gigabit Ethernet (10 Gbps).

10.2.1 MAC Sublayer

In Standard Ethernet, the MAC sublayer governs the operation of the access method. It also frames data received from the upper layer and passes them to the physical layer.

➤ Frame Format

Ethernet does not provide any mechanism for acknowledging received frames, making it what is known as an unreliable medium. Acknowledgments must be implemented at the higher layers. The format of the MAC frame is shown in the figure.

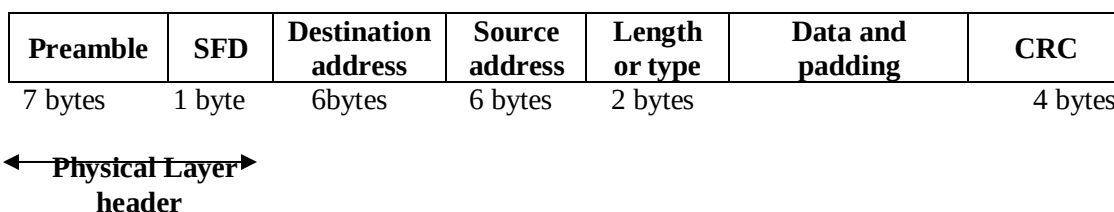


Figure 10.2: Ethernet Frame

- **Preamble.** The first field of the 802.3 frame contains 7 bytes (56 bits) of alternating 0s and 1s that alerts the receiving system to the coming frame and enables it to synchronize its input timing.
- **Start frame delimiter (SFD).** The second field (1 byte: 10101011) signals the beginning of

the frame. The SFD warns the station or stations that this is the last chance for synchronization. The last 2 bits is 11 and alerts the receiver that the next field is the destination address.

- **Destination address (DA).** The DA field is 6 bytes and contains the physical address of the destination station or stations to receive the packet.
- **Source address (SA).** The SA field is also 6 bytes and contains the physical address of the sender of the packet.
- **Length or type.** Define upper layer protocol, length: the number of bytes in data field Data. It is a minimum of 46 and a maximum of 1500 bytes.
- **CRC.** The last field contains error detection information, in this case a CRC-32.

➤ **Frame Length**

Ethernet has imposed restrictions on both the minimum and maximum lengths of a frame. Minimum frame length is 512 bits or 64 bytes and maximum frame length is 12,144 bits or 1518 bytes. It can include payload whose length ranges between 46 bytes and 1500 bytes, as shown in figure.

➤ **Addressing**

- Each station on an Ethernet network (such as a PC, workstation, or printer) has its own **network interface card (NIC)**.
- The NIC fits inside the station and provides the station with a 6-byte physical address.
- As shown in the figure, the Ethernet address is 6 bytes (48 bits), normally written in hexadecimal notation, with a colon between the bytes.

Example: 06:01:02:01:2C:4B

Unicast, Multicast, and Broadcast Addresses:

- A source address is always a unicast address - the frame comes from only one station.
- The destination address, however, can be unicast, multicast, or broadcast.
- If the least significant bit of the first byte in a destination address is 0, the address is unicast; otherwise, it is multicast.

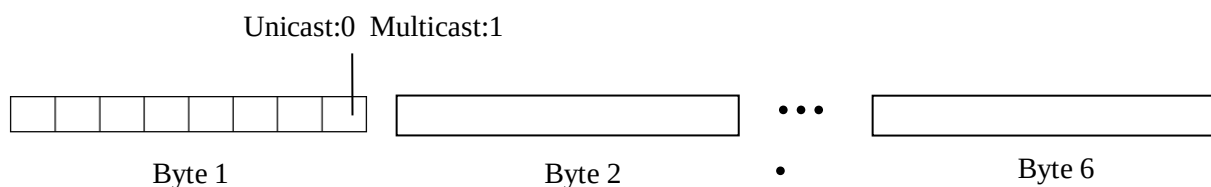


Figure 10.3: Unicast and Multicast Address

- A **unicast** destination address defines only one recipient; the relationship between the sender and the

receiver is one-to-one.

- A **multicast** destination address defines a group of addresses; the relationship between the sender and the receivers is one-to-many.
- The **broadcast** address is a special case of the multicast address; the recipients are all the stations on the LAN. A broadcast destination address is forty-eight bits.

➤ Access Method: CSMA/CD

Standard Ethernet uses 1-persistent CSMA/CD

Slot Time In an Ethernet network, the round-trip time required for a frame to travel from one end of a maximum-length network to the other plus the time needed to send the jam sequence is called the slot time.

Slot time = round-trip time + time required to send the jam sequence

The slot time in Ethernet is defined in bits. It is the time required for a station to send 512 bits. This means that the actual slot time depends on the data rate; for traditional 10-Mbps Ethernet it is 51.2 μs.

Slot Time and Collision The choice of a 512-bit slot time was not accidental. It was chosen to allow the proper functioning of CSMA/CD.

Slot Time and Maximum Network Length There is a relationship between the slot time and the maximum length of the network (collision domain). It is dependent on the propagation speed of the signal in the particular medium.

In most transmission media, the signal propagates at 2×10^8 m/s (two-thirds of the rate for propagation in air). For traditional Ethernet, we calculate

$\text{MaxLength} = \text{Propagation speed} \times (\text{slot time}/2)$ $\text{MaxLength} = (2 \times 10^8) \times (51.2 \times 10^{-6}/2) = 5120\text{m}$
--

Of course, we need to consider the delay times in repeaters and interfaces, and the time required to send the jam sequence. These reduce the maximum-length of a traditional Ethernet network to 2500 m, just 48 percent of the theoretical calculation.

$$\text{MaxLength} = 2500 \text{ m}$$

10.2.2 Physical Layer

The Standard Ethernet defines several physical layer implementations.

Encoding and Decoding

At the **sender**, data are converted to a digital signal using the Manchester scheme; at the **receiver**, the received signal is interpreted as Manchester and decoded into data.

➤ **10Base5: Thick Ethernet**

- Known as Thicknet
- Thick coaxial cable
- Uses bus topology with external transceiver: transceiver is responsible for transmitting, receiving, and detecting collisions.
- Max length of each segment 500m

➤ **10Base2: Thin Ethernet**

- Known as Thin Ethernet
- Uses bus topology with thin and more flexible cable
- Uses internal Transceiver – part of NIC(network interface card) which is installed inside the station.
- Max length of each segment 185m due to the high level of attenuation in thin coaxial cable.
- This implementation is more cost effective than 10Base5 because thin coaxial cable is less expensive than thick coaxial and the tee connections are much cheaper than taps.

➤ **10Base-T: Twisted-Pair Ethernet**

- Known as twisted pair Ethernet
- Uses physical star topology
- Stations connected to hub via two pairs of twisted cable.
- Max length 100m
- Collision here happens in the hub.

➤ **10Base-F: Fiber Ethernet**

- 10Base-F uses a star topology to connect stations to a hub.
- The stations are connected to the hub using two fiber-optic cables.

8.2.3 Changes in the Standard

Bridged Ethernet

Bridges have two effects on an Ethernet LAN: They raise the bandwidth and they separate collision domains.

➤ ***Raising the Bandwidth***

- In an unbridged Ethernet network, the total capacity (10 Mbps) is shared among all stations with a frame to send; the stations share the bandwidth of the network.
- If only one station has frames to send, it benefits from the total capacity (10 Mbps). But if more than

one station needs to use the network, the capacity is shared.

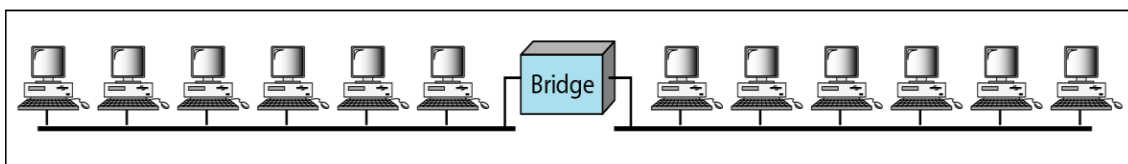
- A **bridge** divides the network into two or more networks. Bandwidth-wise, each network is independent.

For example, in the figure below, a network with 12 stations is divided into two networks, each with 6 stations. Now each network has a capacity of 10 Mbps. The 10-Mbps capacity in each segment is now shared between 6 stations (actually 7 because the bridge acts as a station in each segment), not 12 stations.

- In a network with a heavy load, each station theoretically is offered 10/6 Mbps instead of 10/12 Mbps, assuming that the traffic is not going through the bridge.



a. Without bridging

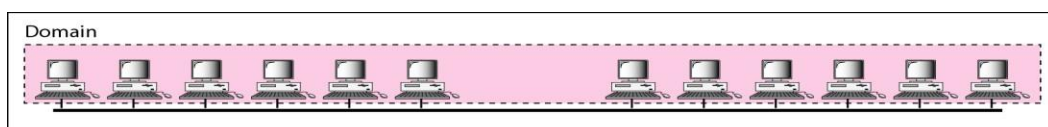


b. With bridging

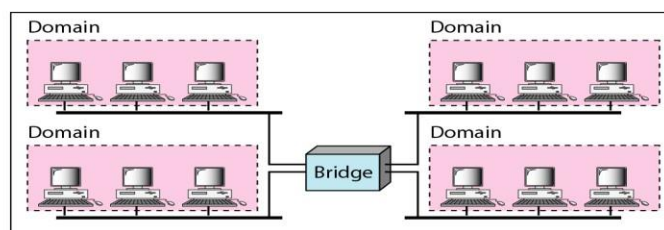
Figure 10.4: Network with and without bridge

➤ **Separating Collision Domains**

- Another advantage of a bridge is the separation of the collision domain.
- The figure below shows the collision domains for an unbridged and a bridged network.
- You can see that the collision domain becomes much smaller and the probability of collision is reduced tremendously. Without bridging, 12 stations contend for access to the medium; with bridging only 3 stations contend for access to the medium.



a. Without bridging



b. With bridging

Figure 10.5: Collision Domain in bridged and unbridged network

Switched Ethernet

- The idea of a bridged LAN can be extended to a switched LAN.
- In other words, if we can have a multiple-port bridge, It has an N-port switch, In this way, the bandwidth is shared only between the station and the switch (5 Mbps each).

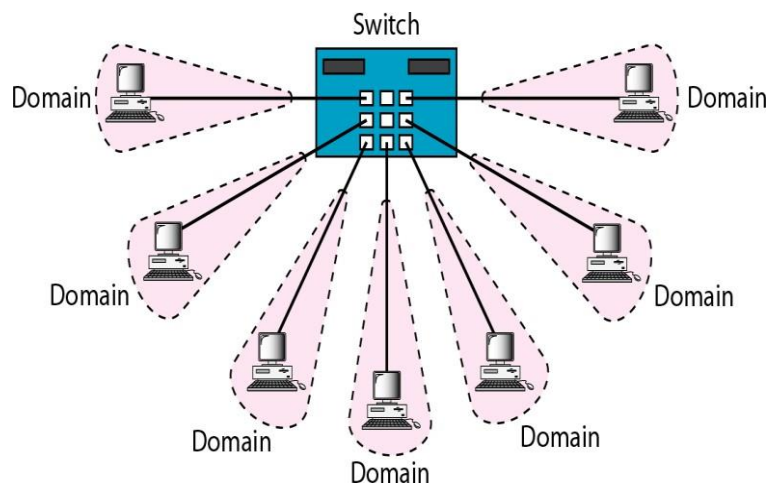


Figure 10.6: Switched Ethernet

Full-Duplex Ethernet

- One of the limitations of 10Base5 and 10Base2 is that communication is half-duplex (10Base-T is always full-duplex); a station can either send or receive, but may not do both at the same time.
- The next step in the evolution was to move from switched Ethernet to full-duplex switched Ethernet. The full-duplex mode increases the capacity of each domain from 10 to 20 Mbps.
- The figure below shows a switched Ethernet in full-duplex mode. Note that instead of using one link between the station and the switch, the configuration uses two links: one to transmit and one to receive.

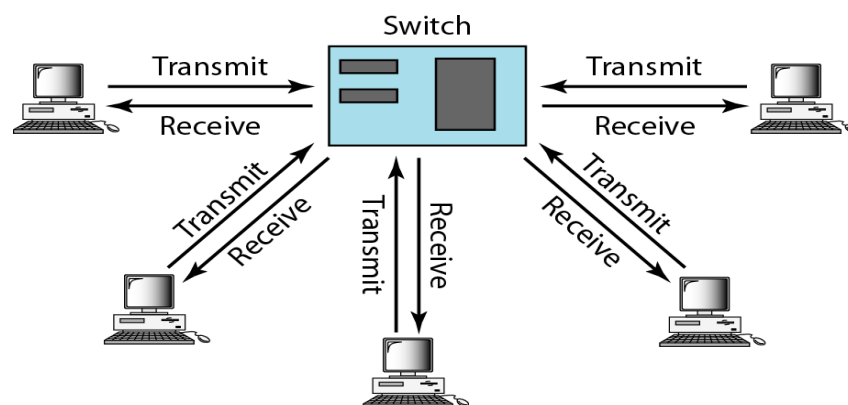


Figure 10.7: Full Duplex Ethernet

10.3 Fast Ethernet

The goals of Fast Ethernet can be summarized as follows:

1. Upgrade the data rate to 100 Mbps.

-
2. Make it compatible with Standard Ethernet.
 3. Keep the same 48-bit address.
 4. Keep the same frame format.
 5. Keep the same minimum and maximum frame lengths.

10.3.1 MAC Sublayer

- For the star topology, there are two choices, as we saw before: half duplex and full duplex.
- In the half-duplex approach, the stations are connected via a hub; in the full-duplex approach, the connection is made via a switch with buffers at each port.
- The access method is the same (CSMA/CD) for the half-duplex approach;
- For full-duplex Fast Ethernet, there is no need for CSMA/CD.

Autonegotiation

A new feature added to Fast Ethernet is called autonegotiation. It allows a station or a hub a range of capabilities. **Autonegotiation allows two devices to negotiate the mode or data rate of operation.**

10.3.2 Physical Layer

Topology

Fast Ethernet is designed to connect two or more stations together.

- If there are only two stations, they can be connected point-to-point.
- Three or more stations need to be connected in a star topology with a hub or a switch at the center

➤ 100Base-TX:

- Uses two pairs of twisted-pair cable (either category 5 UTP or STP).
- MLT-3 scheme was selected since it has good bandwidth performance.
- 4B/5B block coding is used to provide bit synchronization.
- This creates a data rate of 125 Mbps, which is fed into MLT-3 for encoding.

➤ 100Base-FX

- Uses two pairs of fiber-optic cables. Optical fiber can easily handle high bandwidth requirements by using simple encoding schemes.
- NRZ-I encoding scheme for this implementation. However, NRZ-I has a bit synchronization problem for long sequences of 0s (or 1s, based on the encoding).
- To overcome this problem, the designers used 4B/5B block encoding.
- The block encoding increases the bit rate from 100 to 125 Mbps, which can easily be handled by fiber-

optic cable.

➤ **100Base-T4**

- Was designed to use category 3 or higher UTP. The implementation uses four pairs of UTP for transmitting 100 Mbps.
- Encoding/decoding in 100Base-T4 is more complicated. As this implementation uses category 3 UTP, each twisted-pair cannot easily handle more than 25 Mbaud.
- In this design, one pair switches between sending and receiving. Three pairs of UTP category 3, however, can handle only 75 Mbaud (25 Mbaud) each. We need to use an encoding scheme that converts 100 Mbps to a 75 Mbaud signal.
- 8B/6T satisfies this requirement. In 8B/6T, eight data elements are encoded as six signal elements. This means that 100 Mbps uses only $(6/8) \times 100$ Mbps, or 75 Mbaud.

10.4 Gigabit Ethernet

The goals of the Gigabit Ethernet design can be summarized as follows:

1. Upgrade the data rate to 1 Gbps.
2. Make it compatible with Standard or Fast Ethernet.
3. Use the same 48-bit address.
4. Use the same frame format.
5. Keep the same minimum and maximum frame lengths.
6. To support autonegotiation as defined in Fast Ethernet.

10.4.1 MAC Sublayer

Gigabit Ethernet has two distinctive approaches for medium access: half-duplex and full-duplex. Almost all implementations of Gigabit Ethernet follow the full-duplex approach.

Full-Duplex Mode

In full-duplex mode, there is a central switch connected to all computers or other switches. In this mode, each switch has buffers for each input port in which data are stored until they are transmitted. There is no collision in this mode. This means that CSMA/CD is not used.

Half-Duplex Mode

Gigabit Ethernet can also be used in half-duplex mode, although it is rare. In this case, a switch can be replaced by a hub, which acts as the common cable in which a collision might occur. The half-duplex approach uses CSMA/CD.

10.4.2 Physical Layer

Topology

- If there are only two stations, they can be connected point-to-point.
- Three or more stations need to be connected in a star topology with a hub or a switch at the center.

10.5 Ten-Gigabit Ethernet

The goals of the Ten-Gigabit Ethernet design can be summarized as follows:

1. Upgrade the data rate to 10 Gbps.
2. Make it compatible with Standard, Fast, and Gigabit Ethernet.
3. Use the same 48-bit address.
4. Use the same frame format.
5. Keep the same minimum and maximum frame lengths.
6. Allow the interconnection of existing LANs into a metropolitan area network (MAN) or a wide area network (WAN).
7. Make Ethernet compatible with technologies such as Frame Relay and ATM.

10.5.1 MAC Sublayer

Ten-Gigabit Ethernet operates only in full duplex mode which means there is no need for contention; CSMA/CD is not used in Ten-Gigabit Ethernet.

10.5.2 Physical Layer

The physical layer in Ten-Gigabit Ethernet is designed for using fiber-optic cable over long distances. Three implementations are the most common: 10GBase-S, 10GBase-L, and 10GBase-E.

Wireless LANs

11.1 Introduction

11.1.1 Architectural Comparison

- **Medium**

In a wired LAN, wires are used to connect hosts. In a wireless LAN, the medium is air, the signal is generally broadcast. When hosts in a wireless LAN communicate with each other, they are sharing the same medium (multiple access).

- **Hosts**

In a wired LAN, a host is always connected to its network at a point with a fixed link layer address related to its network interface card (NIC).

In a wireless LAN, a host is not physically connected to the network; it can move freely and can use

the services provided by the network.

- **Isolated LANs**

A wired isolated LAN is a set of hosts connected via a link-layer switch. A wireless isolated LAN, called an ad hoc network in wireless LAN terminology, is a set of hosts that communicate freely with each other.

- **Connection to Other Networks**

A wired LAN can be connected to another network or an internetwork such as the Internet using a router. A wireless LAN may be connected to a wired infrastructure network, to a wireless infrastructure network, or to another wireless LAN.

11.1.2 Characteristics

- **Attenuation**

The strength of electromagnetic signals decreases rapidly because the signal disperses in all directions; only a small portion of it reaches the receiver. The situation becomes worse with mobile senders that operate on batteries and normally have small power supplies.

- **Interference**

Another issue is that a receiver may receive signals not only from the intended sender, but also from other senders if they are using the same frequency band.

- **Multipath Propagation**

A receiver may receive more than one signal from the same sender because electromagnetic waves can be reflected back from obstacles such as walls, the ground, or objects.

The result is that the receiver receives some signals at different phases. This makes the signal less recognizable.

- **Error**

With the above characteristics of a wireless network, we can expect that errors and error detection are more serious issues in a wireless network than in a wired network

11.1.3 Access Control

The CSMA/CD algorithm does not work in wireless LANs for three reasons:

1. To detect a collision, a host needs to send and receive at the same time, which means the host needs to work in a duplex mode. Wireless hosts do not have enough power to do so. They can only send or receive at one time.
2. Because of the hidden station problem, in which a station may not be aware of another station's transmission due to some obstacles or range problems, collision may occur but not be detected.

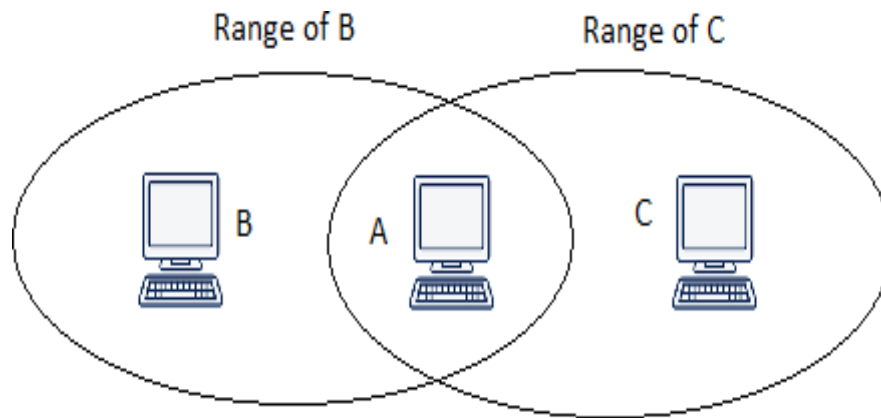


Figure 11.1: Hidden Station Problem

Station C is outside the transmission range of B; likewise, station B is outside the transmission range of C. Station A, however, is in the area covered by both B and C; it can hear any signal transmitted by B or C. Assume that station B is sending data to station A. In the middle of this transmission, station C also has data to send to station A. However, station C is out of B's range and transmissions from B cannot reach C. Therefore C thinks the medium is free. Station C sends its data to A, which results in a collision at A because this station is receiving data from both B and C. In this case, we say that stations B and C are hidden from each other with respect to A. Hidden stations can reduce the capacity of the network because of the possibility of collision.

3. The distance between stations can be great. Signal fading could prevent a station at one end from hearing a collision at the other end.

11.2 IEEE 802.11

IEEE has defined the specifications for a wireless LAN, called IEEE 802.11, which covers the physical and data link layers.

11.2.1 Architecture

The standard defines two kinds of services:

1. The Basic Service Set (BSS)
2. The Extended Service Set (ESS)

Basic Service Set

- IEEE 802.11 defines the basic service set (BSS) as the building block of a wireless LAN.
- A basic service set is made of stationary or mobile wireless stations and an optional central base station, known as the access point (AP).

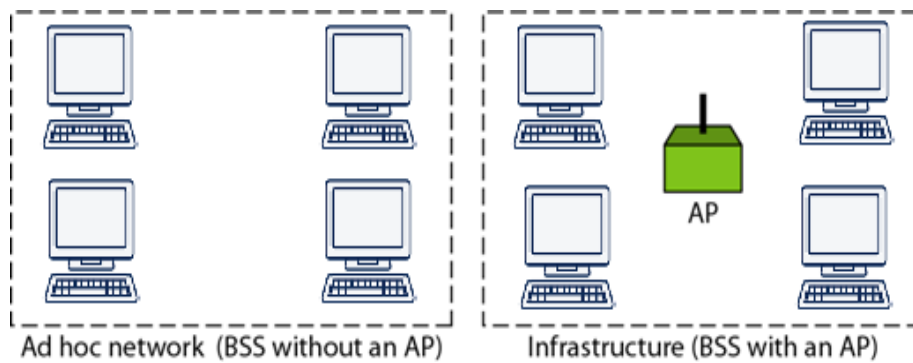


Figure 11.2: Basic Service Set

- The BSS without an AP is a stand-alone network and cannot send data to other BSSs. It is called an *ad hoc architecture*. In this architecture, stations can form a network without the need of an AP; they can locate one another and agree to be part of a BSS.
- A BSS with an AP is sometimes referred to as an *infrastructure* network.

Extended Service Set

- An extended service set (ESS) is made up of two or more BSSs with APs.
- In this case, the BSSs are connected through a distribution system, which is usually a wired LAN.
- The distribution system connects the APs in the BSSs. IEEE 802.11 does not restrict the distribution system; it can be any IEEE LAN such as an Ethernet.
- The extended service set uses two types of stations: mobile and stationary. The mobile stations are normal stations inside a BSS. The stationary stations are AP stations that are part of a wired LAN.

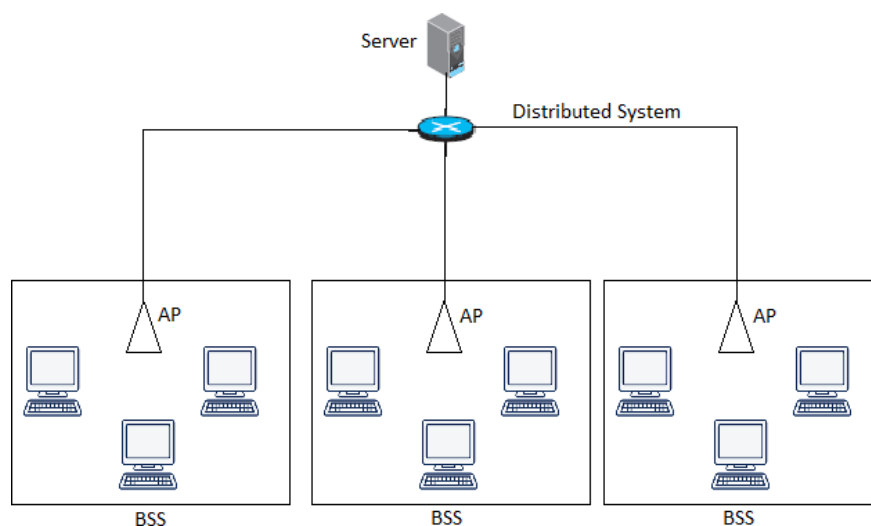


Figure 11.3: Extended Service Set

Station Types

- IEEE 802.11 defines three types of stations based on their mobility in a wireless LAN: no-transition, BSS-transition, and ESS-transition mobility.
- A station with no-transition mobility is either stationary or moving only inside a BSS.
- A station with BSS-transition mobility can move from one BSS to another, but the movement is confined inside one ESS.
- A station with ESS-transition mobility can move from one ESS to another. However, IEEE 802.11 does not guarantee that communication is continuous during the move.

11.2.2 MAC Sub layer

IEEE 802.11 defines two MAC sub layers: the distributed coordination function (DCF) and point coordination function (PCF).

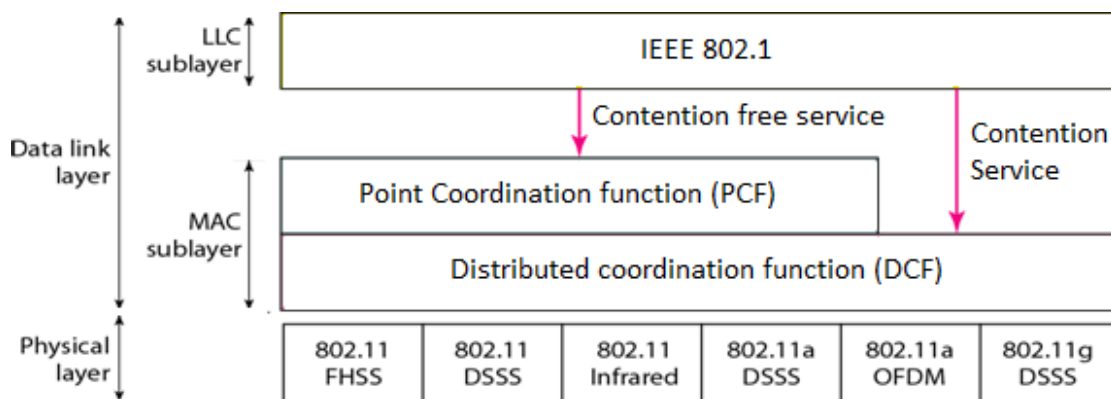


Figure 11.4: MAC Layer in IEEE 802.11

➤ Distributed Coordination Function

One of the two protocols defined by IEEE at the MAC sublayer is called the distributed coordination function (DCF). DCF uses *CSMA/CA* as the access method.

Frame Exchange Time Line The figure shows the exchange of data and control frames in time.

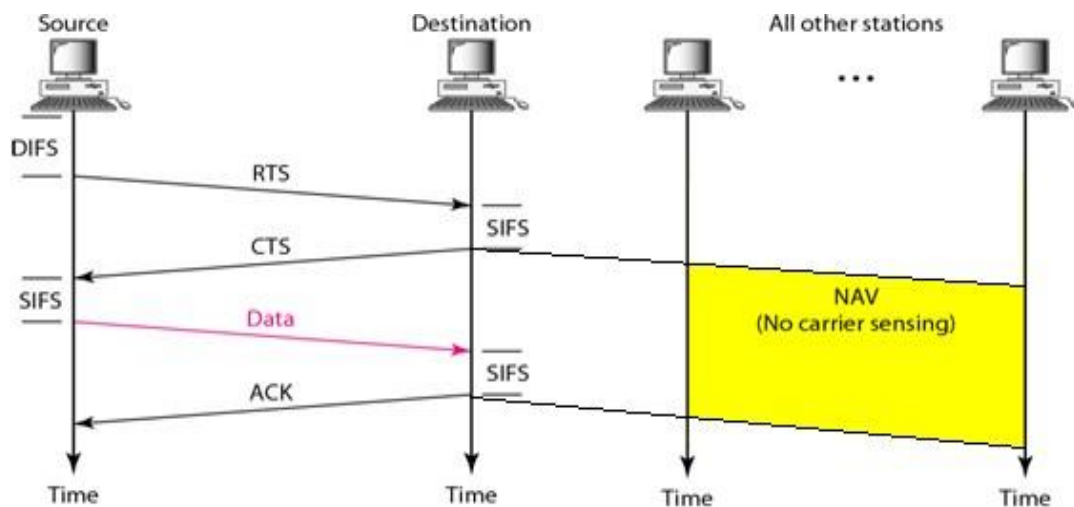


Figure 11.5: CSMA/CA and NAV

1. Before sending a frame, the source station senses the medium by checking the energy level at the carrier frequency.
 - a. The channel uses a persistence strategy with back-off until the channel is idle.
 - b. After the station is found to be idle, the station waits for a period of time called the distributed interframe space (DIFS); then the station sends a control frame called the request to send (RTS).
2. After receiving the RTS and waiting a period of time called the short interframe space (SIFS), the destination station sends a control frame, called the clear to send (CTS), to the source station. This control frame indicates that the destination station is ready to receive data.
3. The source station sends data after waiting an amount of time equal to SIFS.
4. The destination station, after waiting an amount of time equal to SIFS, sends an acknowledgment to show that the frame has been received. Acknowledgment is needed in this protocol because the station does not have any means to check for the successful arrival of its data at the destination. On the other hand, the lack of collision in CSMA/CD is a kind of indication to the source that data have arrived.

Network Allocation Vector:

- When a station sends an RTS frame, it includes the duration of time that it needs to occupy the channel.
- The stations that are affected by this transmission create a timer called a network allocation vector (NAV) that shows how much time must pass before these stations are allowed to check the channel for idleness.
- Each time a station accesses the system and sends an RTS frame, other stations start their NAV.

In other words, each station, before sensing the physical medium to see if it is idle, first checks its NAV to see if it has expired.

Collision during Handshaking

When two or more stations try to send RTS frames at the same time, control frames may collide. However, because there is no mechanism for collision detection, the sender assumes there has been a collision if it has not received a CTS frame from the receiver. The back-off strategy is employed, and the sender tries again.

➤ Point Coordination Function (PCF)

- The point coordination function (PCF) is an optional access method that can be implemented in an infrastructure network (not in an ad hoc network).
- It is implemented on top of the DCF and is used mostly for time-sensitive transmission.
- PCF has a centralized, contention-free polling access method.
- The AP performs polling for stations that are capable of being polled. The stations are polled one after another, sending any data they have to the AP.
- To give priority to PCF over DCF, another set of interframe spaces has been defined: PIFS and SIFS. The SIFS is the same as that in DCF, but the PIFS (PCF IFS) is shorter than the DIFS. This means that if, at the same time, a station wants to use only DCF and an AP wants to use PCF, the AP has priority.
- Due to the priority of PCF over DCF, stations that only use DCF may not gain access to the medium. To prevent this, a repetition interval has been designed to cover both contention-free (PCF) and contention-based (DCF) traffic. T
- The repetition interval, which is repeated continuously, starts with a special control frame, called a beacon frame. When the stations hear the beacon frame, they start their NAV for the duration of the contention-free period of the repetition interval.

Fragmentation

The wireless environment is very noisy; a corrupt frame has to be retransmitted. The protocol, therefore, recommends fragmentation--the division of a large frame into smaller ones. It is more efficient to resend a small frame than a large one.

Frame Format

The MAC layer frame consists of nine fields, as shown.

FC	D	Address 1	Address 2	Address 3	SC	Address 4	Frame body	FCS		
Protocol version	Type	Subtype	To DS	From DS	More Flag	Retry	Power mgmt	More Data	WEP	Rsvd

Figure 11.6: Frame Format

- **Frame control (FC).** The FC field is 2 bytes long and defines the type of frame and some control information. The table describes the subfields.

Field	Explanation
Version	Current Version is 0
Type	Type of Information: Management (00), Control(01), Data(10)
Type	Type of each type
To DS	Distributed System
From DS	From Distributed System
More Flag	More Fragments
Retry	Retransmitted Frame
Power Mgmt	Station is in power management mode
More Data	Station has more data to send
WEP	Wired Equivalence Policy
Rsvd	Reserved

- **ID.** In all frame types except one, this field defines the duration of the transmission that is used to set the value of NAV. In one control frame, this field defines the ID of the frame.
- **Addresses.** There are four address fields, each 6 bytes long. The meaning of each address field depends on the value of the To DS and From DS subfields and will be discussed later.
- **Sequence control.** This field defines the sequence number of the frame to be used in flow control.
- **Frame body.** This field, which can be between 0 and 2312 bytes, contains information based on the type and the subtype defined in the FC field.
- **FCS.** The FCS field is 4 bytes long and contains a CRC-32 error detection sequence.

Frame Types

A wireless LAN defined by IEEE 802.11 has three categories of frames: management frames, control frames, and data frames.

- **Management Frames** Management frames are used for the initial communication between stations and access points.
- **Control Frames** Control frames are used for accessing the channel and acknowledging frames. The figure shows the format.



Figure 11.7: Control Frames

For control frames the value of the type field is 01; the values of the subtype fields for frames we have discussed are shown in the table.

Subtype	Meaning
1011	Request to send (RTS)
1100	Clear to send (CTS)
1101	Acknowledgment (ACK)

- **Data Frames** Data frames are used for carrying data and control information.

11.2.3 Addressing Mechanism

The IEEE 802.11 addressing mechanism specifies four cases, defined by the value of the two flags in the FC field, To DS and From DS. Each flag can be either 0 or 1, resulting in four different situations. The interpretation of the four addresses (address 1 to address 4) in the MAC frame depends on the value of these flags, as shown in the table.

To DS	From DS	Address 1	Address 2	Address 3	Address 4
0	0	Destination	Source	BSS ID	N/A
0	1	Destination	Sending AP	Source	N/A
1	0	Receiving AP	Source	Destination	N/A
1	1	Receiving AP	Sending AP	Destination	Source

- **Case 1:00** In this case, To DS = 0 and From DS = 0. This means that the frame is not going to a distribution system (To DS = 0) and is not coming from a distribution system (From DS = 0). The frame is going from one station in a BSS to another without passing through the distribution

system. The ACK frame should be sent to the original sender.

- **Case 2:01** In this case, To DS = 0 and From DS = 1. This means that the frame is coming from a distribution system (From DS = 1). The frame is coming from an AP and going to a station. The ACK should be sent to the AP.
- **Case 3:10** In this case, To DS = 1 and From DS = 0. This means that the frame is going to a distribution system (To DS = 1). The frame is going from a station to an AP. The ACK is sent to the original station.
- **Case 4:11** In this case, To DS = 1 and From DS = 1. This is the case in which the distribution system is also wireless. The frame is going from one AP to another AP in a wire-less distribution system.

Exposed Station Problem

In the figure, station A is transmitting to station B. Station C has some data to send to station D, which can be sent without interfering with the transmission from A to B. However, station C is exposed to transmission from A; it hears what A is sending and thus refrains from sending. In other words, C is too conservative and wastes the capacity of the channel.

Station C hears the RTS from A, but does not hear the CTS from B. Station C, after hearing the RTS from A, can wait for a time so that the CTS from B reaches A; it then sends an RTS to D to show that it needs to communicate with D. Both stations B and A may hear this RTS, but station A is in the sending state, not the receiving state. Station B, however, responds with a CTS. The problem is here. If station A has started sending its data, station C cannot hear the CTS from station D because of the collision; it cannot send its data to D. It remains exposed until A finishes.

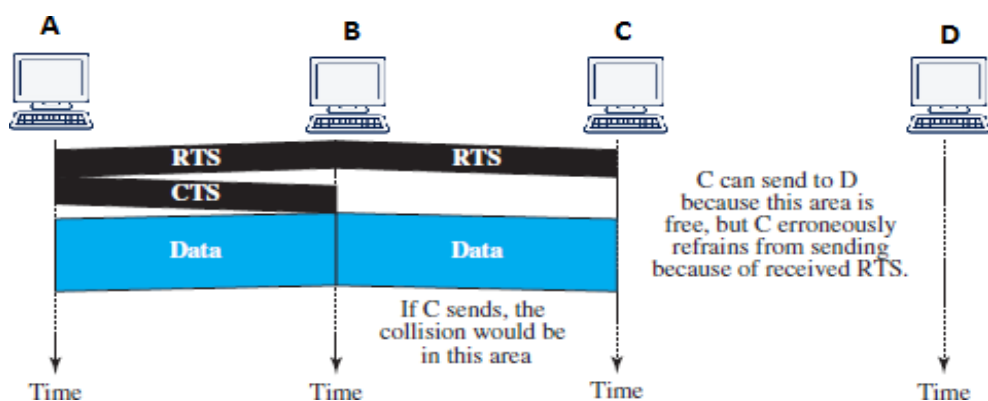


Figure 11.8: Exposed Station Problem

11.2.4 Physical layer:

➤ **IEEE 802.11 FHSS(Frequency-hopping spread spectrum)**

- In Frequency Hopping Spread Spectrum (FHSS) the sender sends on one carrier frequency for a short amount of time, then hops to another carrier frequency for the same amount of time, and so on. After N hop-pings, the cycle is repeated.
- It is Operating in 2.4 GHz ISM band
- Lower cost, power consumption
- Most tolerant to signal interference
- Spreading makes it difficult for unauthorized persons to make sense of the transmitted data

➤ **IEEE 802.11 DSSS (Direct-sequence spread spectrum)**

- In Direct Sequence Spread Spectrum (DSSS) each bit sent by the sender is replaced by a sequence of bits called a chip code.
- To avoid buffering, the time needed to send one chip code must be the same as the time needed to send one original bit.
- DSSS is implemented at the physical layer and uses a 2.4GHz ISM band
- Supports higher data rates
- More range than FH or IR physical layers

➤ **IEEE 802.11 Infrared**

- Lowest cost
- Lowest range compared to spread spectrum
- Doesn't penetrate walls, so no eavesdropping

➤ **IEEE 802.11a**

- IEEE 802.11a describes the orthogonal frequency-division multiplexing (OFDM) method for signal generation in the 5GHz ISM band
- OFDM is the same as FDM with one major difference:
 - All the sub bands are used by one source at a given time
 - Sources contend with one another at the data link layer for access
- OFDM uses PSK (18Mbps) and QAM (54Mbps) for modulation

➤ **IEEE 802.11b**

- IEEE 802.11b describes the high-rate DSSS method for signal generation at 2.4GHz ISM band.
- This is similar to DSSS except for the encoding method, which is called **complementary code keying (CCK)**
- CCK encodes 4 or 8 bits to one CCK symbol

➤ IEEE 802.11g

- 802.11g operates in 2.4 GHz band
- Provides data rates of 22 and 54 Mbps
- Uses orthogonal frequency division multiplexing (OFDM)

11.3 Bluetooth

Bluetooth is a wireless LAN technology designed to connect devices of different functions such as telephones, notebooks, computers, cameras, printers, coffee makers, and so on. A Bluetooth LAN is an ad hoc network, which means that the network is formed spontaneously.

11.3.1 Architecture

Bluetooth defines two types of networks: piconet and scatternet.

Piconet:

- A Bluetooth network is called a piconet, or a small net.
- It can have up to eight stations, one of which is called the master; the rest are called slaves.
- Maximum of seven slaves. Only one master.
- Slaves synchronize their clocks and hopping sequence with the master.

But an additional eight slaves can stay in parked state, which means they can be synchronized with the master but cannot take part in communication until it is moved from the parked state

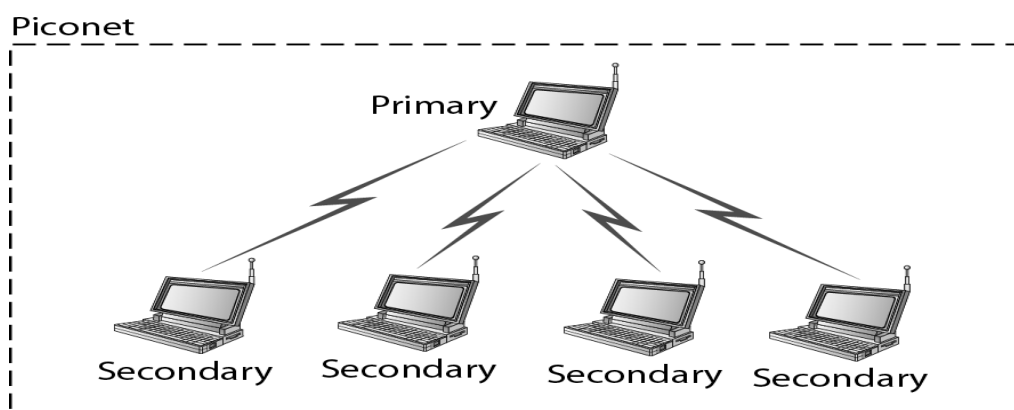
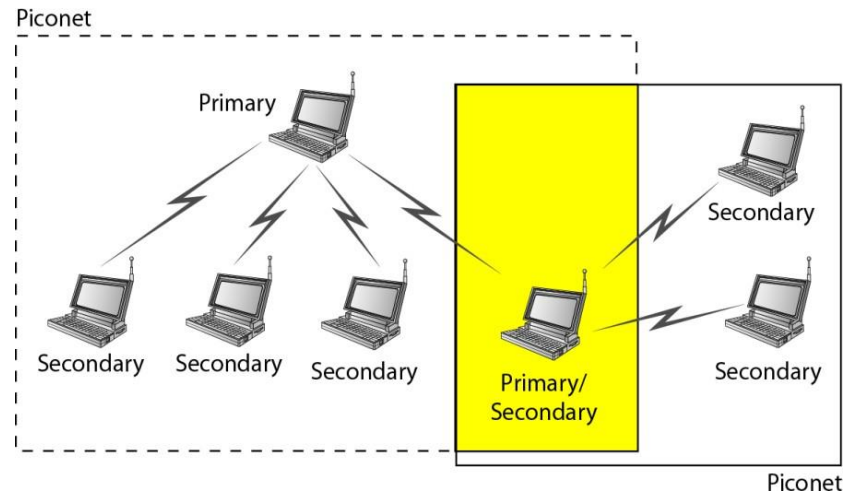


Figure 11.9: Piconet

Scatternet

- Piconets can be combined to form what is called a scatternet.
- A slave station in one piconet can become the master in another piconet.
- Bluetooth devices has a built-in short-range radio transmitter.

**Figure 11.10:** Scatternet**11.3.2 Bluetooth layers**

- **Radio Layer:** Roughly equivalent to physical layer of the Internet model. Physical links can be synchronous or asynchronous.
 - Uses Frequency-hopping spread spectrum [Changing frequency of usage]. Changes its modulation frequency 1600 times per second.
 - Uses frequency shift keying (FSK) with Gaussian bandwidth filtering to transform bits to a signal.
- **Baseband layer:** Roughly equivalent to MAC sublayer in LANs. Access is using Time Division (Time slots).
 - Length of time slot = dwell time = 625 microsec. So, during one frequency, a sender sends a frame to a slave, or a slave sends a frame to the master.
 - Time division duplexing TDMA (TDD-TDMA) is a kind of half-duplex communication in which the slave and receiver send and receive data, but not at the same time (half-duplex). However, the communication for each direction uses different hops, like walkie-talkies.

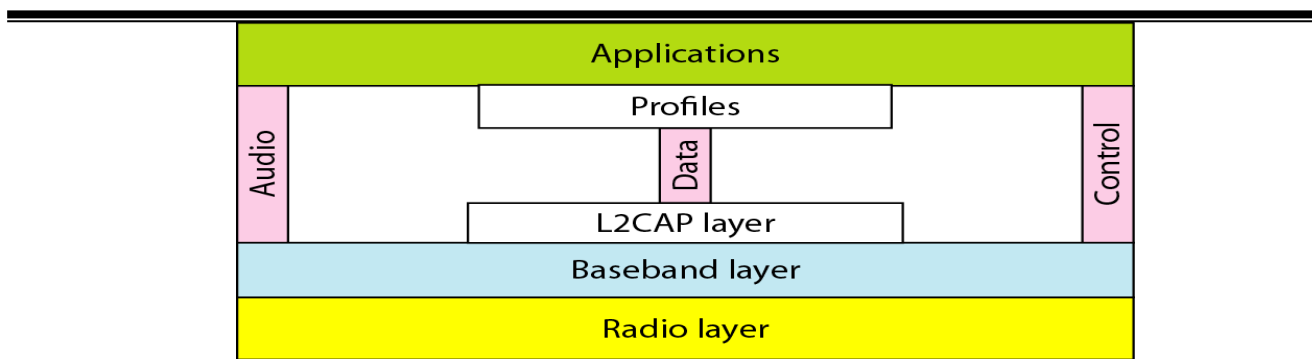


Figure 11.11: Bluetooth Layer

Single-secondary communication

- Master uses even-numbered slots
- Slave uses odd-numbered slots

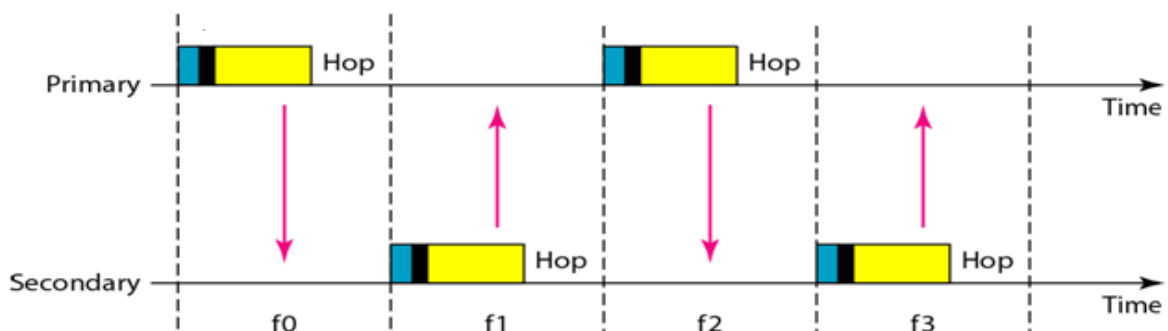


Figure 11.12: Single-secondary communication

Multiple-secondary communication also called Multiple-slave communication

- Master uses even-numbered slots
- Slave sends in the next odd-numbered slot if the packet in the previous slot was addressed to it.

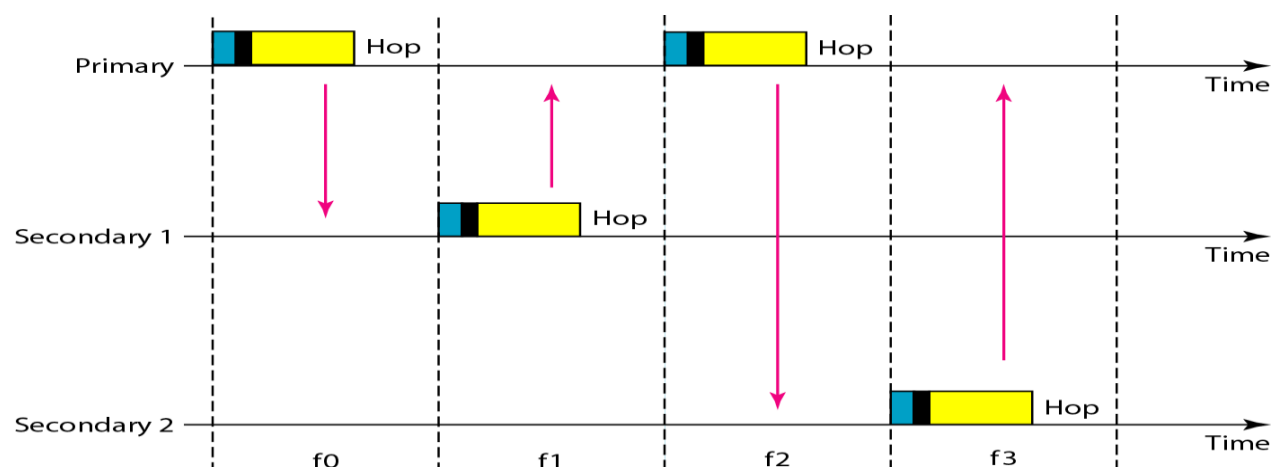


Figure 11.13: Multiple-secondary communication

Physical Links

- Synchronous connection-oriented (SCO)
 - Latency is important than integrity.
 - Transmission using slots.
 - No retransmission.
- Asynchronous connectionless link (ACL)
 - Integrity is important than latency.
 - Does like multiple-slave communication.
 - Retransmission is done.

11.3.3. Frame format

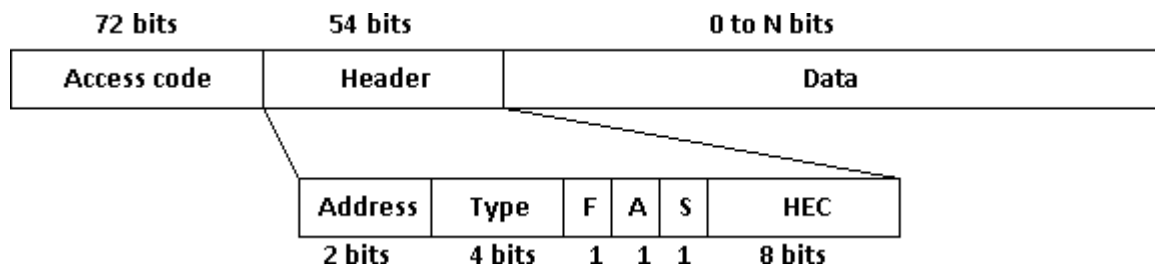


Figure 11.14: Bluetooth Frame format

- **Access code.** This 72-bit field normally contains synchronization bits and the identifier of the primary to distinguish the frame of one piconet from another.
- **Header.** This 54-bit field is a repeated 18-bit pattern. Each pattern has the following subfields:
 1. **Address.** The 3-bit address subfield can define up to seven secondaries (1 to 7). If the address is zero, it is used for broadcast communication from the primary to all secondaries.
 2. **Type.** The 4-bit type subfield defines the type of data coming from the upper layers.
 3. **F.** This 1-bit subfield is for flow control. When set (1), it indicates that the device is unable to receive more frames (buffer is full).
 - A. This 1-bit subfield is for acknowledgment. Bluetooth uses Stop-and-Wait ARQ; 1 bit is sufficient for acknowledgment.
 4. **S.** This 1-bit subfield holds a sequence number. Bluetooth uses Stop-and-Wait ARQ; 1 bit is sufficient for sequence numbering.
 5. **HEC.** The 8-bit header error correction subfield is a checksum to detect errors in each 18-bit header section.

The header has three identical 18-bit sections. The receiver compares these three sections, bit by bit. If each of the corresponding bits is the same, the bit is accepted; if not, the majority opinion rules. This is a form

of forward error correction (for the header only). This double error control is needed because the nature of the communication, via air, is very noisy. Note that there is no retransmission in this sublayer.

- **Payload.** This subfield can be 0 to 2740 bits long. It contains data or control information coming from the upper layers.

L2CAP (Logical Link Control and Adaptation Protocol)

- Equivalent to LLC sublayer in LANs.
- Used for data exchange on ACL Link. SCO channels do not use L2CAP.
- Frame format has 16-bit length [Size of data coming from upper layer in bytes], channel ID, data and control.
- Can do Multiplexing, segmentation and Reassembly, QoS [with no QoS, best-effort delivery is provided] and Group management [Can do like multicast group, using some kind of logical addresses].

L2CAP data packet format

2 bytes	2 bytes	0 to 65,535 bytes
Length	Channel ID	Data and Control

Figure 11.15: L2CAP data packet format

Other Wireless Networks

12.1 Cellular Telephony

- Cellular telephony is designed to provide communications between two moving units, called mobile stations (MSs), or between one mobile unit and one stationary unit, often called a land unit.
- A service provider must be able to locate and track a caller, assign a channel to the call, and transfer the channel from base station to base station as the caller moves out of range.
- To make this tracking possible, each cellular service area is divided into small regions called cells.
- Each cell contains an antenna and is controlled by a solar or AC powered network station, called the base station (BS).
- Each base station, in turn, is controlled by a switching office, called a mobile switching center (MSC).
- The MSC coordinates communication between all the base stations and the telephone central

office. It is a computerized center that is responsible for connecting calls, recording call information, and billing.

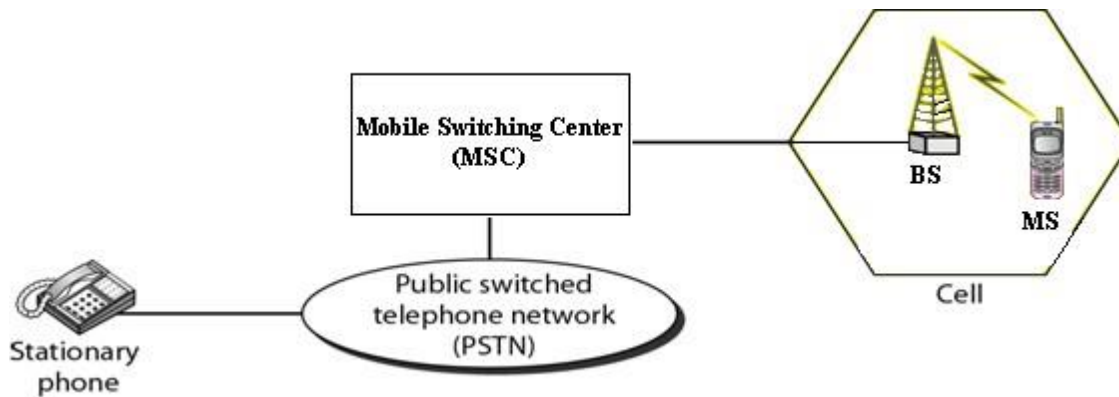


Figure 12.1: Cellular System

- Cell size is not fixed and can be increased or decreased depending on the population of the area. The typical radius of a cell is 1 to 12 mi. High-density areas require more, geographically smaller cells to meet traffic demands than do low-density areas.
- The transmission power of each cell is kept low to prevent its signal from interfering with those of other cells.

12.1.1 Operation

Frequency-Reuse Principle

- Neighboring cells cannot use the same set of frequencies for communication because it may create interference for the users located near the cell boundaries.
- The set of frequencies available is limited, and frequencies need to be reused.
- A frequency reuse pattern is a configuration of N cells, N being the **reuse factor**, in which each cell uses a unique set of frequencies.
- When the pattern is repeated, the frequencies can be reused. There are several different patterns.

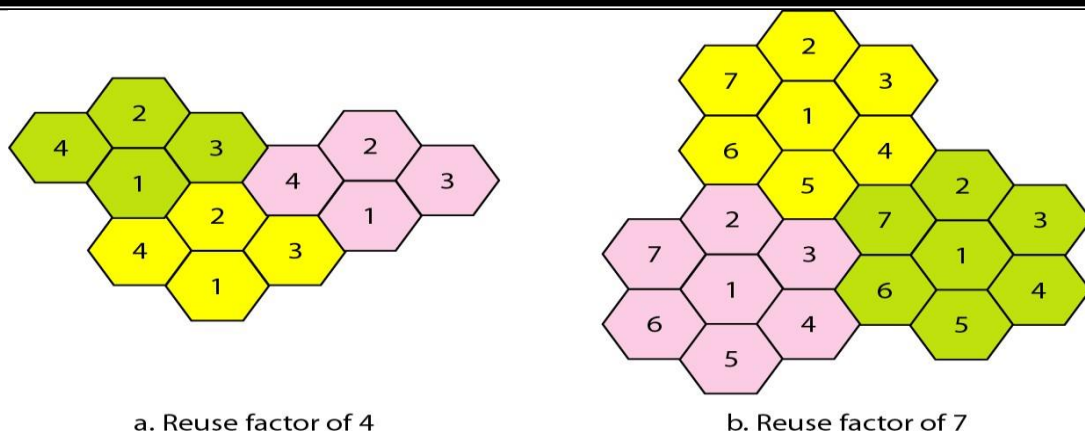


Figure 12.2: Frequency reuse patterns

Transmitting

- To place a call from a mobile station, the caller enters a code of 7 or 10 digits (a phone number) and presses the send button.
- The mobile station then scans the band, seeking a setup channel with a strong signal, and sends the data (phone number) to the closest base station using that channel.
- The base station relays the data to the MSC. The MSC sends the data on to the telephone central office.
- If the called party is available, a connection is made and the result is relayed back to the MSC. At this point, the MSC assigns an unused voice channel to the call, and a connection is established.
- The mobile station automatically adjusts its tuning to the new channel, and communication can begin.

Receiving

- When a mobile phone is called, the telephone central office sends the number to the MSC.
- The MSC searches for the location of the mobile station by sending query signals to each cell in a process called *paging*.
- Once the mobile station is found, the MSC transmits a ringing signal and, when the mobile station answers, assigns a voice channel to the call, allowing voice communication to begin.

Handoff

- It may happen that, during a conversation, the mobile station moves from one cell to another. When it does, the signal may become weak. To solve this problem, the MSC monitors the level of the signal every few seconds.

-
- If the strength of the signal diminishes, the MSC seeks a new cell that can better accommodate the communication.
 - The MSC then changes the channel carrying the call
 - **Hard Handoff:** Early systems used a hard handoff. In a hard handoff, a mobile station only communicates with one base station. When the MS moves from one cell to another, communication must first be broken with the previous base station before communication can be established with the new one. This may create a rough transition.
 - **Soft Handoff:** New systems use a soft handoff. In this case, a mobile station can communicate with two base stations at the same time. This means that, during handoff, a mobile station may continue with the new base station before breaking off from the old one.

Roaming

One feature of cellular telephony is called roaming. Roaming means, in principle, that a user can have access to communication or can be reached where there is coverage. A service provider usually has limited coverage. Neighboring service providers can provide extended coverage through a roaming contract. The situation is similar to snail mail between countries. The charge for delivery of a letter between two countries can be divided upon agreement by the two countries.

12.1.2 First Generation

AMPS - Advanced Mobile Phone System

- AMPS operates in the ISM 800-MHz band. The system uses two separate analog channels, one for forward (base station to mobile station) communication and one for reverse (mobile station to base station) communication. The band between 824 and 849 MHz carries reverse communication; the band between 869 and 894 MHz carries forward communication.
- Each band is divided into 832 channels.
- AMPS uses FM and FSK for modulation.

12.1.3 Second Generation

To provide higher-quality (less noise-prone) mobile voice communications, the second generation of the cellular phone network was developed.

While the first generation was designed for analog voice communication, the second generation was mainly designed for digitized voice.

1) Digital –AMPS

- D-AMPS was designed to be backward-compatible with AMPS.
- D-AMPS uses the same bands and channels as AMPS.
- Each voice channel is digitized using a very complex PCM and compression technique.
- D-AMPS uses TDMA and FDMA

2) GSM - Global System for Mobile Communication

- GSM uses two bands for duplex communication. Each band is 25 MHz in width, shifted toward 900 MHz.
- Each band is divided into 124 channels of 200 kHz separated by guard bands.
- Each voice channel is digitized and compressed to a 13-kbps digital signal.
- GSM uses TDMA and FDMA.
- GSM allows a reuse factor as low as 3.

3) IS – 95: Interim Standard-95

- It is based on CDMA and DSSS.
- IS-95 uses two bands for duplex communication. The bands can be the traditional ISM 800-MHz band or the ISM 1900-MHz band.
- Each band is divided into 20 channels of 1.228 MHz separated by guard bands.
- All base channels need to be synchronized to use CDMA. To provide synchronization, bases use the services of GPS.
- IS-95 has two different transmission techniques: one for use in the forward (base to mobile) direction and another for use in the reverse (mobile to base) direction.
- In the forward direction, communications between the base and all mobiles are synchronized; the base sends synchronized data to all mobiles.
- The synchronization is not used in the reverse direction because we need an entity to do that, which is not feasible. Instead of CDMA, the reverse channels use DSSS.

12.1.4 Third Generation

- The third generation of cellular telephony refers to a combination of technologies that provide a variety of services.
- Using a small portable device, a person should be able to talk to anyone else in the world with a

voice quality similar to that of the existing fixed telephone network. A person can download and watch a movie, can download and listen to music, can surf the Internet or play games, can have a video conference, and can do much more. One of the interesting characteristics of a third generation system is that the portable device is always connected.

- Voice quality comparable to that of the existing public telephone network.
- Data rate of 144 kbps for access in a moving vehicle (car), 384 kbps for access as the user walks (pedestrians), and 2 Mbps for the stationary user (office or home).
- Support for packet-switched and circuit-switched data services.
- It uses Bandwidths of 2 MHz.
- Interface to the Internet.
- The main goal of third-generation cellular telephony is to provide universal personal communication.

IMT-2000 Radio Interface:

- **IMT-DS** This approach uses a version of COMA called wideband COMA or W-COMA. W-COMA uses a 5-MHz bandwidth.
- **IMT-MC** It is an evolution of COMA technology used in IS-95 channels. It combines the new wideband (15-MHz) spread spectrum with the narrowband (1.25-MHz) COMA of IS-95. It is backward-compatible with IS-95. It allows communication on multiple 1.25-MHz channels (1, 3, 6, 9, 12 times), up to 15 MHz. The use of the wider channels allows it to reach the 2-Mbps data rate defined for the third generation.
- **IMT-TC** This standard uses a combination of W-COMA and TDMA. The standard tries to reach the IMT-2000 goals by adding TMA multiplexing to W-COMA.
- **IMT-SC** This standard only uses TMA.
- **IMT-FT** This standard uses a combination of FDMA and TMA.

12.1.5 Fourth Generation (4G)

Some of the objectives defined by the 4G working group are as follows:

- A spectrally efficient system.
- High network capacity.
- Data rate of 100 Mbit/s for access in a moving car and 1 Gbit/s for stationary users.
- Data rate of at least 100 Mbit/s between any two points in the world.
- Smooth handoff across heterogeneous networks.

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- Seamless connectivity and global roaming across multiple networks.
 - High quality of service for next generation multimedia support (quality of service will be discussed in Chapter 30).
 - Interoperability with existing wireless standards.
 - All IP, packet-switched, networks.

The fourth generation is only packet-based (unlike 3G) and supports IPv6. This provides better multicast, security, and route optimization capabilities.

Access Scheme

Orthogonal FDMA (OFDMA) and interleaved FDMA (IFDMA) are being considered respectively for the downlink and uplink of the next generation Universal Mobile Telecommunications System (UMTS). Similarly, multicarrier code division multiple access (MC-CDMA) is proposed for the IEEE 802.20 standard.

Modulation

More efficient quadrature amplitude modulation (64-QAM) is being proposed for use with the Long Term Evolution (LTE) standards.

Radio System

The fourth generation uses a Software Defined Radio (SDR) system.

Antenna

The multiple-input multiple-output (MIMO) and multiuser MIMO (MU-MIMO) antenna system, a branch of intelligent antenna, is proposed for 4G. Using this antenna system together with special multiplexing, 4G allows independent streams to be transmitted simultaneously from all the antennas to increase the data rate into multiple folds. MIMO also allows the transmitter and receiver coordinates to move to an open frequency when interference occurs.